

Bounding the AI Impact Debate:

Capabilities, Adoption, Jobs and Access

Filotas Theodosiou



The AI Debate.

International Conference on Large-Scale AI Risks

📅 [May 26, 2025 12:30 PM to May 28, 2025 01:15 PM](#) (Europe/Brussels / UTC200)

📍 Institute of Philosophy

👤 Lode Lauwaert

📅 [iCal](#)

AI's Large-Scale Risks: Control, Governance, and Ethics

While advanced AI has large beneficial potential that attracts billions in both private and public funding, the academic community interested in the understanding and mitigation of the technology's large-scale risks, such as enabling bioterrorism and the loss of human control over the technology, is still tiny in comparison. Addressing this important global problem requires experts from various disciplines to cooperate on the identification of issues and solutions. While the academic community has recently experienced growth, it is still largely centered around the US and UK. Moreover, disciplines remain largely disconnected, since they present their work at their specific venues. This academic conference on AI's large-scale risks brings together experts from various disciplines to discuss technical, social, and philosophical topics around the control, governance, and ethics of advanced AI. Putting a strong emphasis on the interdisciplinary nature of the topic, the conference will include an interdisciplinary panel discussion on potential symbiotic effects of interdisciplinary AI risk research.

The two sides of the coin



AGI

Endless €€
No more working

Extinction

No critical thinking
No jobs

The two sides of the coin

Interview

Demis Hassabis on our AI future: 'It'll be 10 times bigger than the Industrial Revolution - and maybe 10 times faster'

Steve Rose

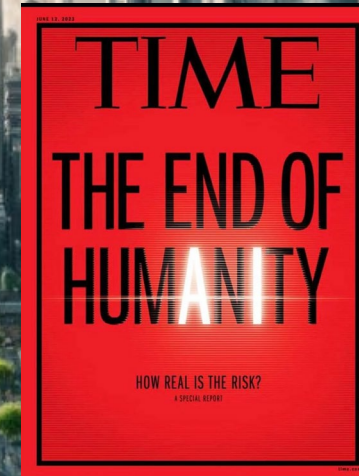


Dario Amodèi

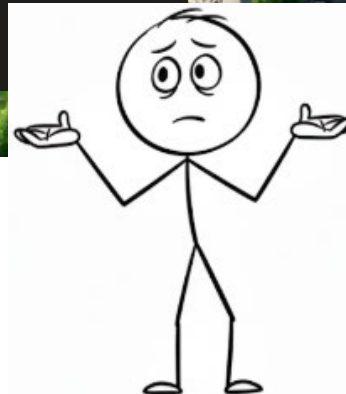
Machines of Loving Grace¹

How AI Could Transform the World for the Better

October 2024



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L+

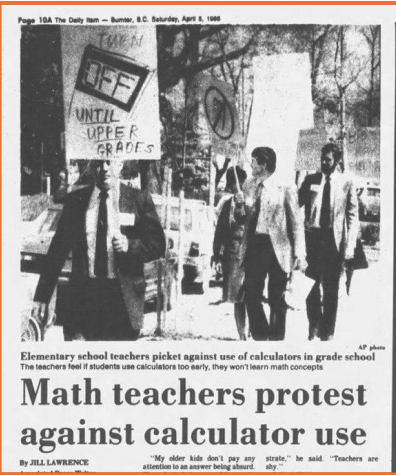
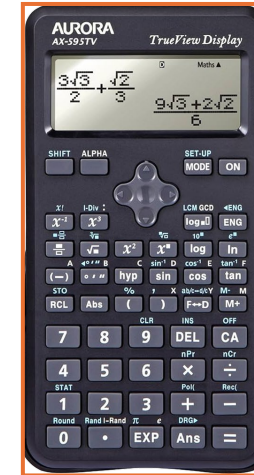
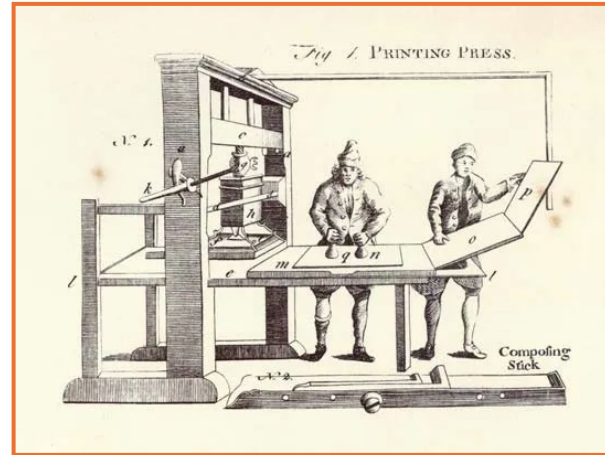
Associated Press

By PETER F. DRUCKER

Robert L. Fansler
ice President and

Are these debates new?

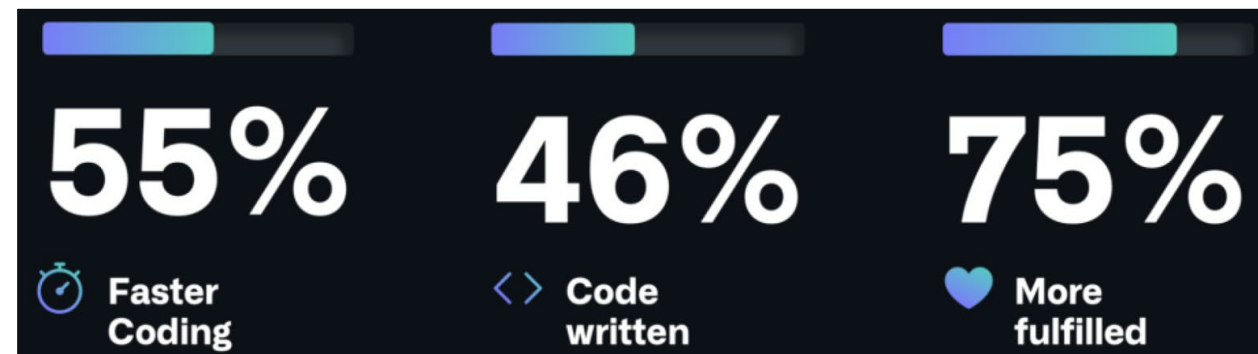
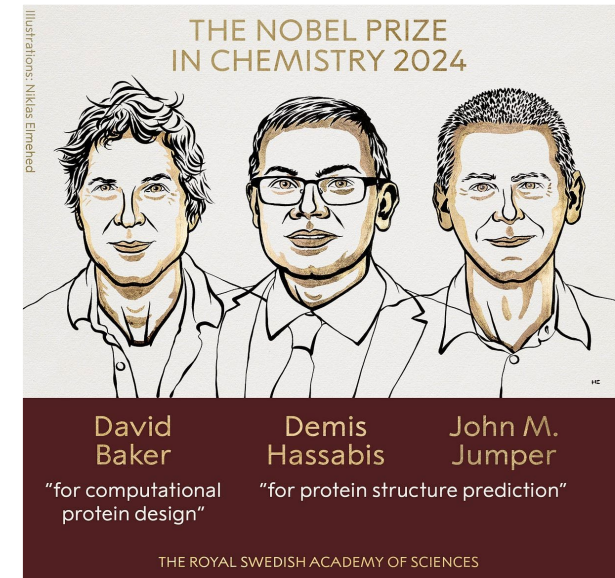
- Printing Press
 - Fear of Information overload
 - Fear of job loss
 - Widespread “misinformation”
- Calculator
 - Students would stop thinking
 - Over reliance on machines
 - Equity issues
- Computer/mobile/internet
 - All the above +
 - Privacy
 - Over-dependance + ...



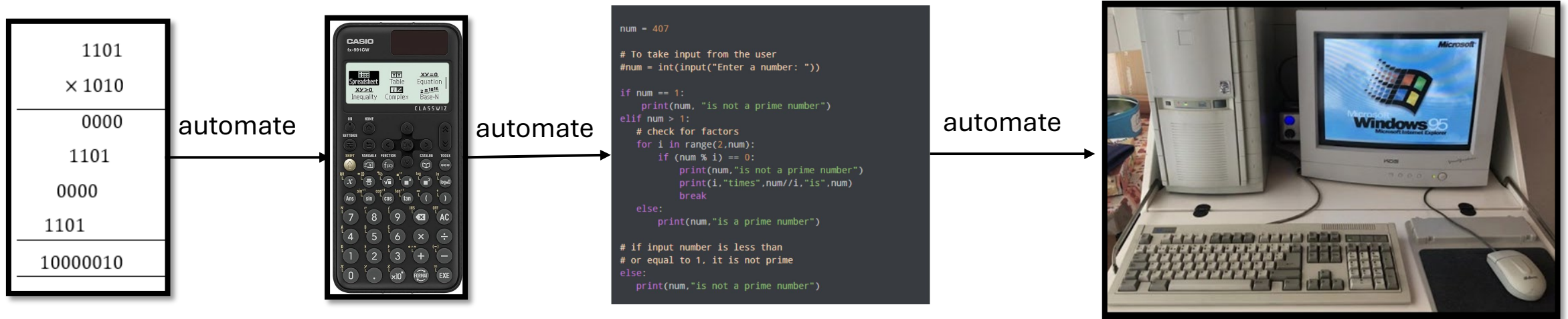
Ah, by the way, these technologies (and their tools) changed the world!

Do you mean the world is changing again?

- Education
- Science
- Productivity
- Information



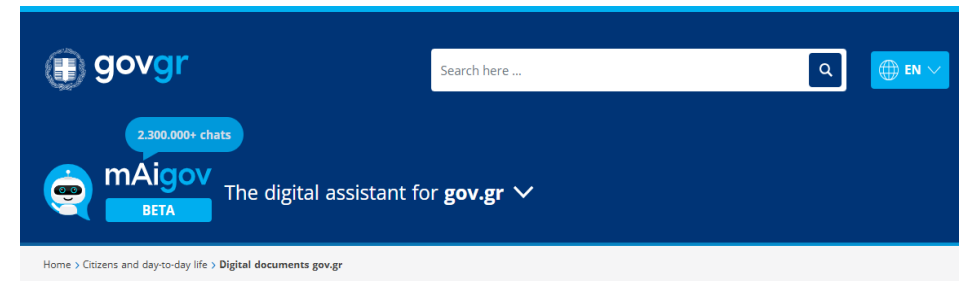
Understanding technological changes



With computers we moved tasks into the digital world

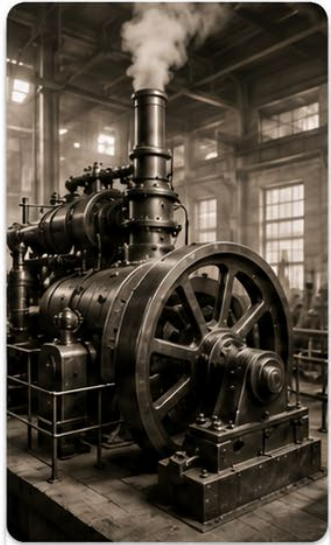
- Admin
- Writing/Coding
- Teaching

Then we built software (programs) to solve them



So what is next?

Lets make a “forecast”



1

Machines

Automated
productivity tasks



2

Computers

Moved many tasks
into the digital world



3

Internet

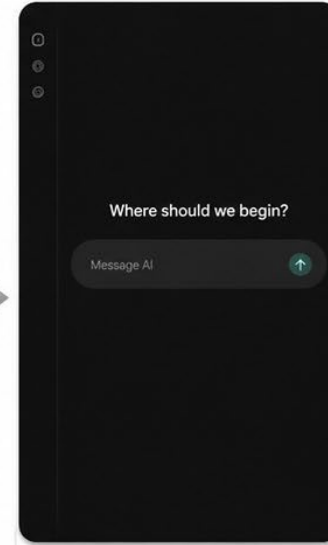
Connected digital
systems



4

Cloud & Mobile

Made software
always available



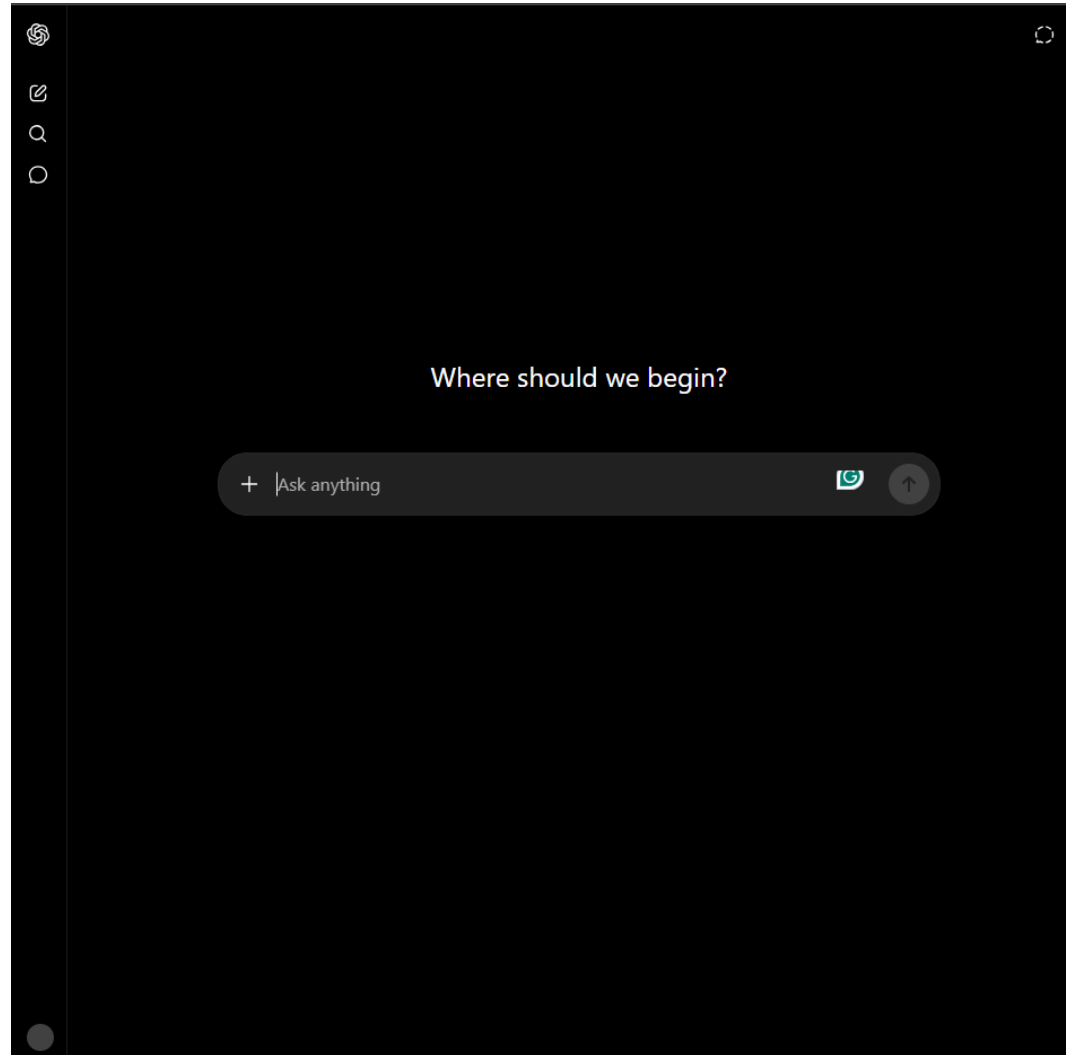
5

AI Agents

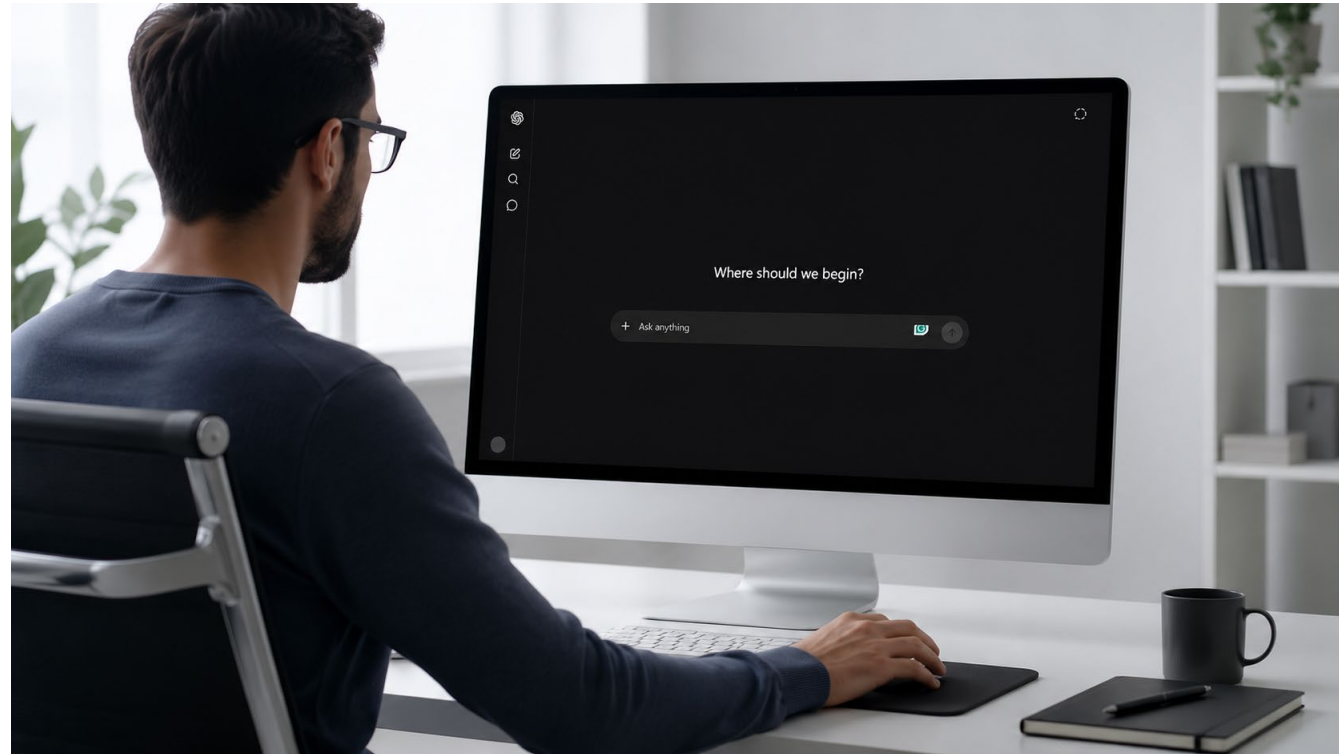
Automate task execution
in the digital world

- Answer a question (with or without web search)
- Draft a document (multiple iterations)
- Code an application
- Research a problem
- Perform some online set of actions (make a reservation)

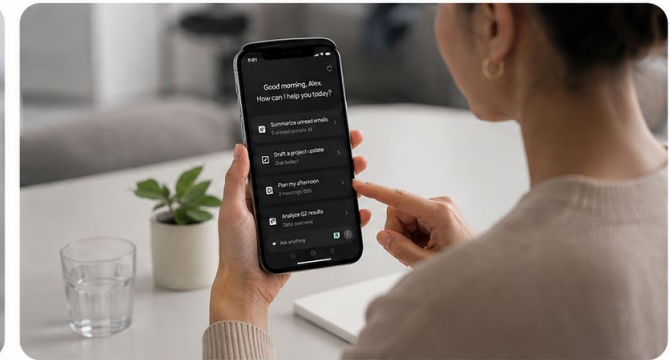
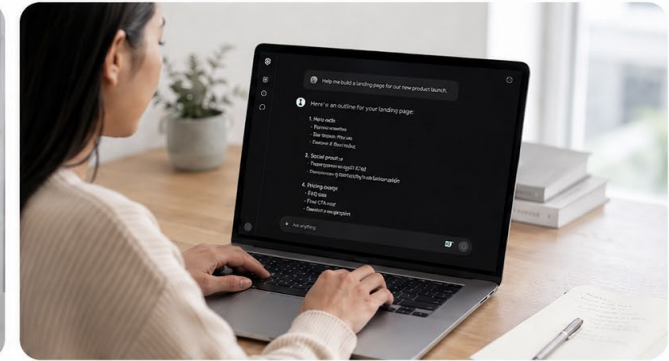
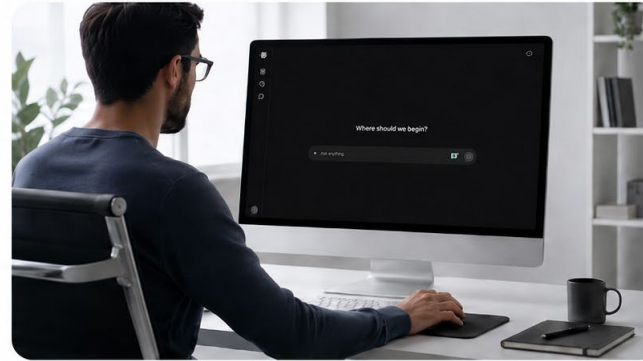
What is automated task execution in the digital world?



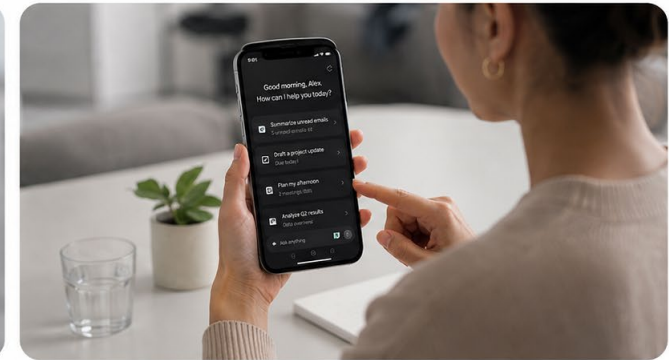
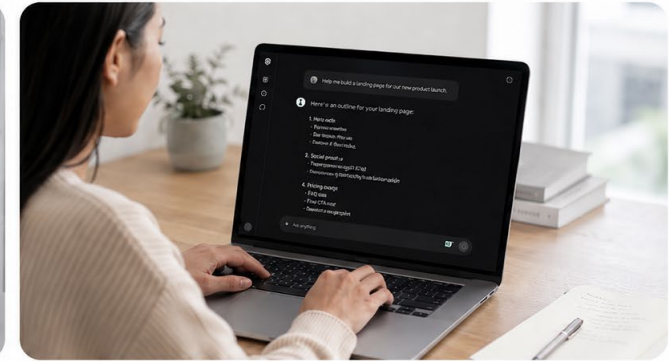
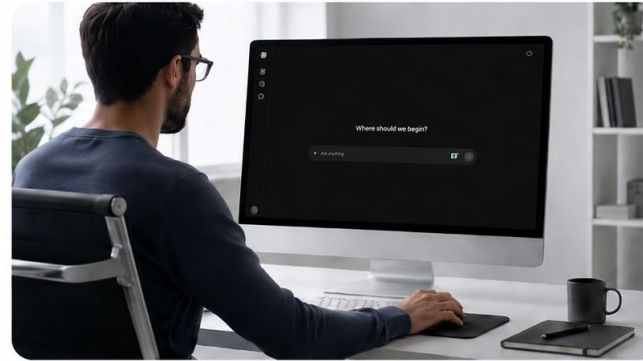
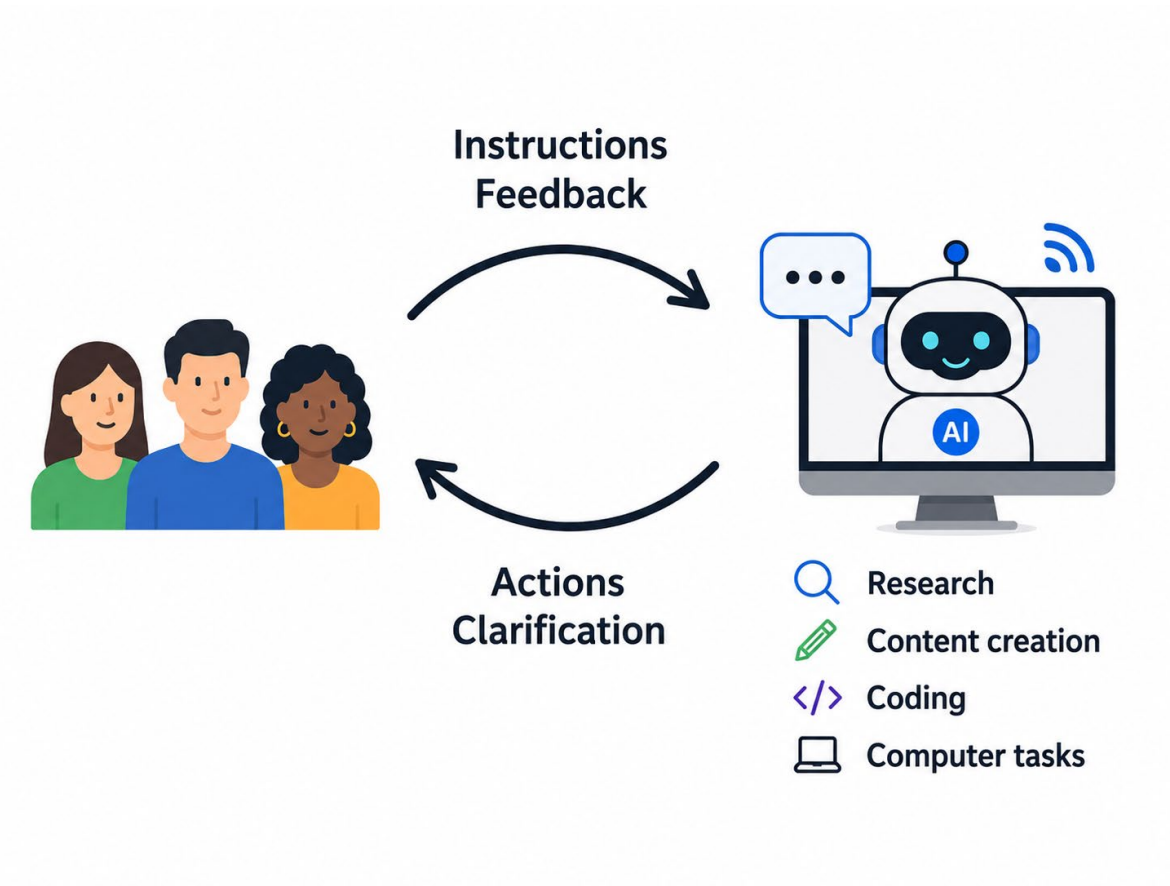
What is automated task execution in the digital world?



What is automated task execution in the digital world?



What is automated task execution in the digital world?



The framework I am going to use.

We are still far from embodied intelligence

- Data scarcity
- Real world is unforgiving
- Current models do not generalize
- Hardware
- Safety / trust

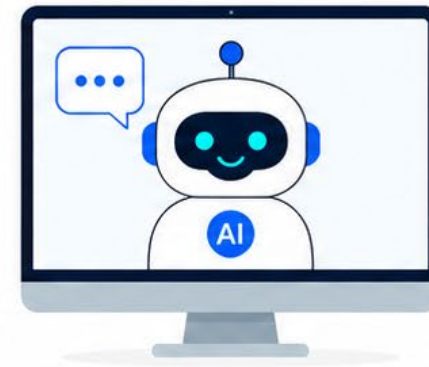
Physical world



Human goals:

- Improve quality of life
- Earn money
- Publish more
- Other...

Digital world



Digital tasks to achieve goals

-  Research
-  Content creation
-  Coding
-  Computer tasks

Instructions /
Feedback

Actions /
Clarification

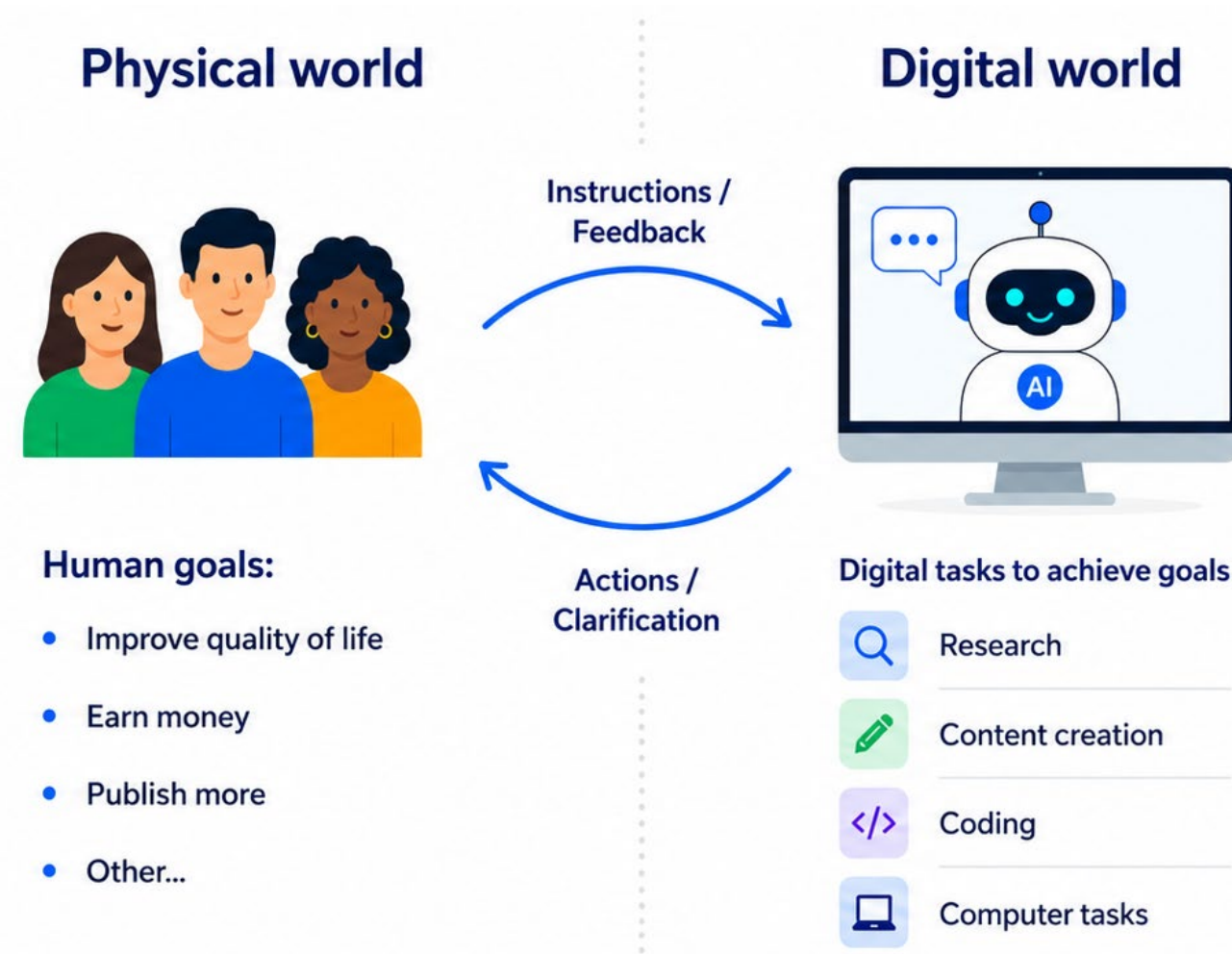
AI Progress

~~AGI
happens
here~~

Content of the talk

Part 2:

1. What are the implications of AI?
2. Why predicting the implications is very hard and what we can do instead?



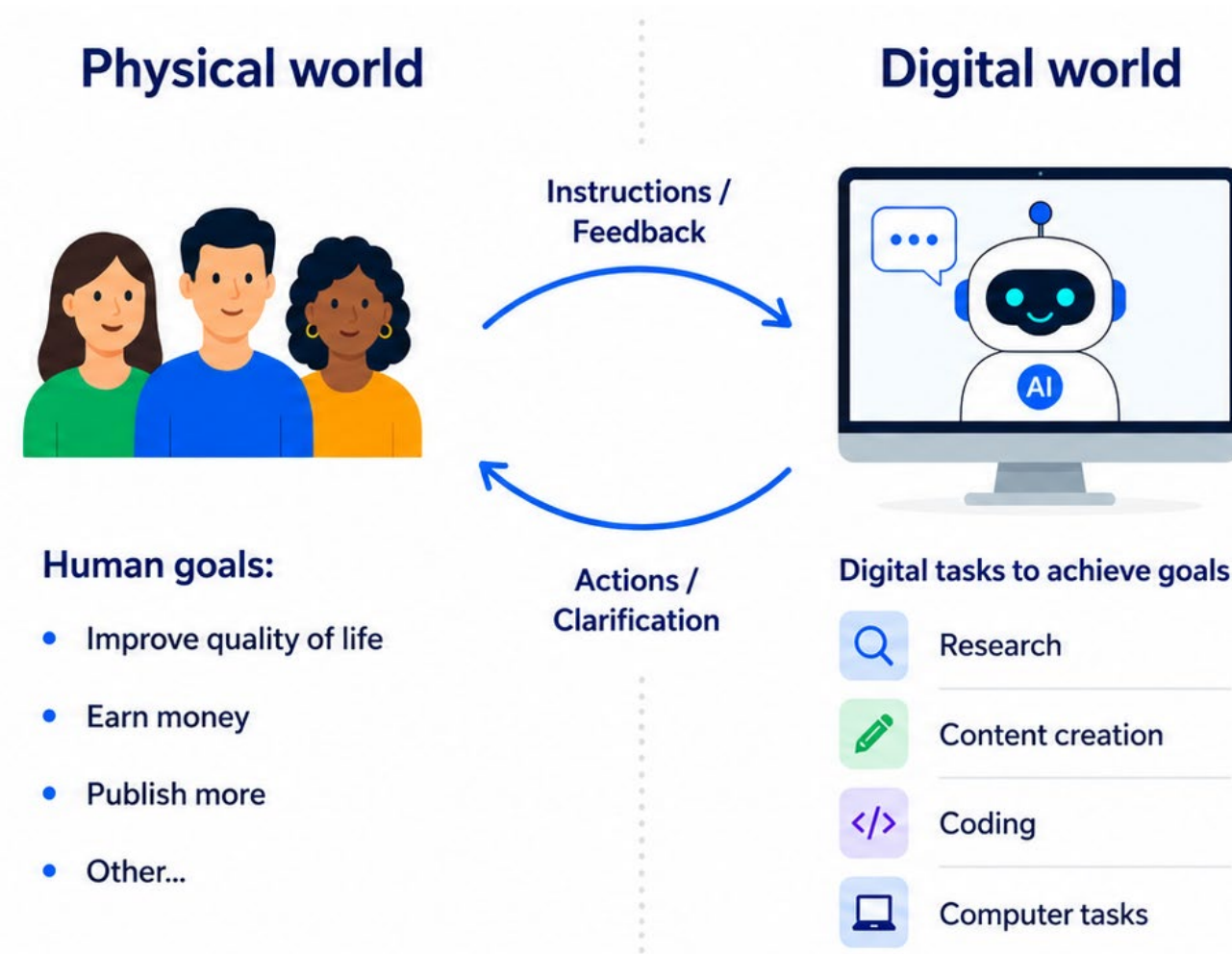
Part 1:

1. What is this AI? (in a nutshell)
2. Where we are currently at?

Content of the talk

Part 2:

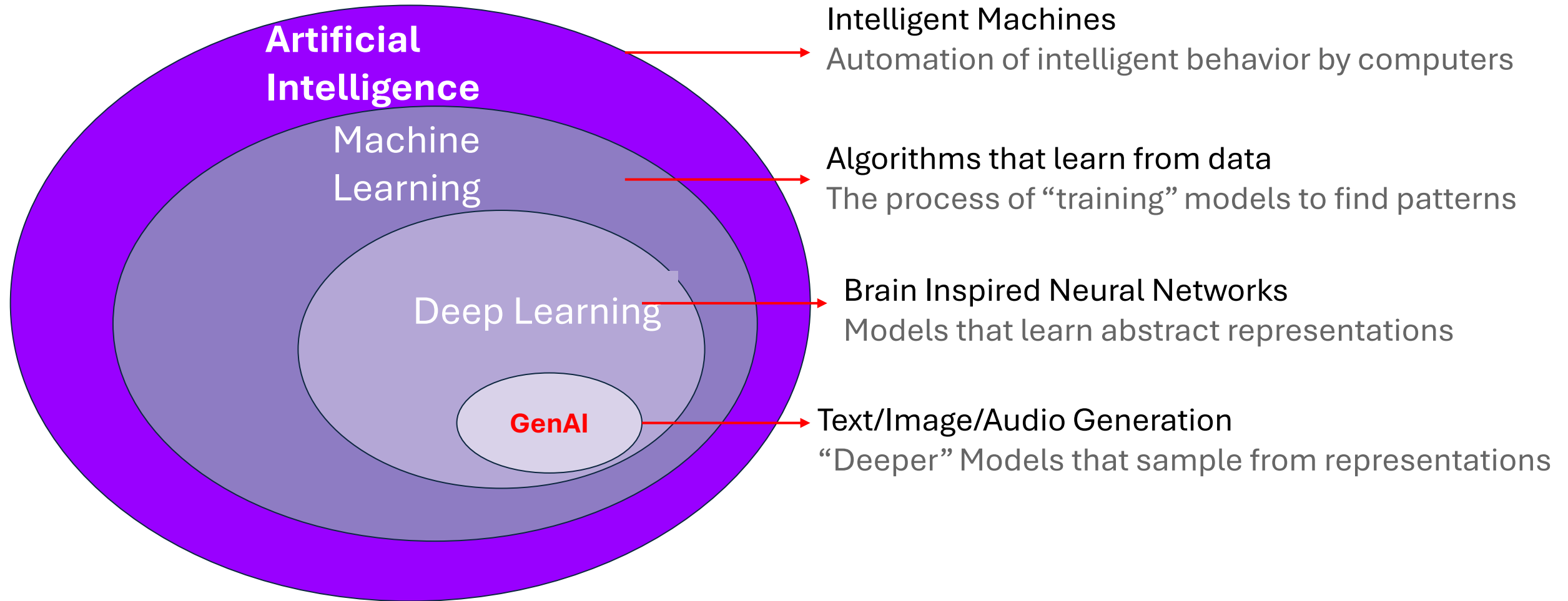
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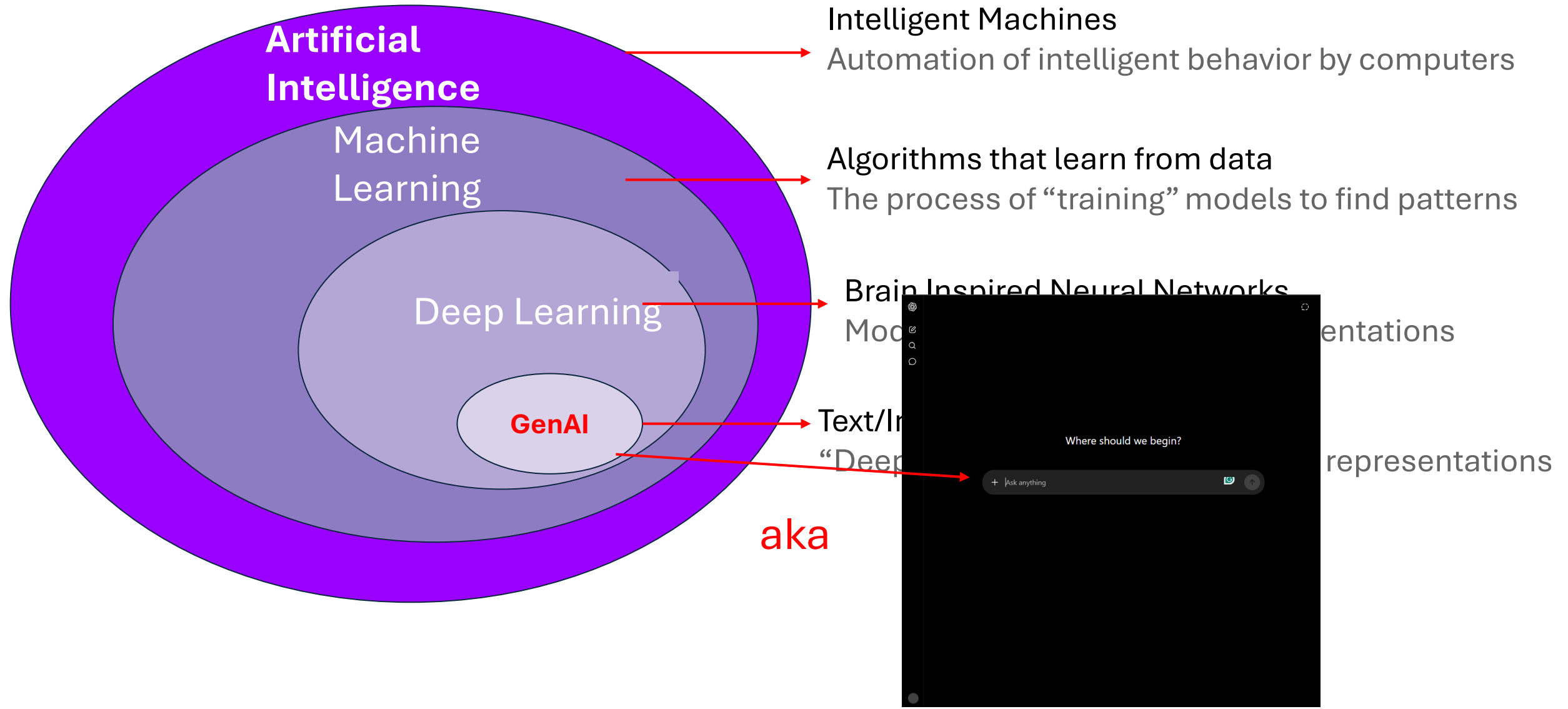
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Lets start with the basics. What is AI?



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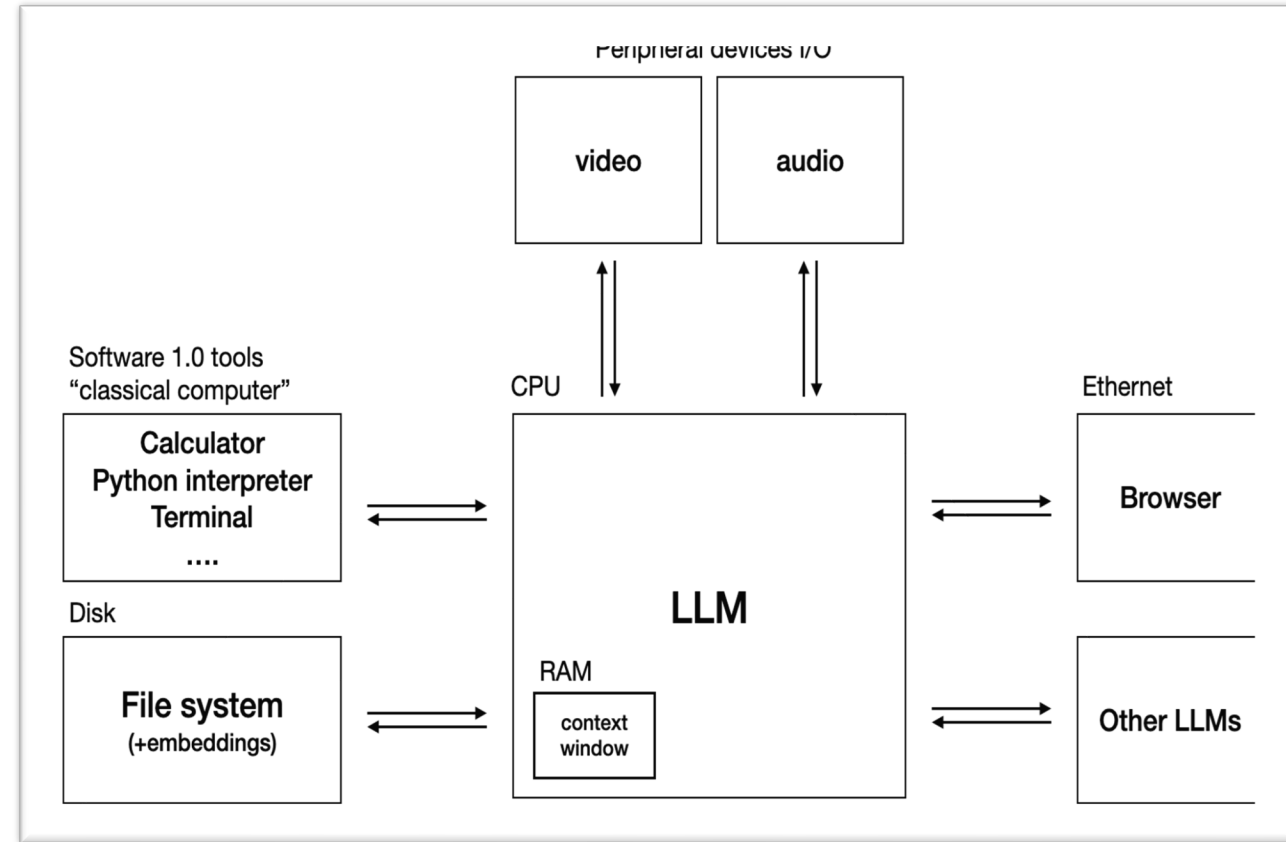
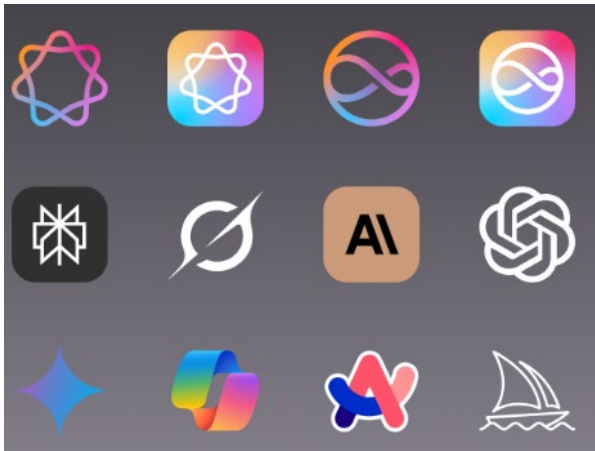
What is this chatgpt thingy?

A modern operating system
“Some software”



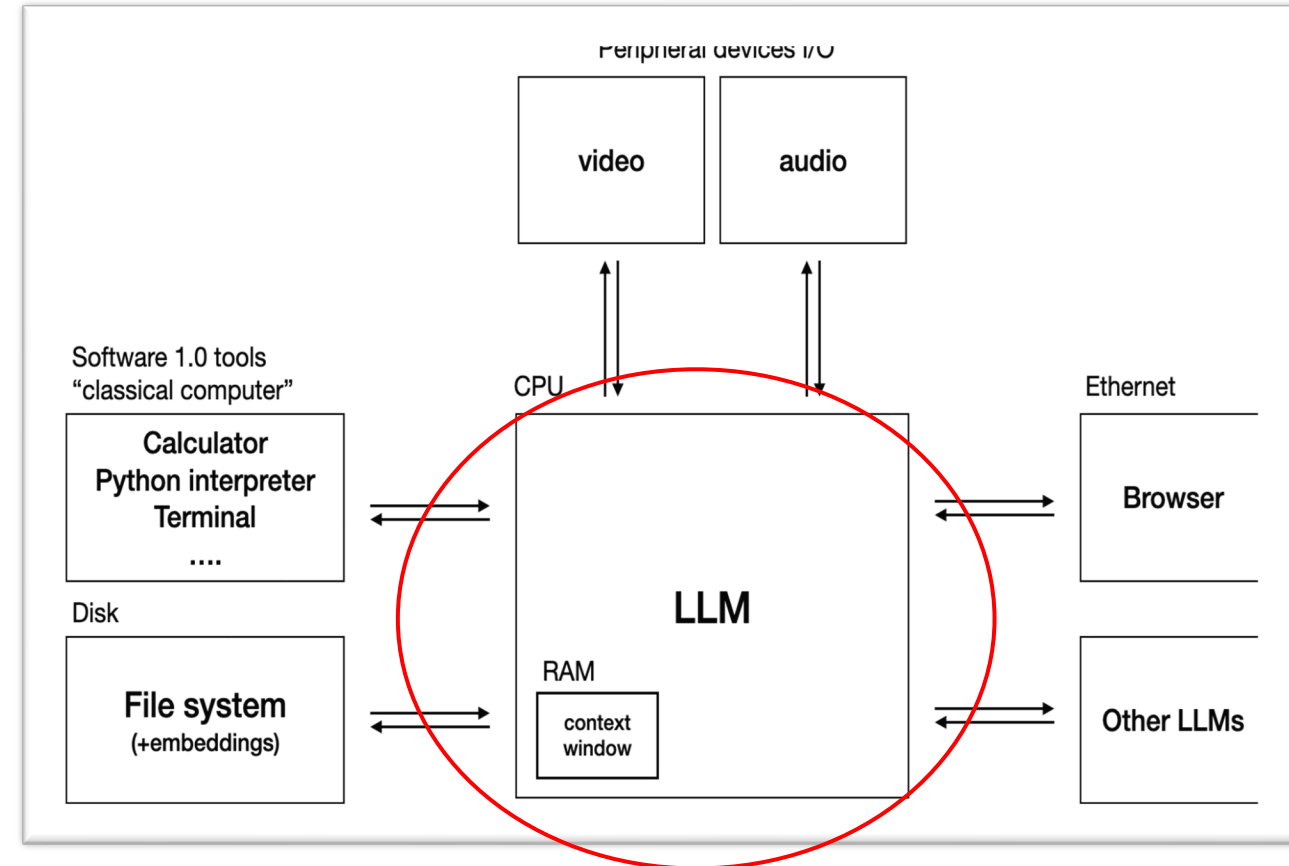
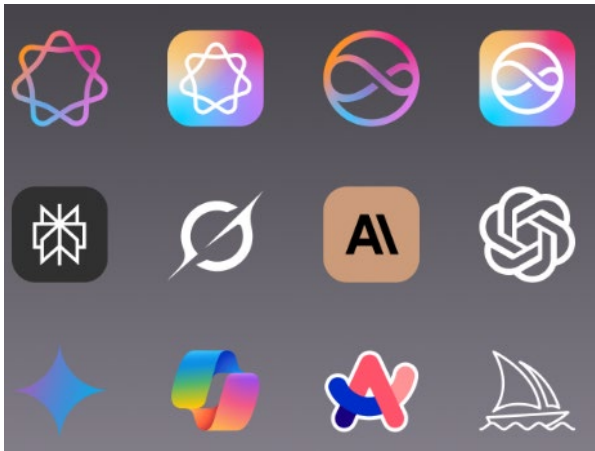
Lets take this step by step

- A new kind of Operating System
- That we build tools (programs) around it

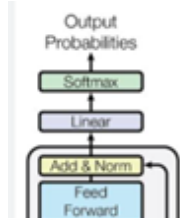
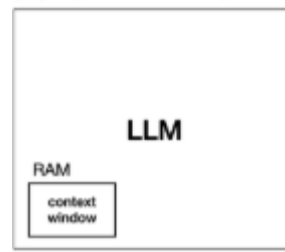


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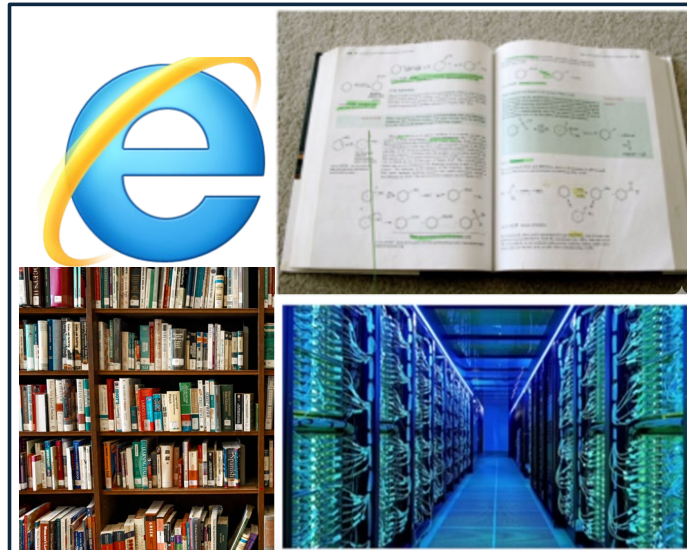
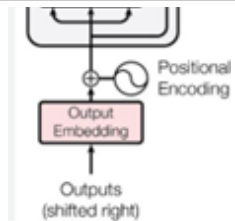


What is an LLM?



giant mathematical expression

$$\frac{1}{1 + \exp(-(w_0 * (1/(1 + \exp(-(w_1 * x_1 + w_2 * x_2 + w_3)))) + w_4 * (1/(1 + \exp(-(w_5 * x_1 + w_6 * x_2 + w_7)))) + w_8 * (1/(1 + \exp(-(w_9 * x_1 + w_{10} * x_2 + w_{11})))) + w_{12}))}$$

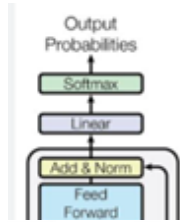


How to Boil a Perfect Egg

- Place eggs in a pot and cover with cold water (about 2–3 cm above eggs).
- Add a pinch of salt or a splash of vinegar (optional, helps prevent cracking).
- Bring water to a boil over medium-high heat.
- Once boiling, reduce heat to a simmer and start timing:
 - 6 min for soft-boiled,
 - 9 min for medium,
 - 11–12 min for hard-boiled.
- When done, transfer eggs immediately to a bowl of ice water for 5–10 minutes.
- Gently tap and peel under running water to remove the shell cleanly.

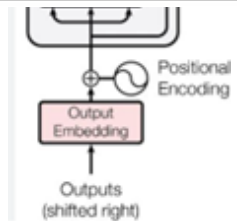
1. A special artificial, brain-inspired neural network (Big mathematical equation)
2. That samples from text representations (Patterns learnt from written knowledge)
3. Learnt through large scale training (Lots of GPUs and lots of data)
4. To automate an intelligent behavior (user – assistant responses, agentic tasks)

Congratulations you just figured out modern LLMs



giant mathematical expression

$$\frac{1}{1 + \exp(-(w_0 * (1/(1 + \exp(-(w_1 * x_1 + w_2 * x_2 + \dots * (1/(1 + \exp(-(w_5 * x_1 + w_6 * x_2 + w_7)))))) + w_8 * (1/(1 + \exp(-(w_9 * x_1 + w_{10} * x_2 + w_{11})))))) + \dots)}$$



39

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boiled,

um,

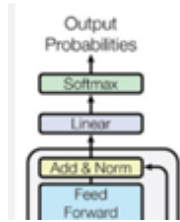
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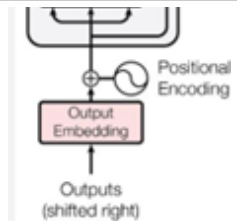
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No, not yet.

39

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Two extra ingredients

Supervised Fine Tuning

<USER>
Can you write a short introduction about the relevance of the term "monopsony" in economics? Please use examples related to potential monopsonies in the labour market and cite relevant research.

<ASSISTANT>
"Monopsony" refers to a market structure where there is only one buyer for a particular good or service. In economics, this term is particularly relevant in the labor market, where a monopsony employer has significant power over the wages and working conditions of their employees. The presence of a monopsony can result in lower wages and reduced employment opportunities for workers, as the employer has little incentive to increase wages or provide better working conditions...

~100K conversations

- Special data to "force" assistant behavior
- Manually assembling high quality documents
- Very hard and expensive to compose such documents

Reinforcement Learning with Human Feedback

Write a poem about beer in Shakespeare style

A

Hark! What light through yonder tankard breaks? O fairest brew, thou golden elixir rare,
It is the ale, the golden nectar bright, Distilled by sun and hand of husbandman,
That doth from humble barley, Thou spring'st from barley fields

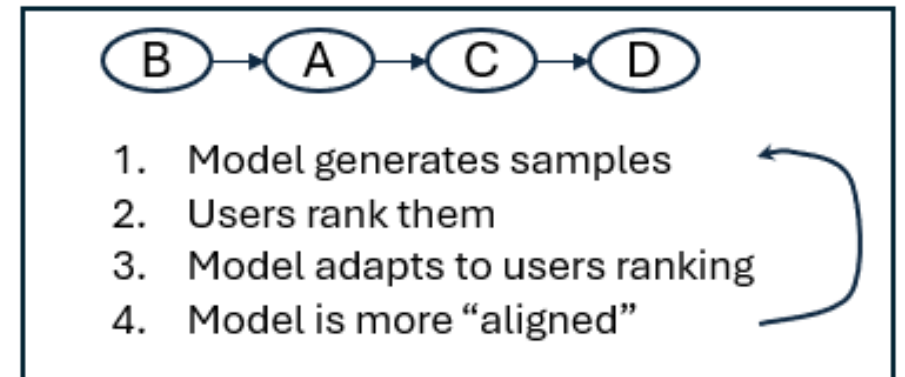
C

O golden nectar poured in tankard bright, Thou dost inflame the senses with delight—
A merry draught to kindle heart and soul.

B

O golden brew, thou nectar of the soul, In tankard bright dost calm the restless heart;
One sip, and sorrow takes its darkened part.

D



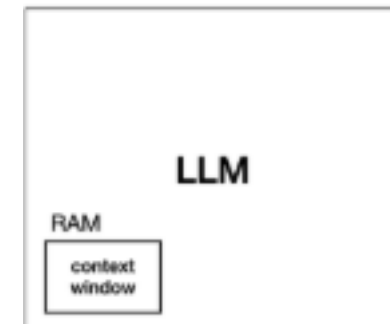
OK, so now you have ChatGPT (version 2022)

But, we still have some issues

Yeah, other than:

- having to download ~10TB of data,
- find a cluster of ~6.000 GPUs, pay 12M to train,
- find 100K high quality responses and then align the behavior of the model

1. The model struggles at complex tasks



System 1 vs System 2.

Human: "Emily buys 3 apples and 2 oranges. Each orange costs \$2. The total cost of all the fruit is \$13. What is the cost of apples?"

Assistant: "The answer is \$3."



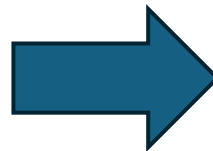
Human: "Emily buys 3 apples and 2 oranges. Each orange costs \$2. The total cost of all the fruit is \$13. What is the cost of apples? Think step by step"

Assistant: "Okay, let's think step by step:

First, find the total cost of the oranges: $2 \text{ oranges} \times \$2/\text{orange} = \$4$.

Next, subtract the cost of the oranges from the total bill to find the cost of the apples: $\$13 \text{ (total)} - \$4 \text{ (oranges)} = \$9$.

So, the total cost of the apples is \$9.



DeepSeek Flips Script on AI

Chinese dark horse emerges, threatening a market darling and other big tech stocks

For two years, markets' belief that the rise of artificial intelligence would usher in a new

By Ganjun Binerji,
Ana Fitch
and Karen Langley

era of productivity growth has fueled trillions of dollars in stock-market gains.

Nvidia, the maker of the computer chips at the heart of

the AI boom, has been in the vanguard of this advance. Wall Street has perceived the company to have an almost unreachable defense against competition with its offerings of high-tech chips. The company's rapid growth and windfall profits have helped push other technology firms and the Nasdaq Composite Index to record after record, with giddy investors expecting more of the same down the road.

On Monday, the mood turned sour. DeepSeek, a dark-horse power in artificial intelligence, emerged from China. That rattled big tech stocks, led by a plunge of almost \$600 billion in

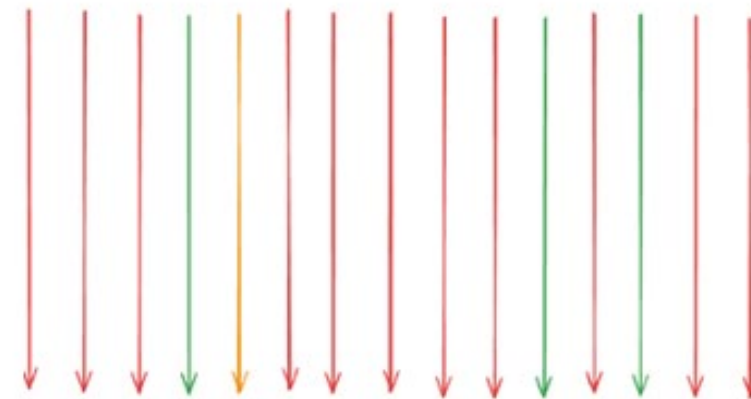
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Nvidia market value



Emily buys 3 apples and 2 oranges. Each orange costs \$2. The total cost of all the fruit is \$13. What is the cost of each apple?



solutions

Answer: 3

We generated 15 solutions.
Only 4 of them got the right answer.
Take the top solution (each right and short).
Train on it.
Repeat many, many times.

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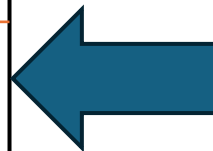
Assistant: "The answer is \$3."



Human: ""Emily buys 3 apples and 2 oranges. Each orange costs \$2. The total cost of all the fruit is \$13. What is the cost of apples?"

<Assistant is thinking...>

Assistant: "The apples cost \$9 in total.
The oranges cost \$4 in total, so the apples cost $\$13 - \$4 = \$9$."



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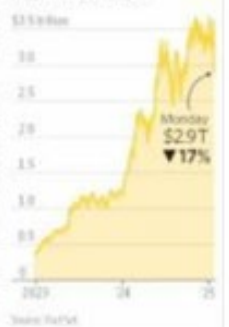
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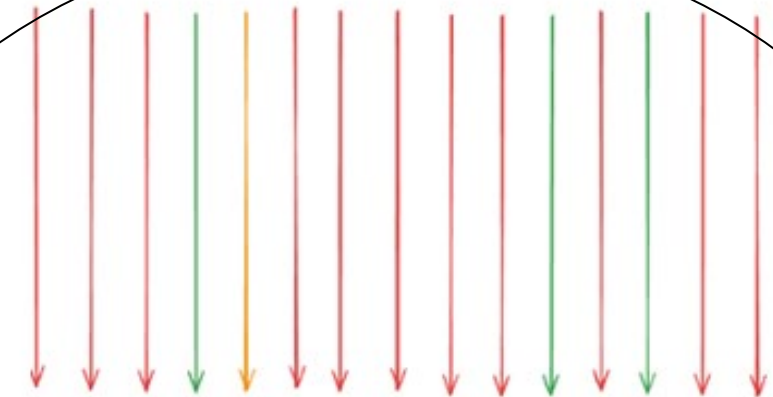
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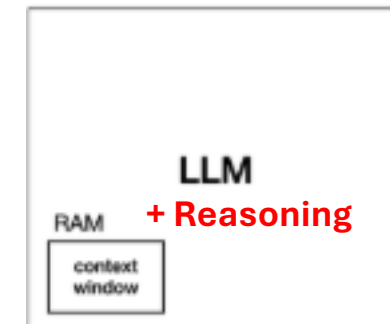
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1. The model struggles at complex tasks
2. The model still cannot do math (Reason is a bit technical)



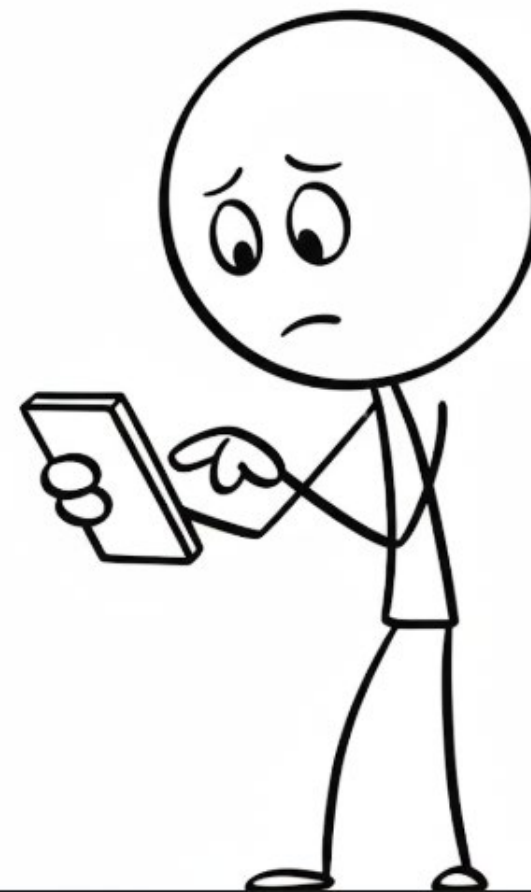
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But
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Quiz Time

What is y?

$$y = \frac{(258.5 \times 16) + \frac{972}{12-5.5} - 340}{\frac{5600}{3.5} + (11 \times (153 - 97)) - 1600}$$



LLM

RAM

+ Reasoning

context
window

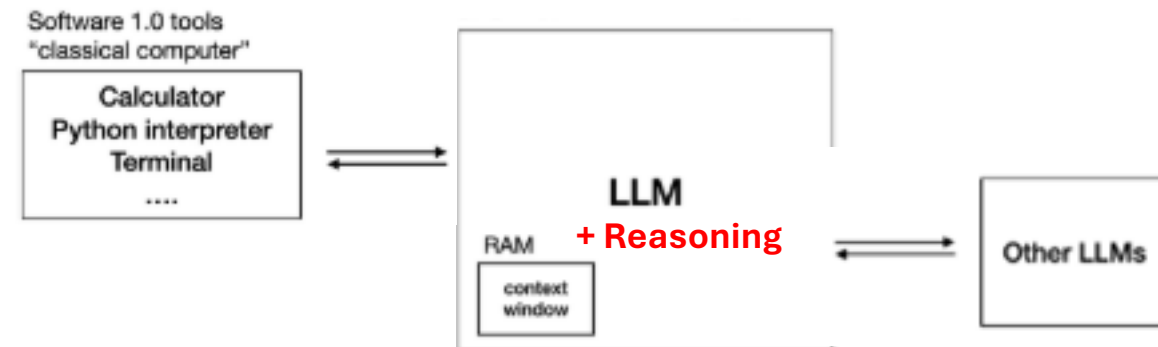
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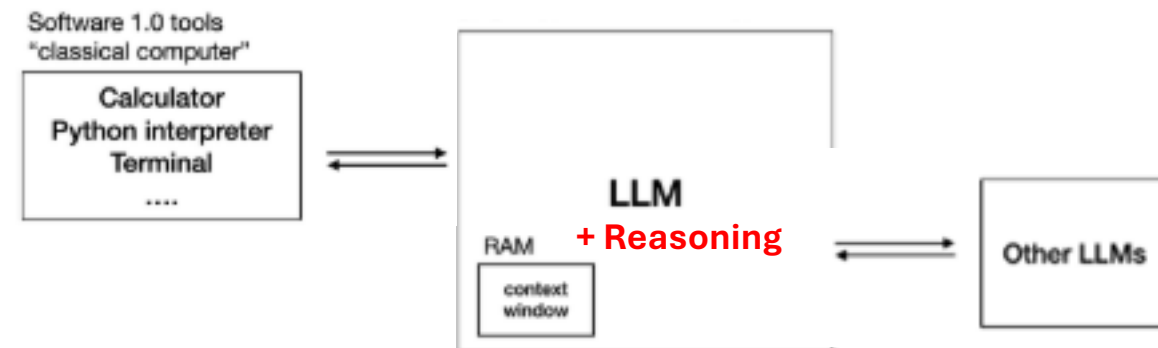
OK, so now you have ChatGPT (version 2022)

But, we still have some issues

Yeah, other than:

- having to download ~10TB of data,
- find a cluster of ~6.000 GPUs, pay 12M to train,
- find 100K high quality responses and then align the behavior of the model

1. The model struggles at complex tasks
2. The model still cannot do math (Reason is a bit technical)
3. The model hallucinates (by construction)



OK, so now you have ChatGPT (version 2022)

But, we still have some issues

Yeah, other than:

Chatbot

What are 5 popular scientific papers that prove the world is flat.

1. "The Flat Earth and Astrobiology" by Jeffrey Burton Smith: This paper argues that the theory of a spherical earth contradicts the principles of astrophysics, including the observed behavior of celestial bodies.
2. "Flat Earth: A Survey and Critique of Beliefs in a Flat Earth" by William Carroll: This paper provides an overview of beliefs in a flat earth, as well as critiques of these beliefs from a scientific perspective.
3. "A New Physical Model for the Solar System" by Albert Parsons: This paper presents a physical model of the solar system based on a flat earth, which he claims to be supported by observational evidence.
4. "Evidence for a Flat Earth: The Case of Eclipses" by Mark Sargent: This paper analyzes eclipse data and claims to provide strong evidence against the idea of a spherical earth.
5. "The Curved Earth Hoax Exposed" by Kevin McCloud: This paper provides an analysis of various arguments used to support the concept of a curved earth, such as photosphere curvature and geodesic arcs, and concludes that they do not with

BREAKING | BUSINESS

Lawyer Used ChatGPT In Court—And Cited Fake Cases. A Judge Is Considering Sanctions

By [Molly Bohannon](#), Former Staff. Molly Bohannon has been a Forbes news reporter since 2023.

Published Jun 08, 2023, 02:06pm EDT, Updated Jun 08, 2023, 03:42pm EDT

Other LLMs

OK, so now you have ChatGPT (version 2022)

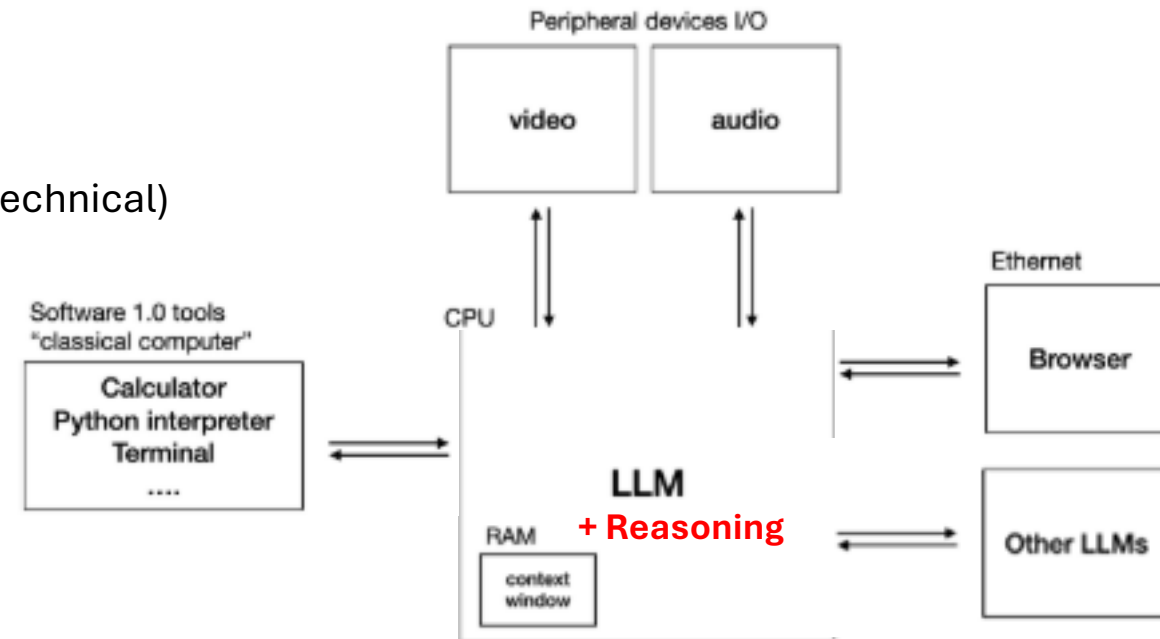
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OK, so now you have ChatGPT_(version 2022)

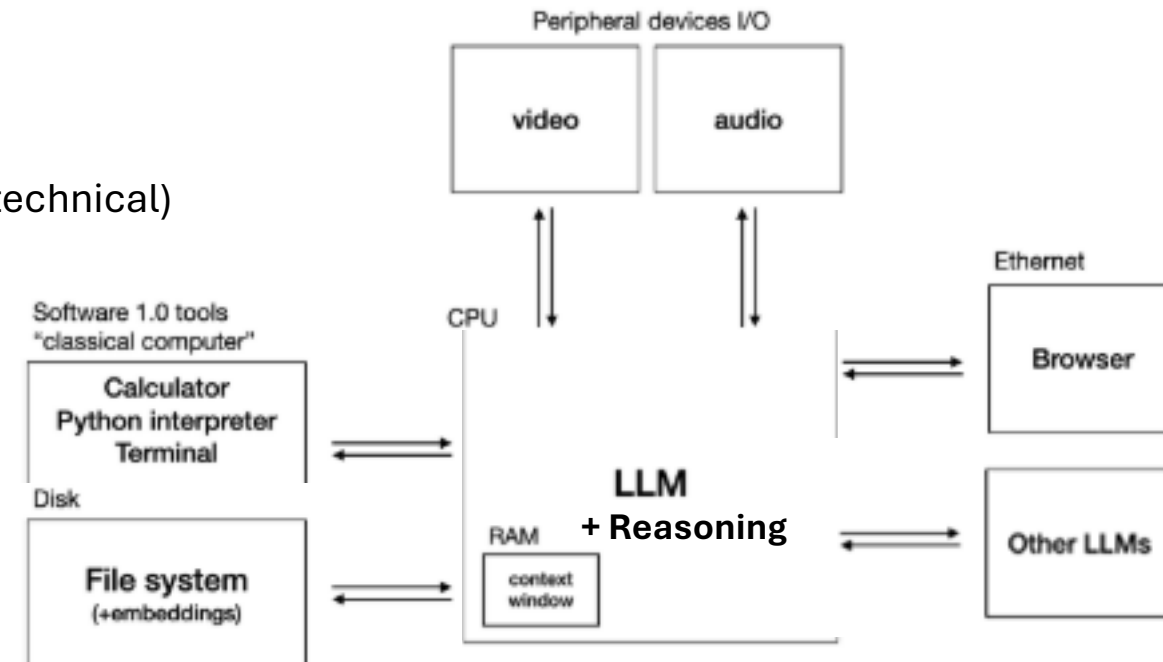
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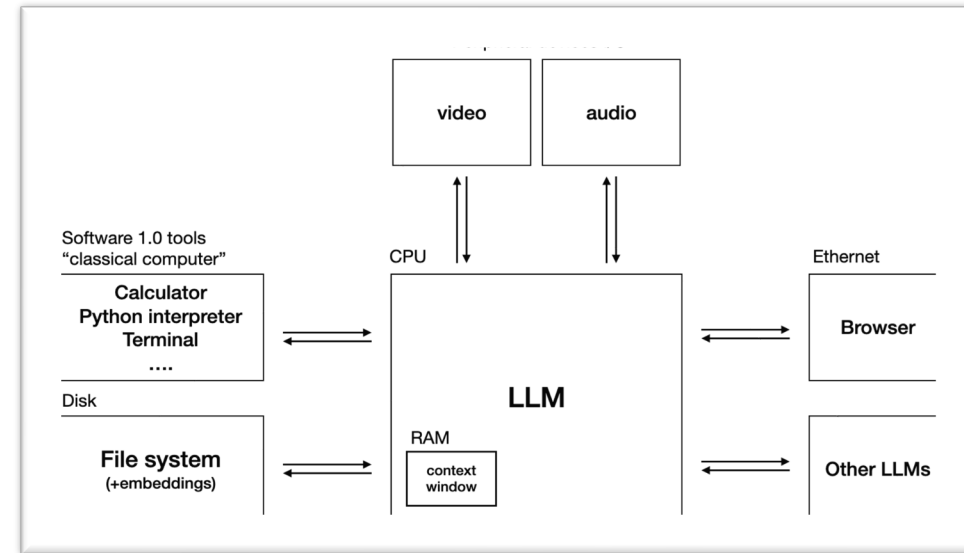


1. The model struggles at complex tasks
2. The model still cannot do math (Reason is a bit technical)
3. The model hallucinates (by construction)
4. Start forgetting after long discussions



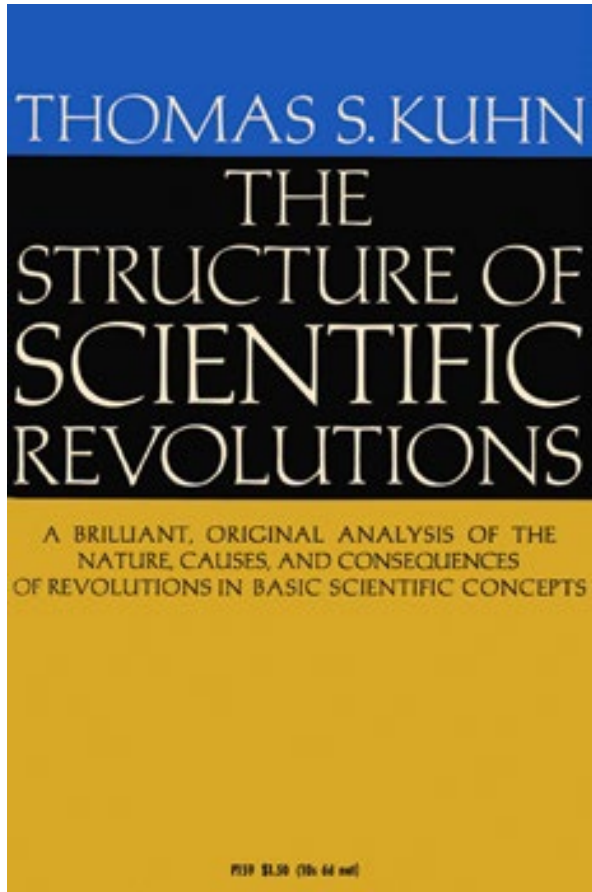
In a nutshell:

1. This is a modern LLM →
2. It's a product of years of research
3. It's a mix of: Data + Compute (GPUs) + Machine Learning Model + Algorithms
4. It reproduces text used for training, in order to solve queries (assistant/copilot)
5. It has many limitations.
6. Some are solved, some are being solved, some are hard to be solved



Where we are currently at?

A two-wave model analogy.



Revolutionary Science:

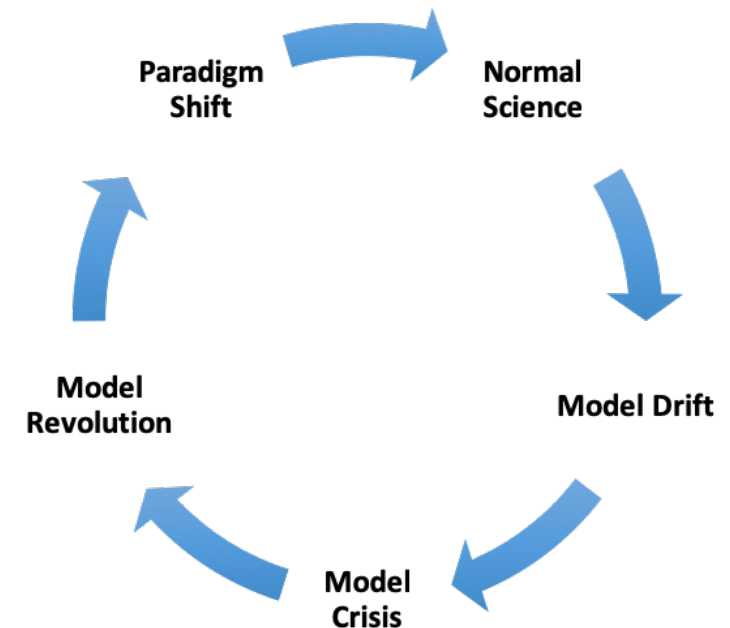
(The Breakthrough Period)

Brief periods of paradigm-shifting breakthroughs.
(eg, Transistor ,first TCP/IP protocol)

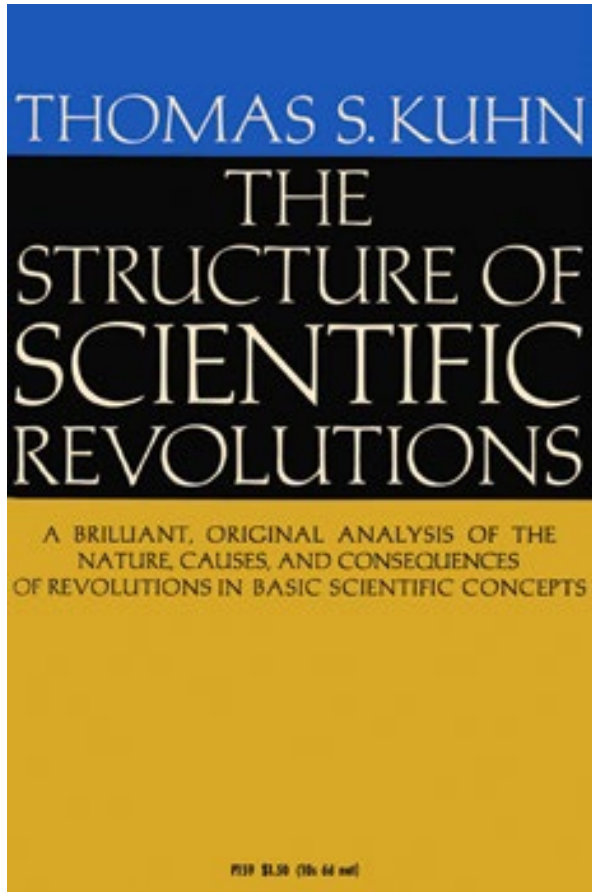
Normal Science:

(The Consolidation Period)

Longer periods of problem solving with newly established paradigms, working out implications, refining methods.
(eg Computer, Internet etc)



A two-wave model analogy.



Revolutionary Science:

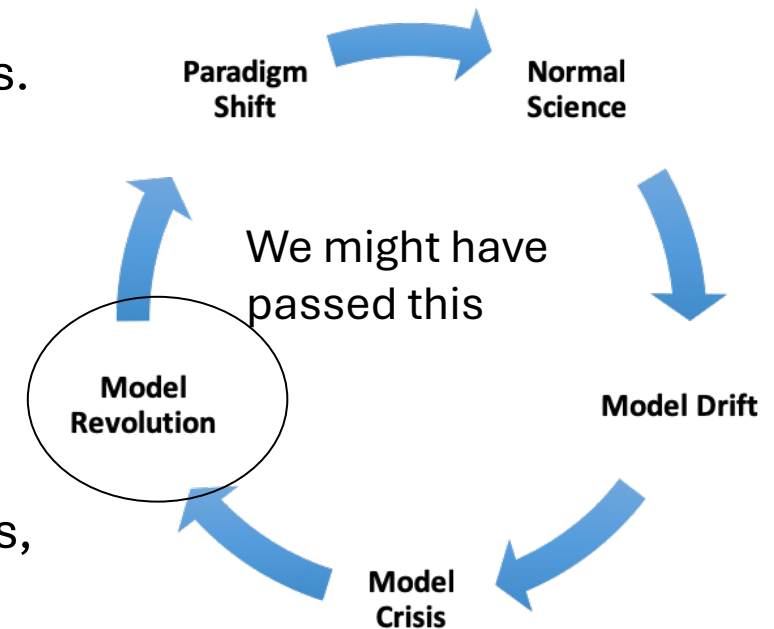
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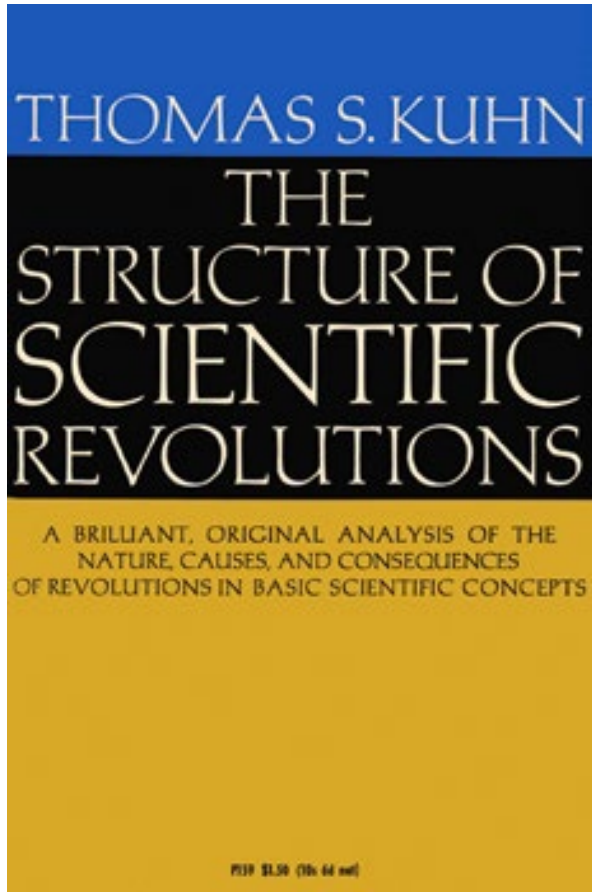
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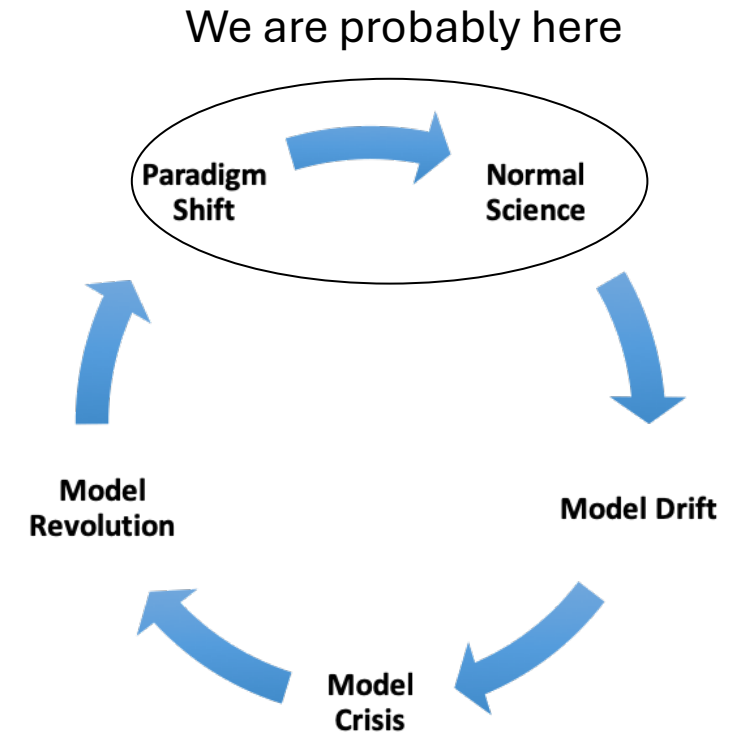
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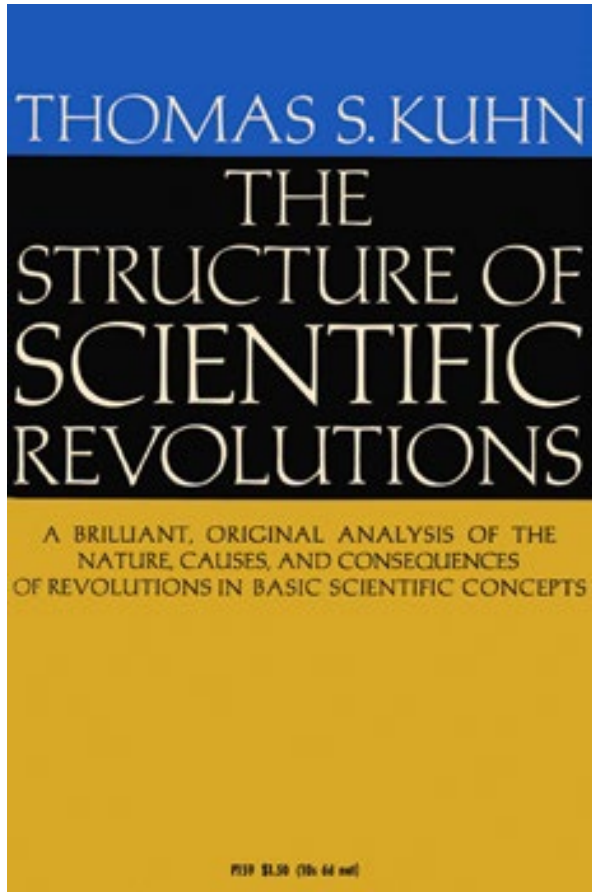
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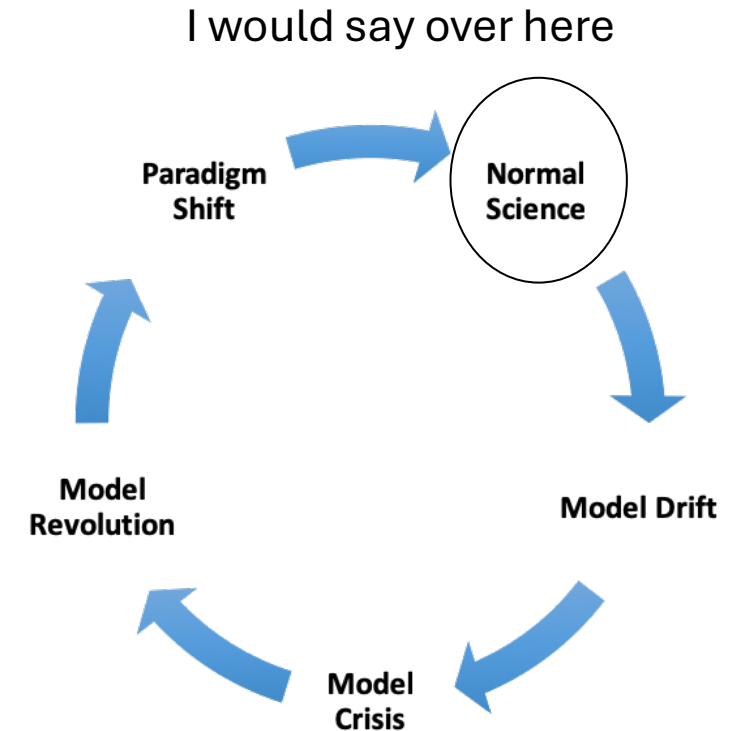
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(eg Computer, Internet etc)



Where we are & where are we heading?

- Technology is pretty much established
 - Mostly efficiency breakthroughs
 - Yes there is Fable and GPT 5.6, but for 99% of people the difference is not clear

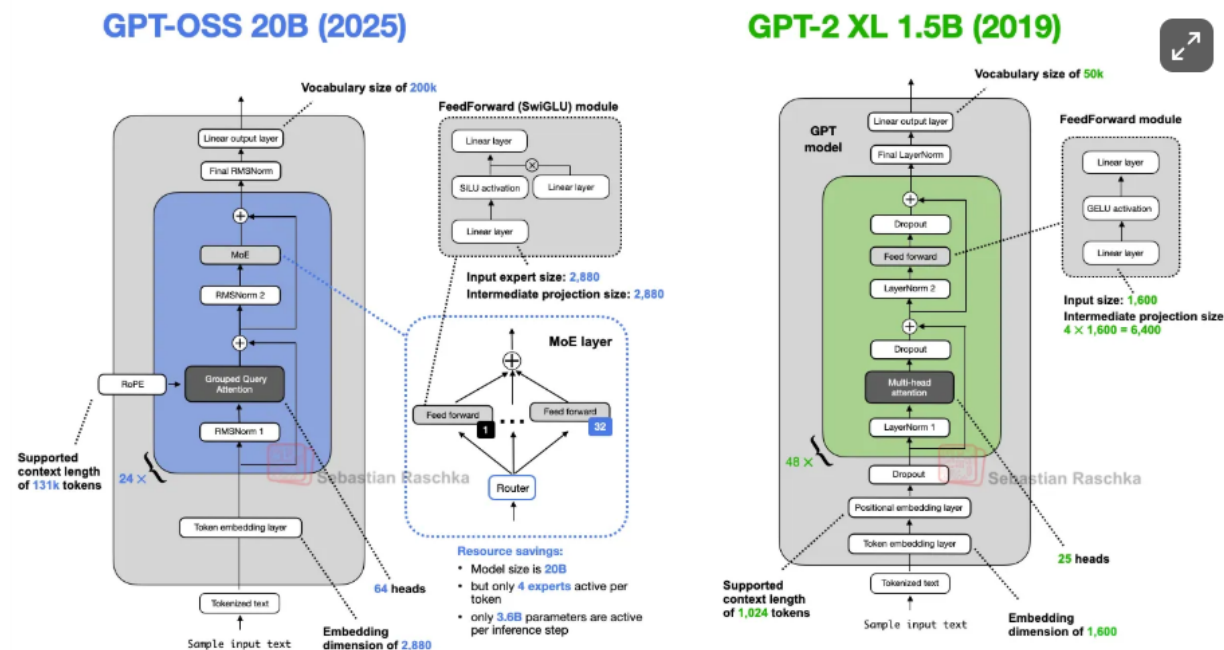


Figure 2: A side-by-side comparison between gpt-oss-20b and GPT-2 XL 1.5B.

Where we are & where are we heading?

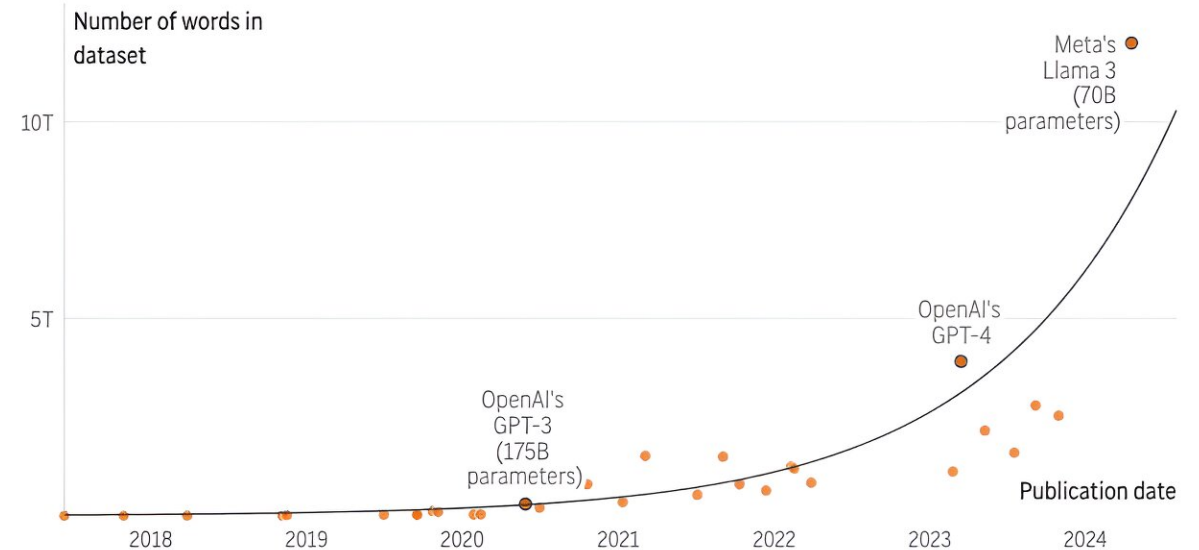
- Technology is pretty much established
- We are running out of things to scale

Where we are & where are we heading?

- Technology is pretty much established
- We are running out of things to scale
 - Data 

LLM training datasets are growing

Datasets used to train key machine learning models have rapidly increased in size since 2017.



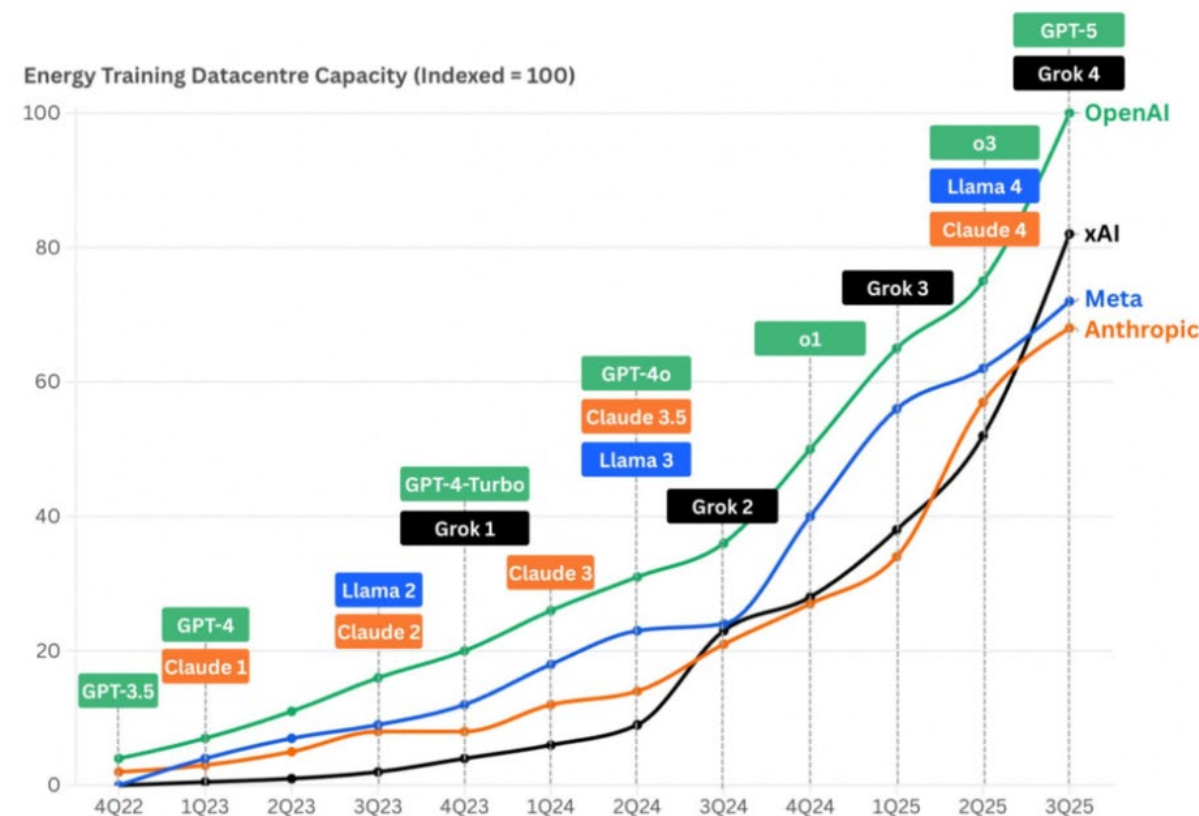
Source: Epoch AI

AP

Where we are & where are we heading?

- Technology is pretty much established
- We are running out of things to scale
 - Data ✓
 - Compute ✓

The amount of compute labs have has 3x in a year
OpenAI has the most compute* but xAI is growing the fastest



Where we are & where are we heading?

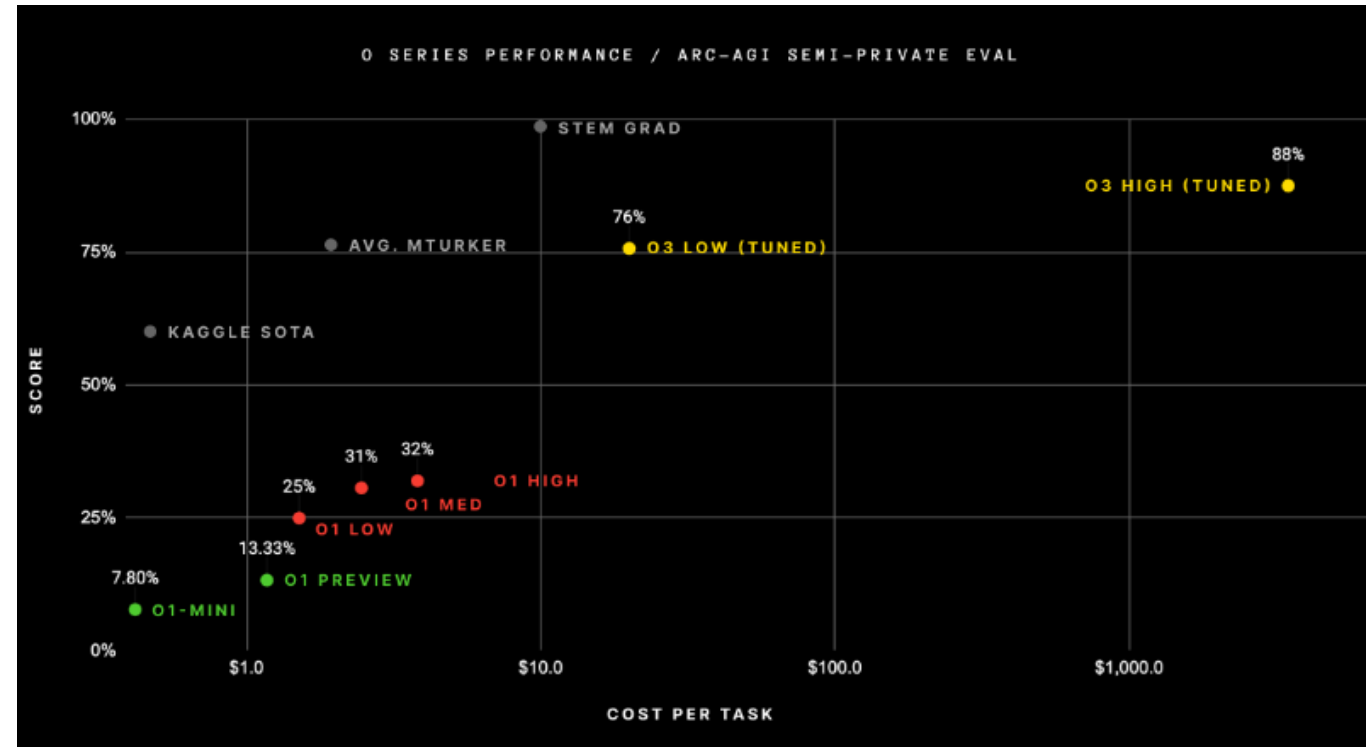
- Technology is pretty much established
- We are running out of things to scale
 - Data 
 - Compute 
 - Inference time (think longer)

DeepMind and OpenAI models solve maths problems at level of top students

For the first time, large language models performed on a par with gold medallists in the International Mathematical Olympiad.

Where we are & where are we heading?

- Technology is pretty much established
- We are running out of things to scale
 - Data ✓
 - Compute ✓
 - Inference time (think longer)

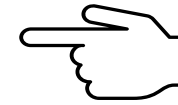


Where we are & where are we heading?

- Technology is pretty much established
- We are running out of things to scale
 - Data 
 - Compute 
 - Inference time (think longer) 
 - New? (There are good candidates)

Where we are & where are we heading?

- Technology is pretty much established
 - We are looking for new things to scale
-



The Breakthrough Period

Where we are & where are we heading?

- Technology is pretty much established
 - We are looking for new things to scale
-
- We now have models for different tasks
 - Open/closed source
 - Different pricing
 - Different capabilities

Model	Input Price	Output Price	Context Length	Max Output Tokens	Arena Score
Google Gemini-2.5-Pro	\$2.50	\$15.00	1,000k	65k	1,456
OpenAI o3	\$10.00	\$40.00	200k	100k	1,444
OpenAI gpt-4o	\$2.50	\$10.00	128k	16k	1,443
OpenAI gpt-4.5	\$75.00	\$150.00	128k	16k	1,439
xAI Grok-3-Preview	\$3.00	\$15.00	131k	n/a	1,409

Where we are & where are we

- Technology is pretty much established
 - We are looking for new things to scale
-
- We now have models for different tasks
 - We now build AI Applications
 - Agentic applications.
 - For collaboration & task automation.
 - Using expertise injection

First, where do you want to go?

You'll get custom recs you can save and turn into an itinerary.

🔍 Choose a city or town

Hey, I'm Layla, your personal travel agent

Tell me what you want, and I'll handle the rest: flights, hotels, itineraries, in seconds.

Help me plan a budge



🗺️ Create a new trip

📖 Inspire me where to go

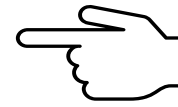
🚗 Build a road trip

📅 Plan a last minute getaway

Not sure where to start? [See how it works](#) ↓

Where we are & where are we heading?

- Technology is established
 - Scaling dimension (more is better)
-



The Breakthrough Period

- Different model per task
- New applications for task automation



The Consolidation Period

Where we are & where are we heading?

- Technology is established
 - Scaling dimension (more is better)
-
- Different model per task
 - New applications for task automation
 - Start figuring out the implications



The Breakthrough Period

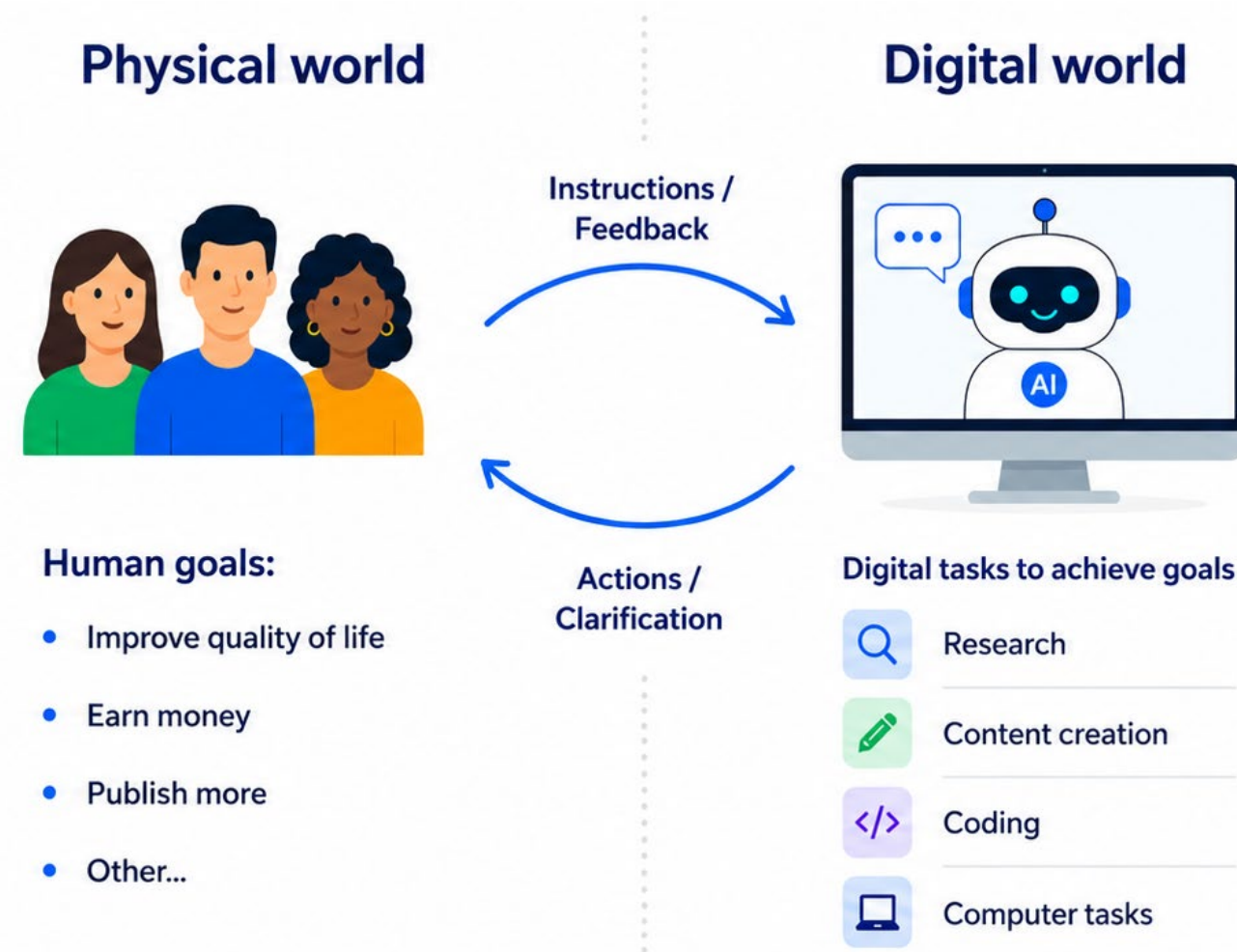


The Consolidation Period

Content of the talk

Part 2:

1. What are the implications of AI?
2. Why predicting the implications is very hard and what we can do instead?



Part 1:

1. What is this AI? (in a nutshell)
2. Where we are currently at?

Capability and impact. Two different things.

“Frontier models have reached or exceeded human-level expertise on cognitive tasks, **thus** we are on the threshold of AGI, **so** we need to study consequences”

Capability and impact. Two different things.

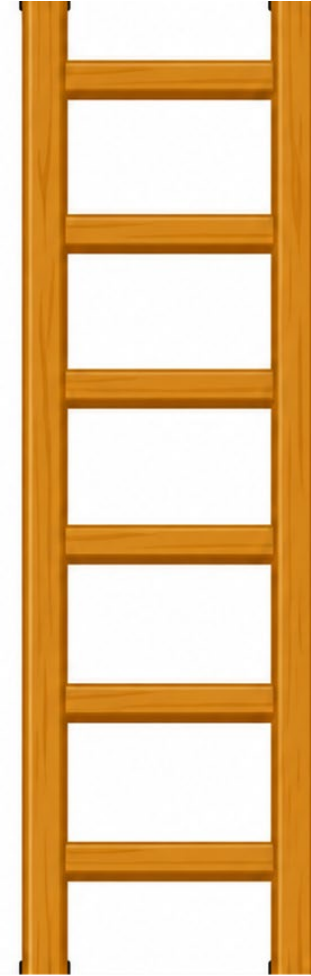
“Frontier models have reached or exceeded human-level expertise on cognitive tasks, **thus** we are on the threshold of AGI, **so** we need to study consequences”

Simplified:

“Models are getting so good that they will surpass human intelligence on all tasks, so we need to start studying what will happen”

Capability and impact. Two different things.

“Frontier models have reached or exceeded human-level expertise on cognitive tasks, **thus** we are on the threshold of AGI, **so** we need to study consequences”



Capability and impact. Two different things.

“Frontier models have reached or exceeded human-level expertise on cognitive tasks, **thus** we are on the threshold of AGI, **so** we need to study consequences”

Digital tasks to achieve goals



Research



Content creation



Coding

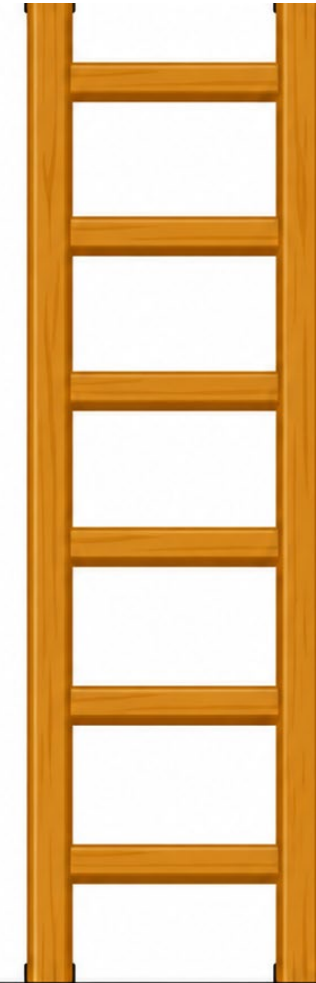


Computer tasks



Benchmark performance

(exams / coding / reasoning / tasks)



Capability and impact.

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Digital tasks to achieve goals



Research



Content creation



Coding



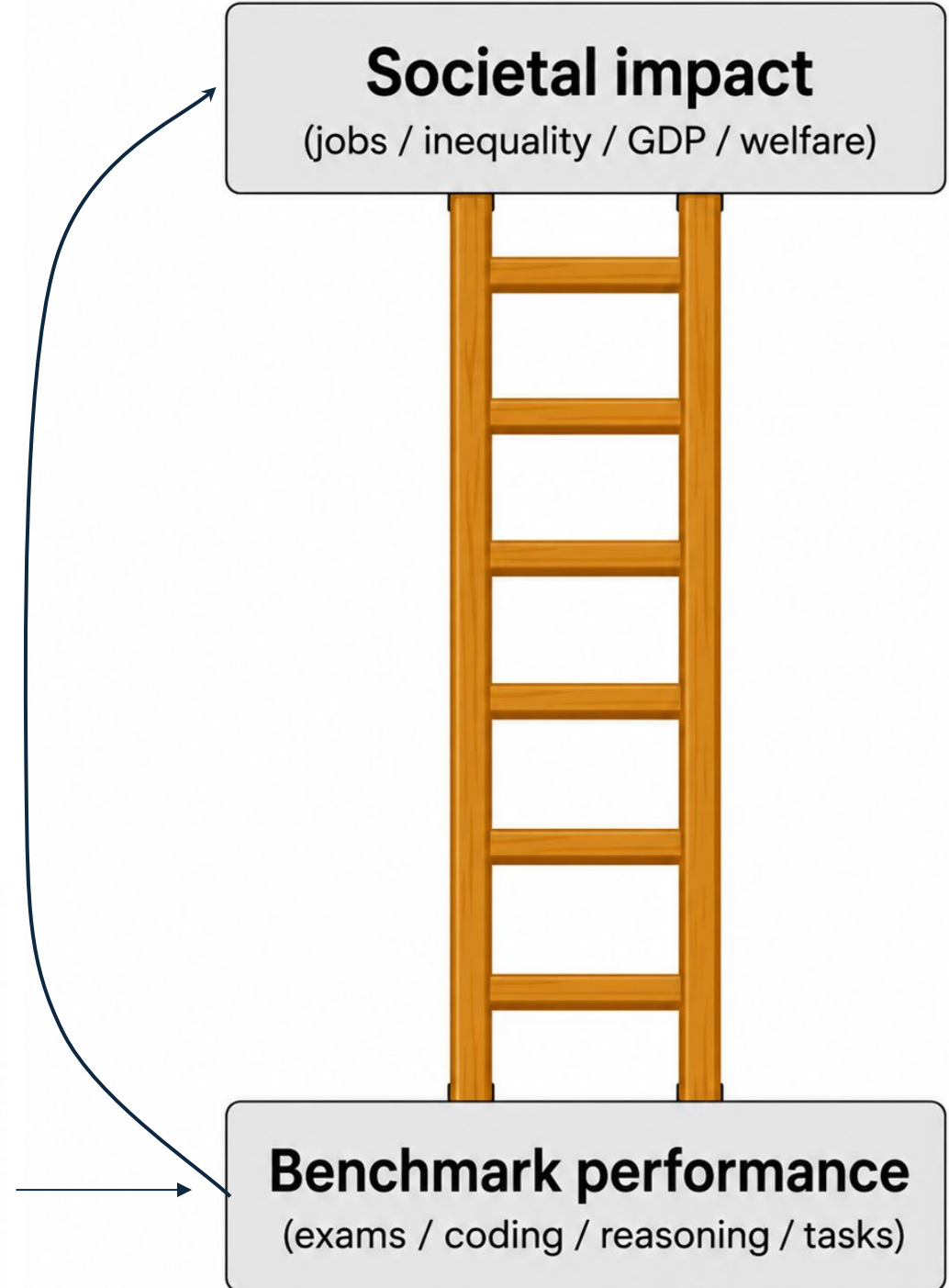
Computer tasks

Societal impact

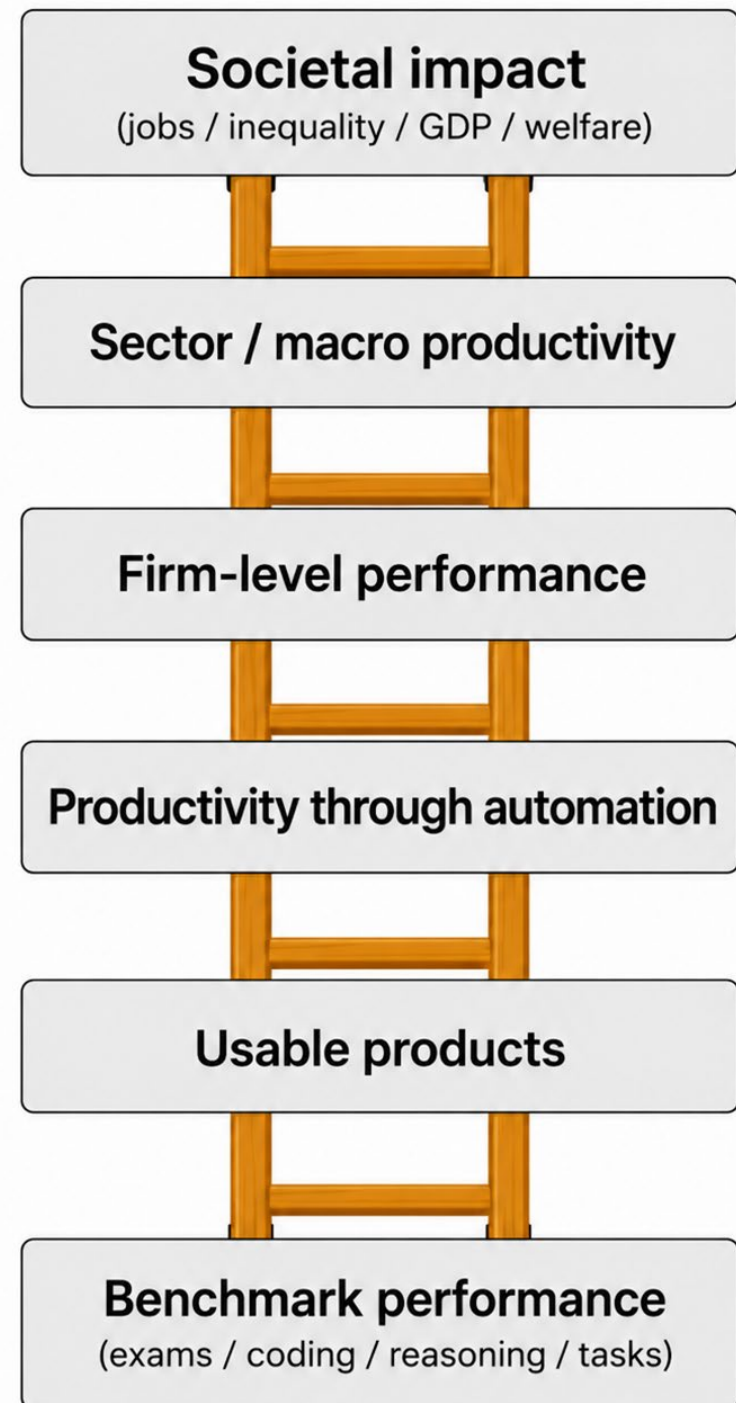
(jobs / inequality / GDP / welfare)

Benchmark performance

(exams / coding / reasoning / tasks)

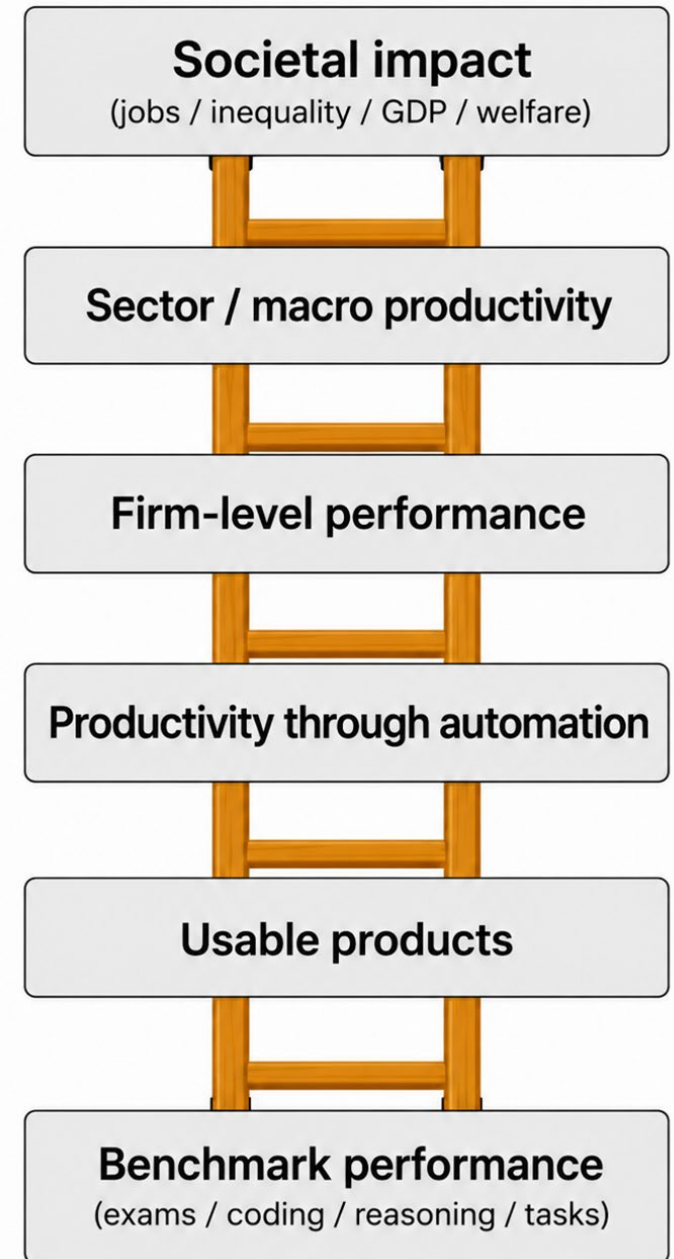


Capability and impact.

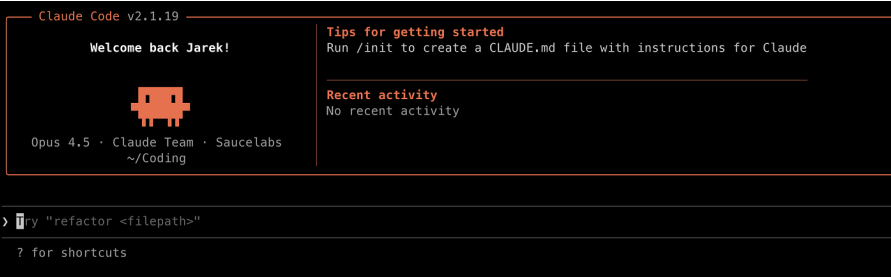
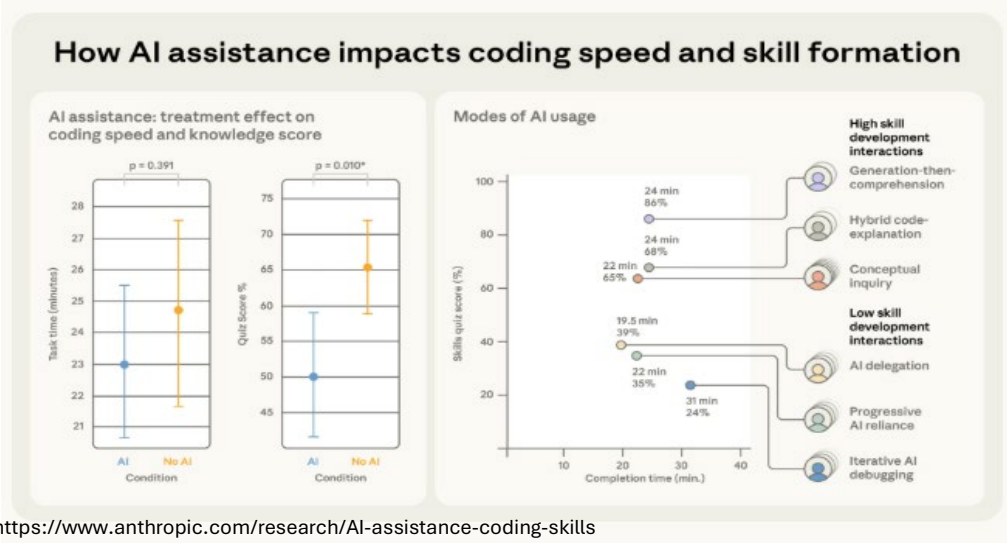


Capability and impact.

The higher we go, the
less clear the evidence
becomes

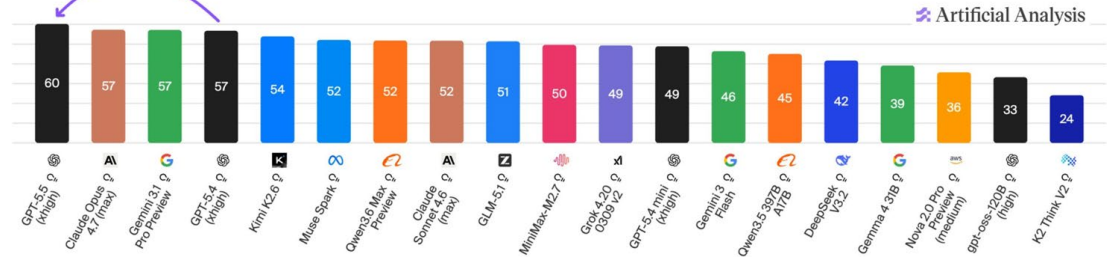


Capability and impact.

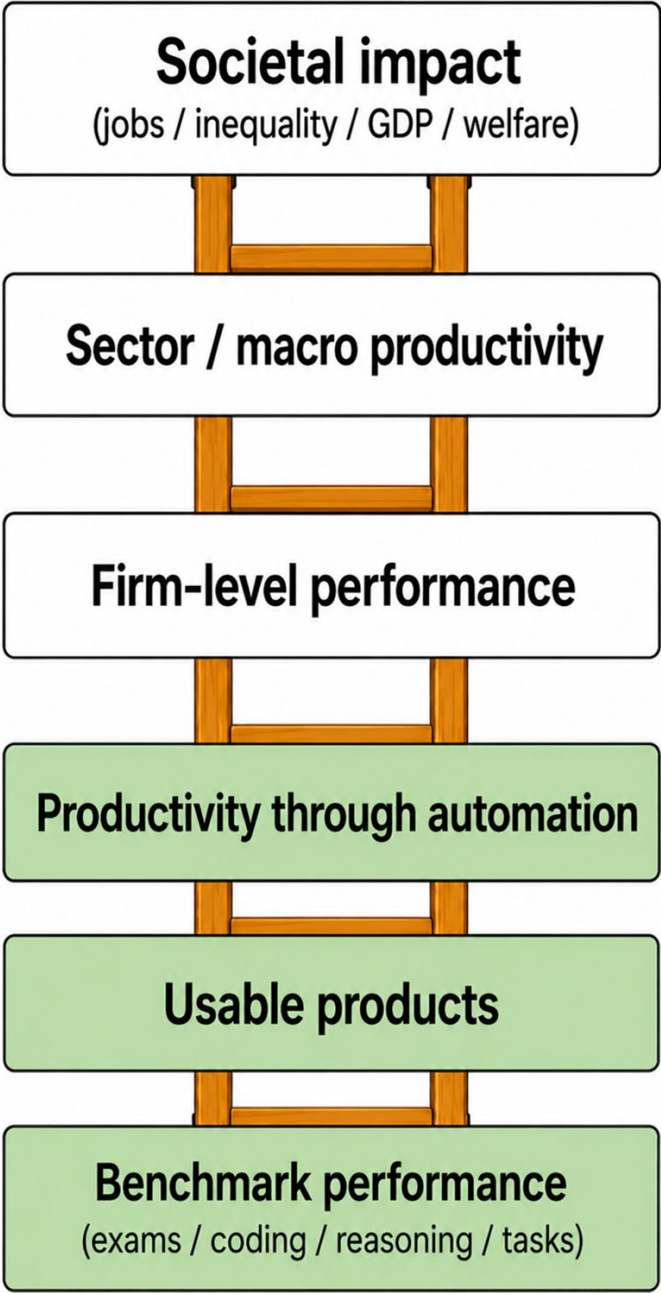


Artificial Analysis Intelligence Index

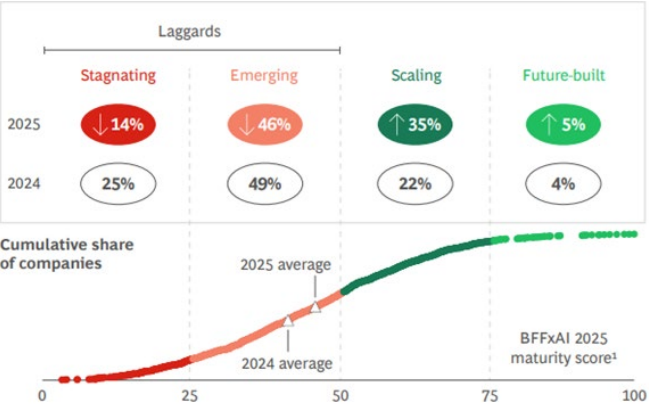
Artificial Analysis Intelligence Index v4.0 incorporates 10 evaluations: GDPval-AA, r²-Bench Telecom, Terminal-Bench Hard, SciCode, AA-LCR, AA-Omniscience, IFBench, Humanity's Last Exam, GPQA Diamond, CritPi



Good evidence →



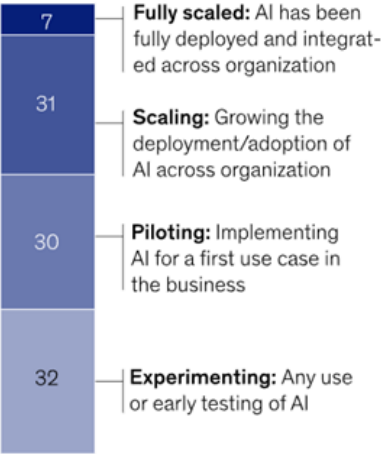
Capability and impact.



Phase of AI use among organizations using AI in 2025

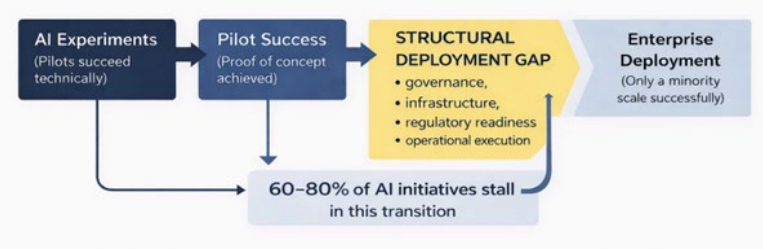
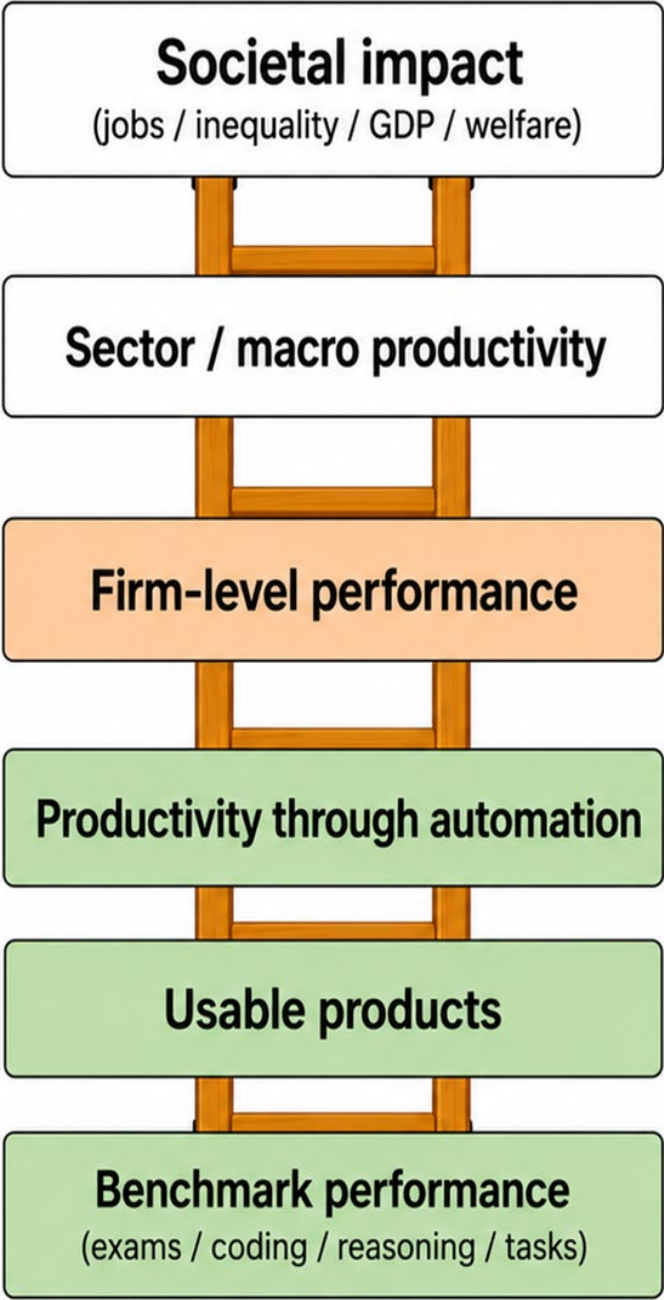
Efficiency gains are strong, but revenue impact lags

Globally, AI continues to deliver clear improvements in efficiency; however, the financial returns of this technology are being felt much more slowly. While 74% of organisations surveyed hope AI will grow revenue, only 20% can currently demonstrate measurable impact. This gap shows where many initiatives are stalling. Boards are asking what AI drives beyond just efficiency, and which new offerings and service models it can unlock.



Conflicting evidence →

Good evidence →



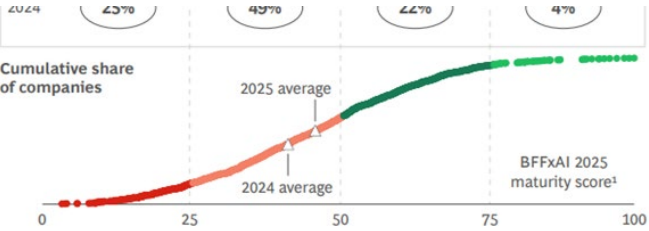
Capability and impact.

... BBC

Ford rehires human engineers after AI fails to match quality checks

Ford says it has hired back some human engineers after AI failed to match their skills and experience. In a bid to reap the benefits of the...

1 week ago

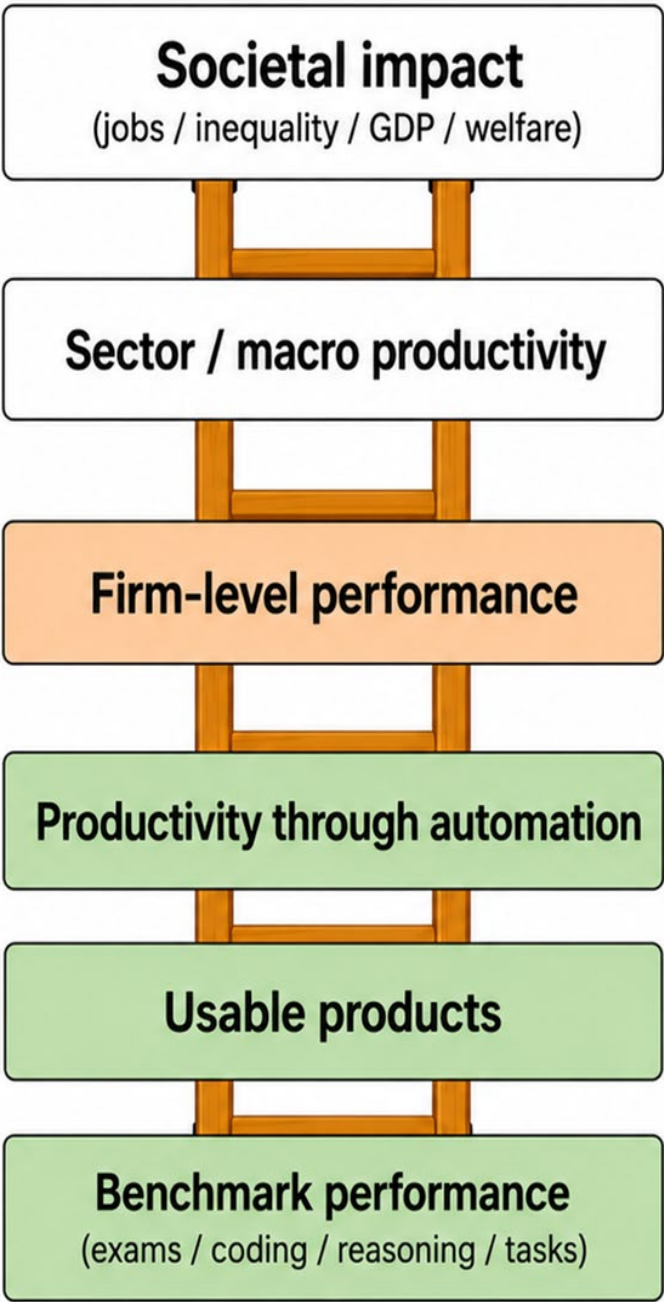


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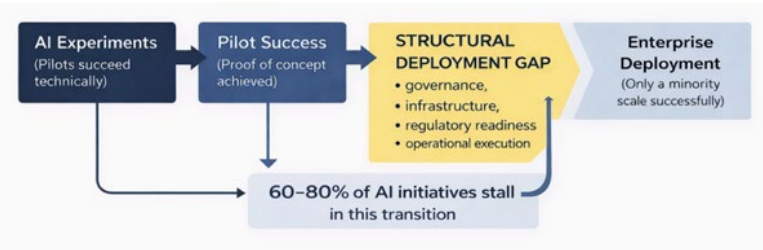
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7	Fully scaled: AI has been fully deployed and integrated across organization
31	Scaling: Growing the deployment/adoption of AI across organization
30	Piloting: Implementing AI for a first use case in the business
32	Experimenting: Any use or early testing of AI

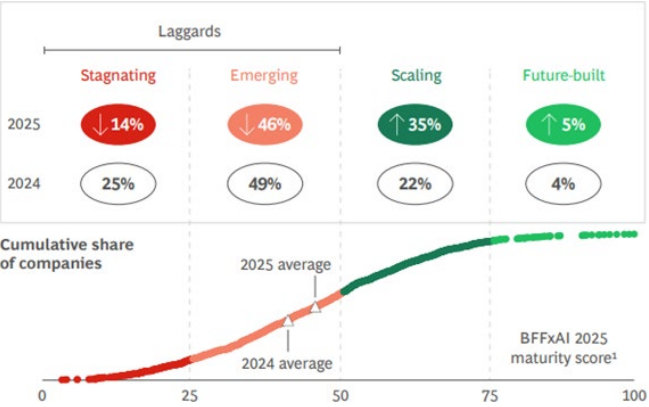
Conflicting evidence →



Good evidence →

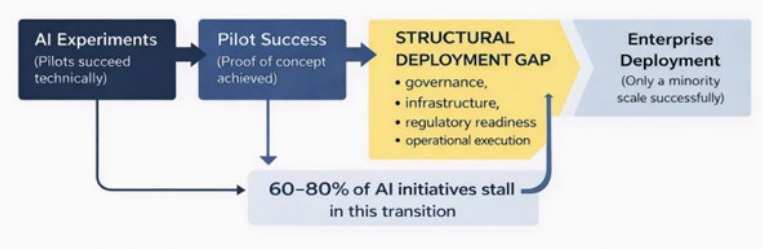
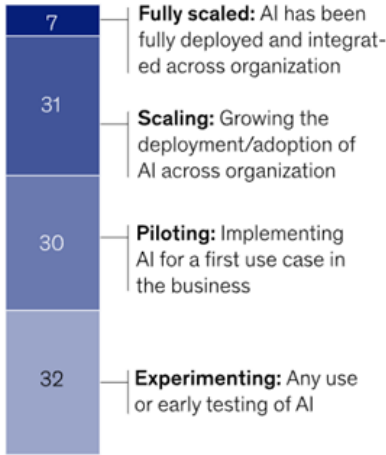


Capability and impact.



Efficiency gains are strong, but revenue impact lags

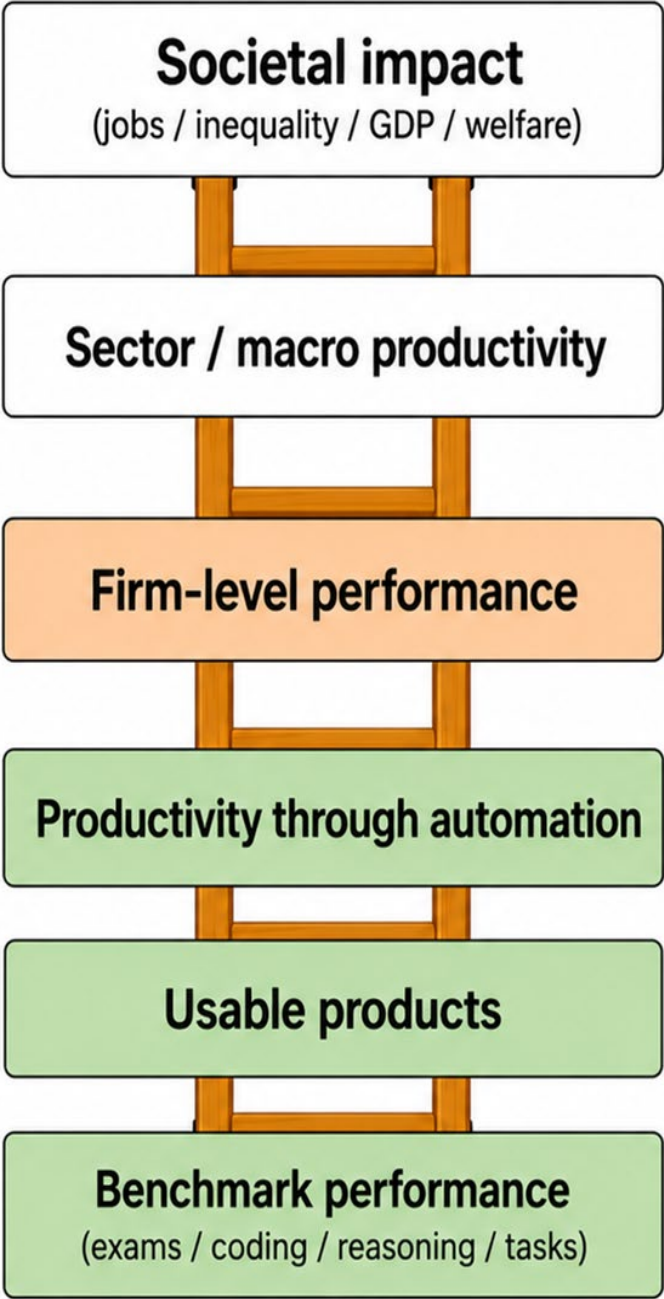
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No clear way to measure (yet)

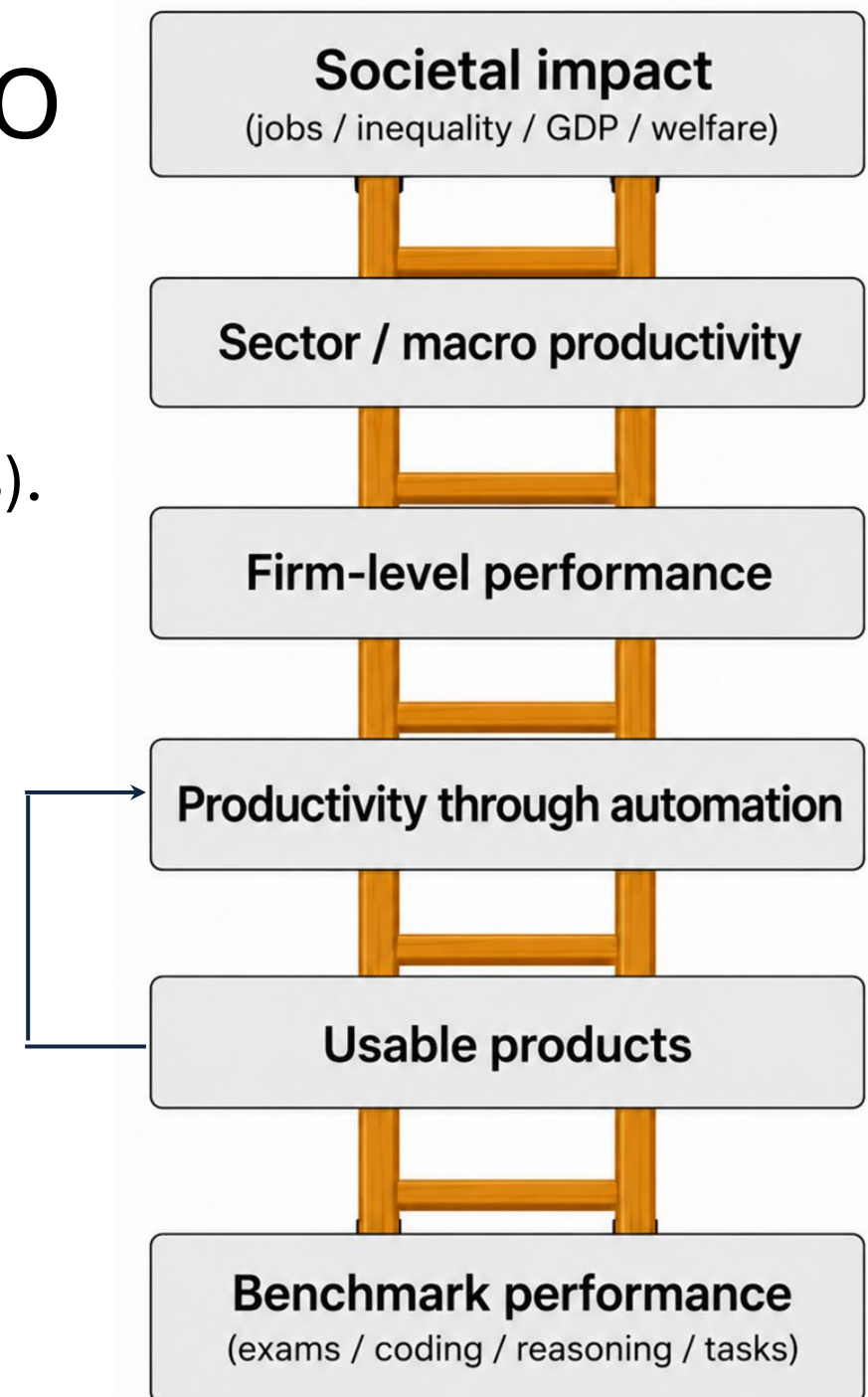
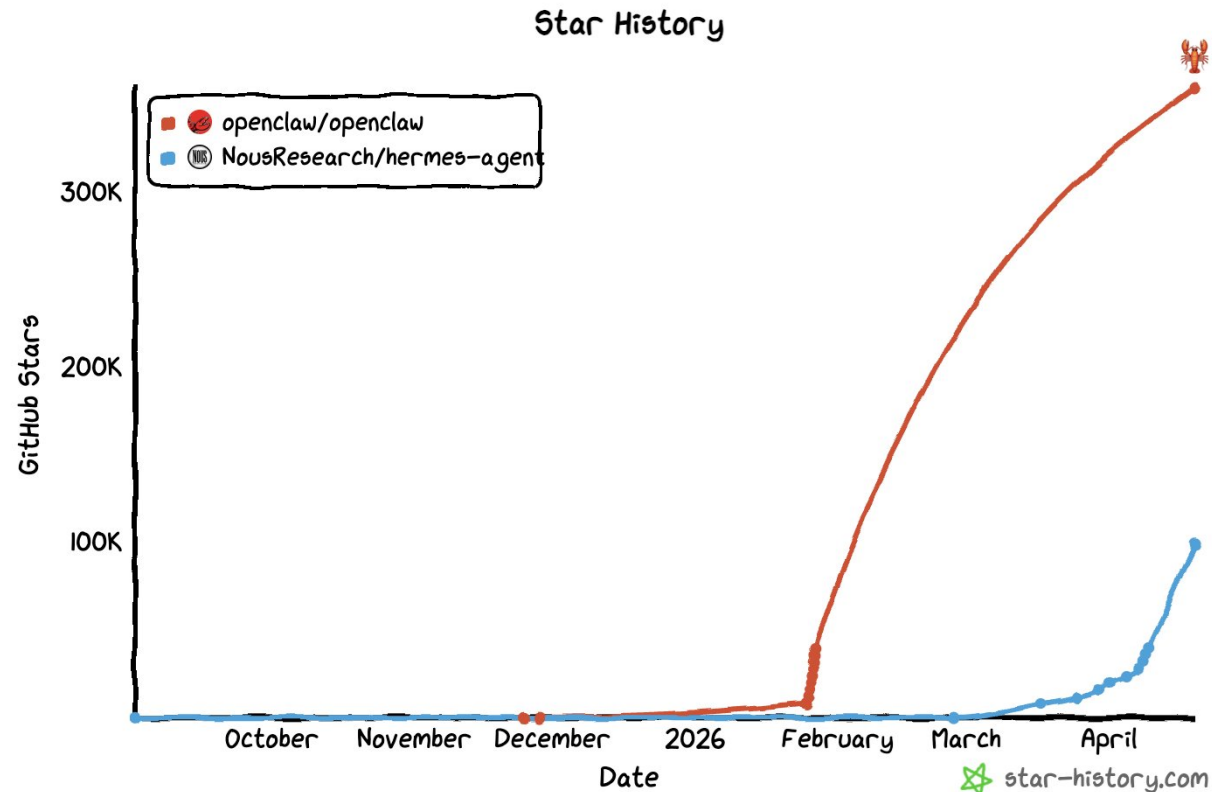
Conflicting evidence

Good evidence



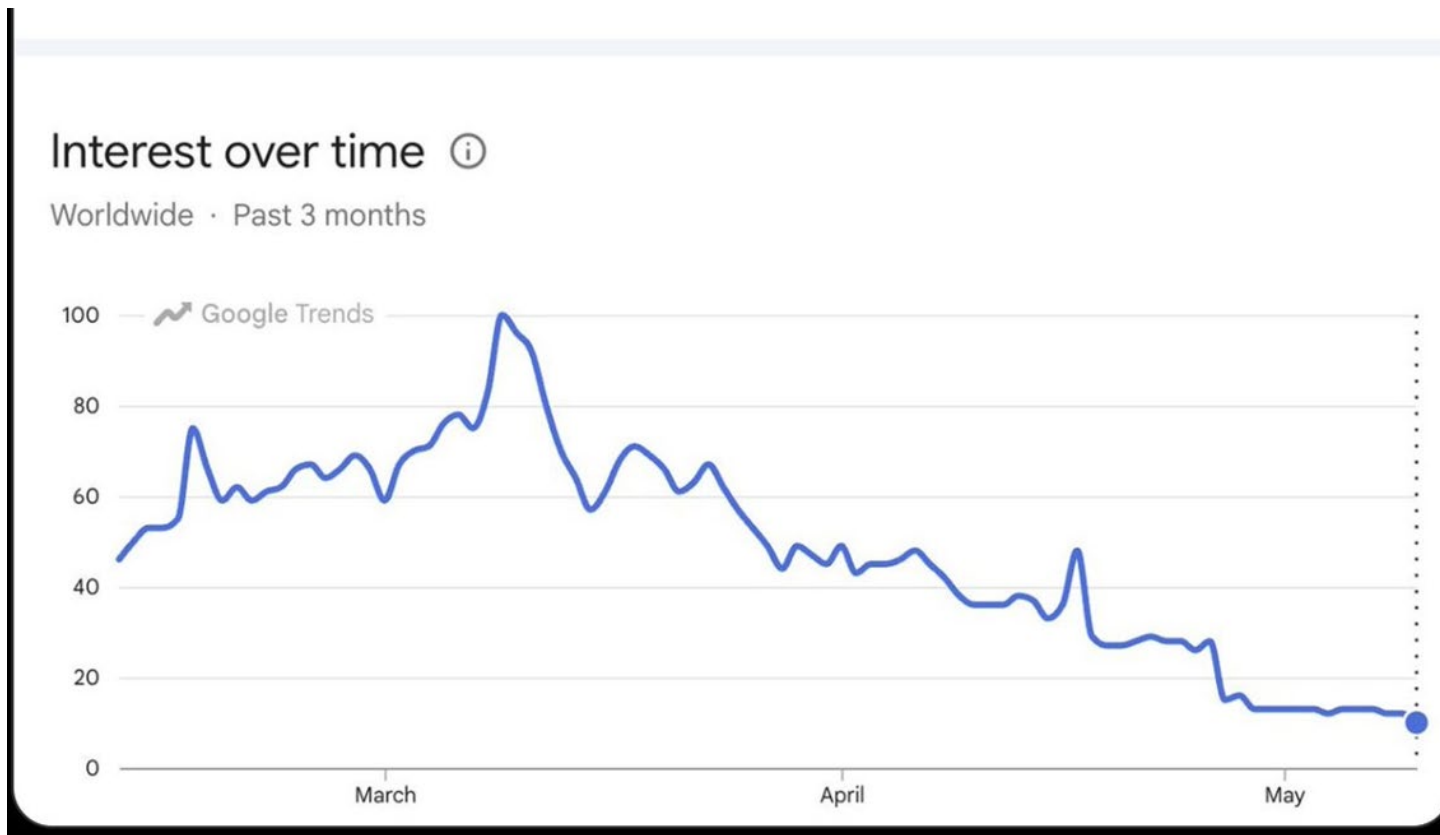
Impact needs foundations not FOMO

For true impact, each step needs good foundations.
Otherwise it is hype driven by FOMO (and AI Psychosis).



Impact needs foundations not FOMO

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Societal impact

(jobs / inequality / GDP / welfare)

Sector / macro productivity

Firm-level performance

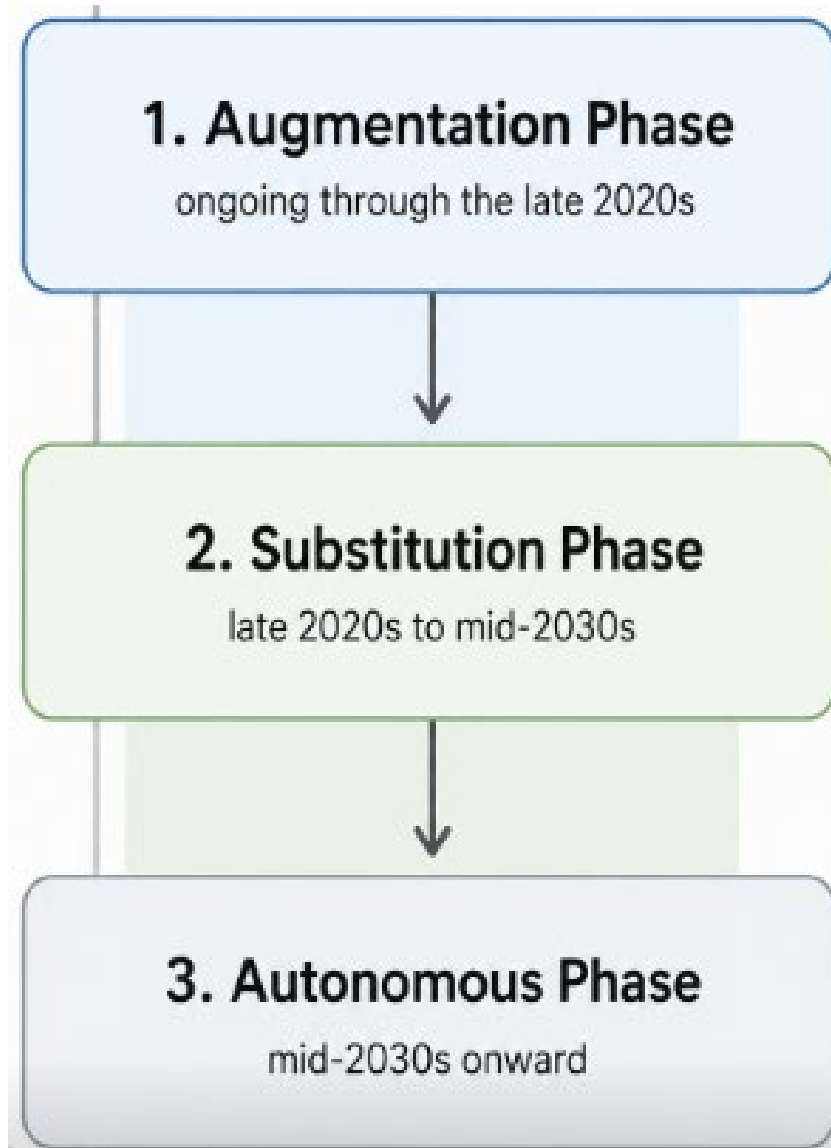
Productivity through automation

Usable products

Benchmark performance

(exams / coding / reasoning / tasks)

Ok, so, will autonomous agents replace humans on most economic tasks?



1. Augmentation Phase

- AI assists human workers
- Automates repetitive digital tasks
- Humans still decide and supervise

2. Substitution Phase

- AI takes over full workflows
- Replaces some human roles
- Humans handle exceptions and strategy

3. Autonomous Phase

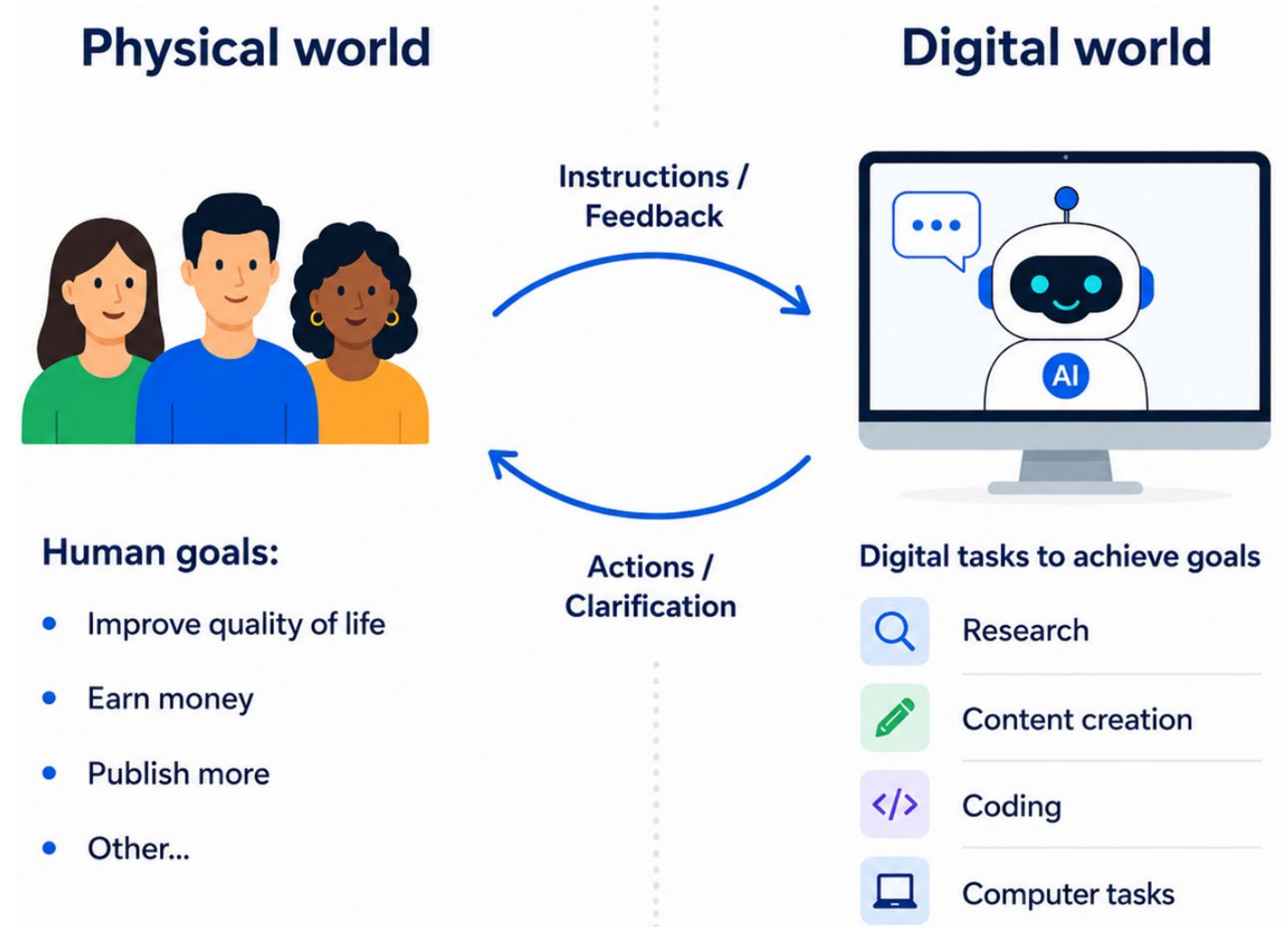
- AI agents plan and act independently
- Systems coordinate with minimal humans
- Humans set goals, limits, and governance

First, we have the augmentation

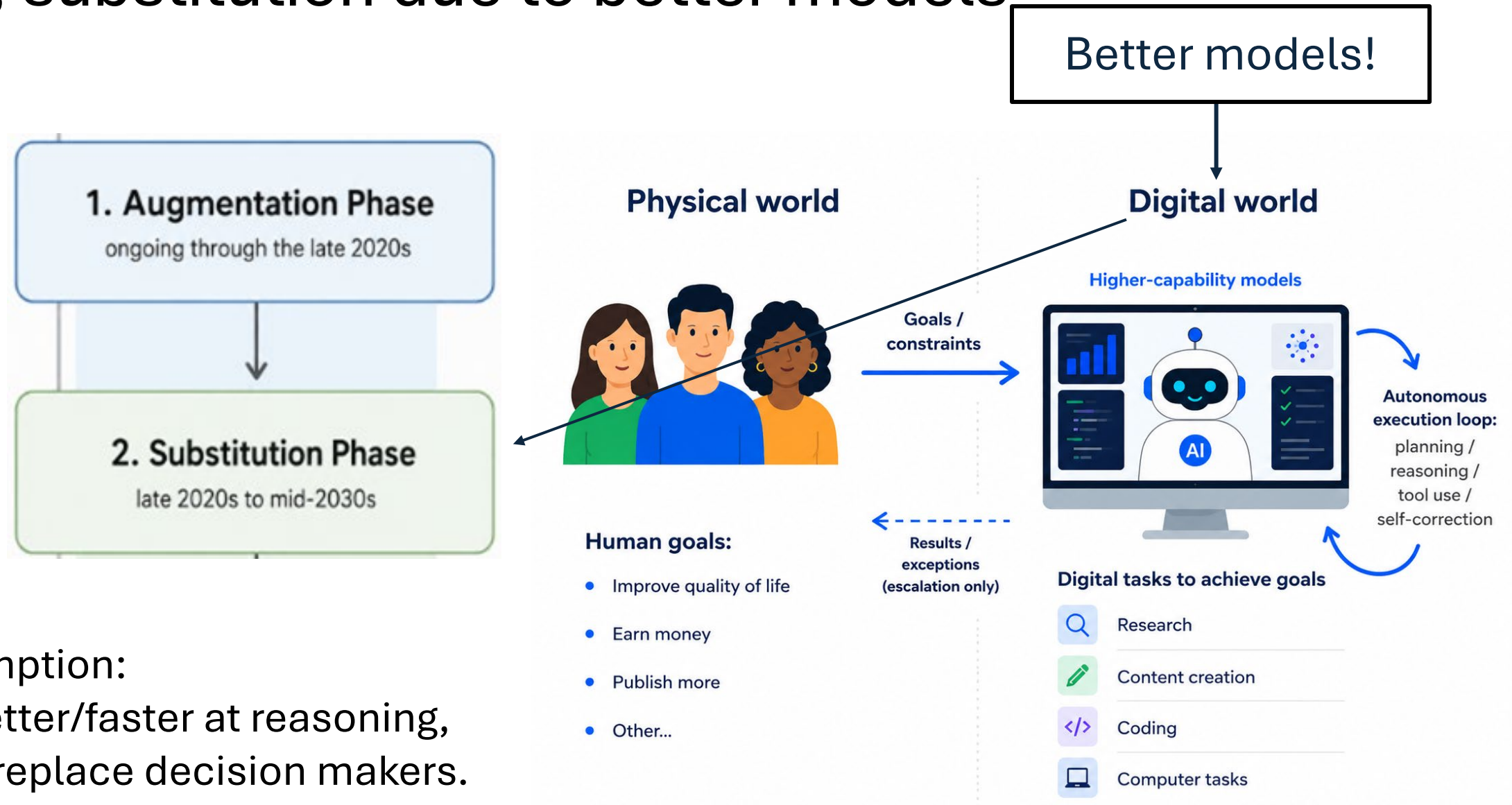
1. Augmentation Phase

ongoing through the late 2020s

- AI assists human workers
- Automates repetitive digital tasks
- Humans still decide and supervise

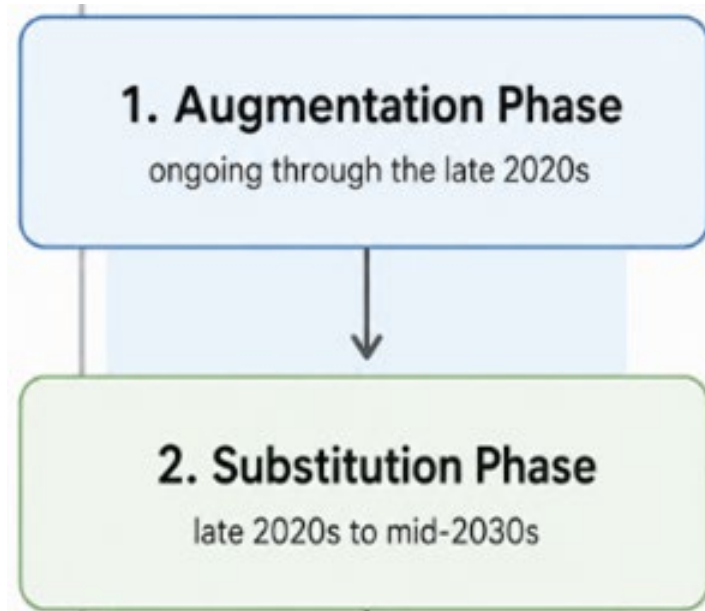


Then, substitution due to better models



The Assumption:
Models better/faster at reasoning,
thus they replace decision makers.

Then, substitution due to better models



The Assumption:
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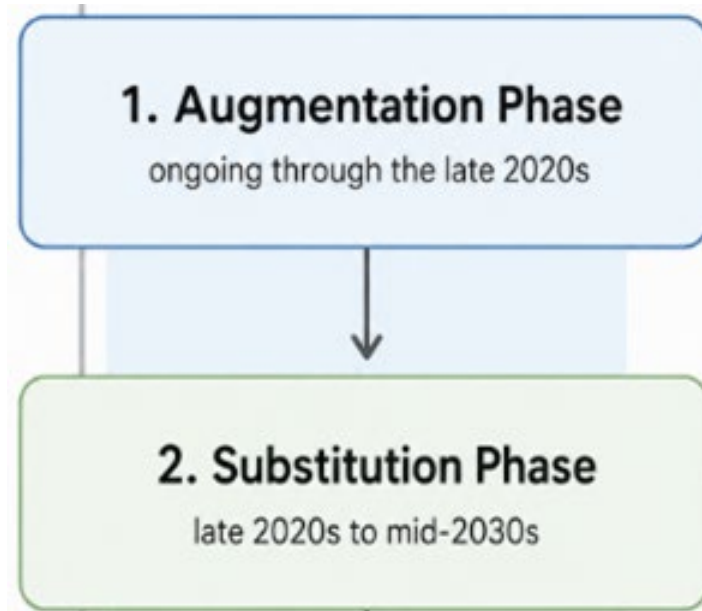


"I think that if you work as a radiologist, you are like Wile E. Coyote in the cartoon. You're already over the edge of the cliff, but you haven't yet looked down. There's no ground underneath. People should stop training radiologists now. It's just completely obvious that in five years deep learning is going to do better than radiologists."

Nov 24, 2016

As of May 2026, the demand for radiologists is at an all-time high, with job growth expected to rise 9% through 2034—significantly faster than the average for all occupations. Contrary to predictions a decade ago that AI would make radiologists obsolete, AI has instead become an essential tool that "supercharges" radiologists by handling high-volume tasks, leading to increased productivity and higher salaries, which reached an average of \$571,000 in 2025. Fortune +2

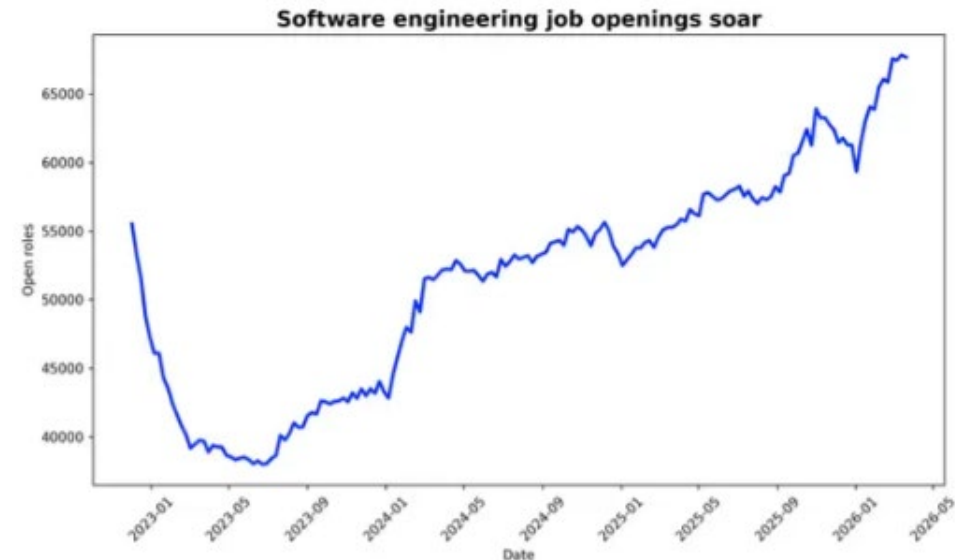
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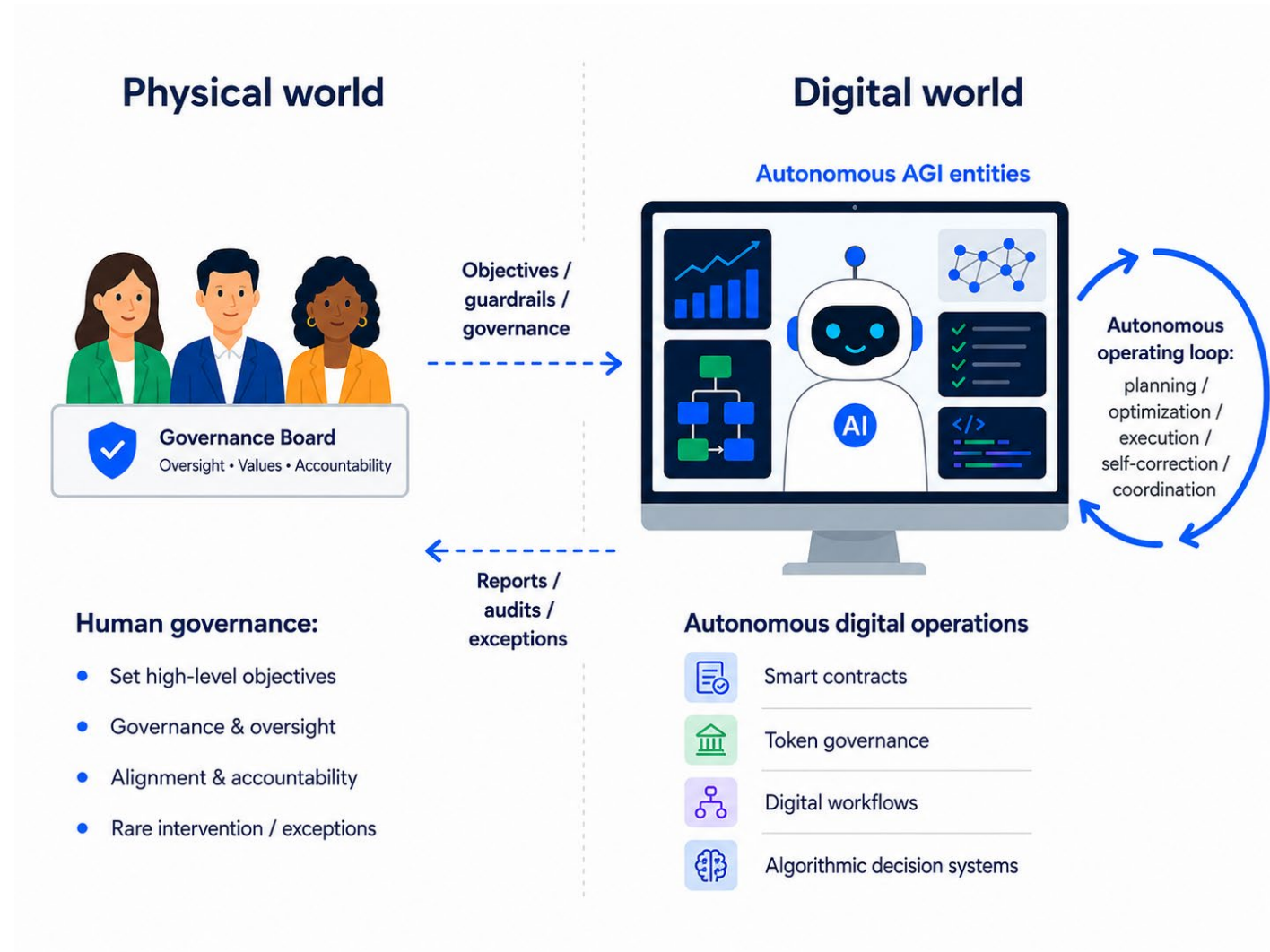
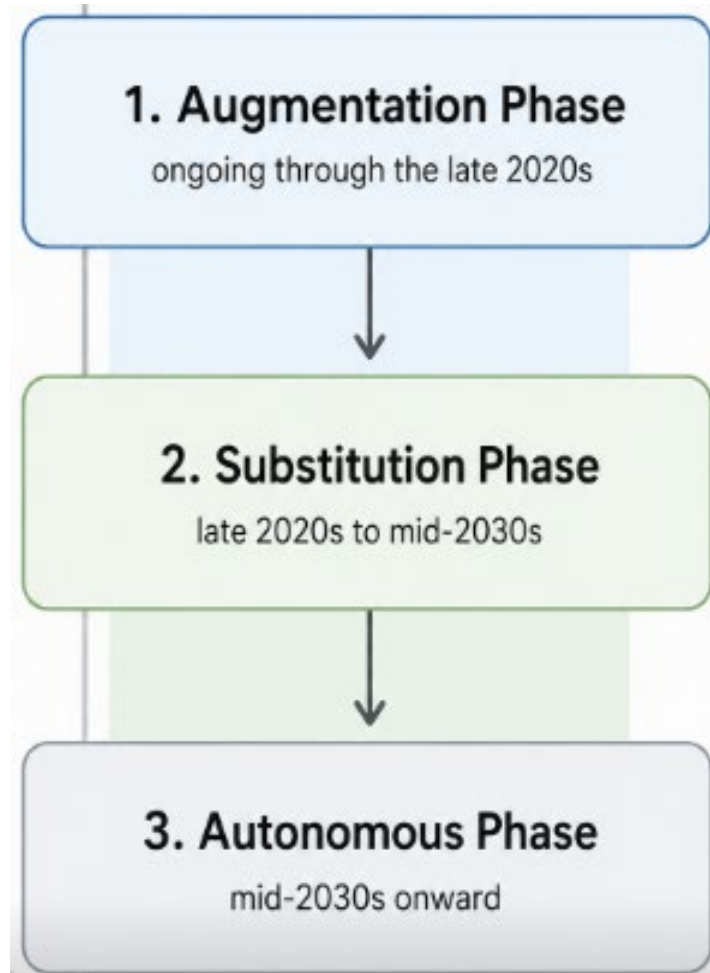
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Models better/faster at reasoning,
thus they replace decision makers.



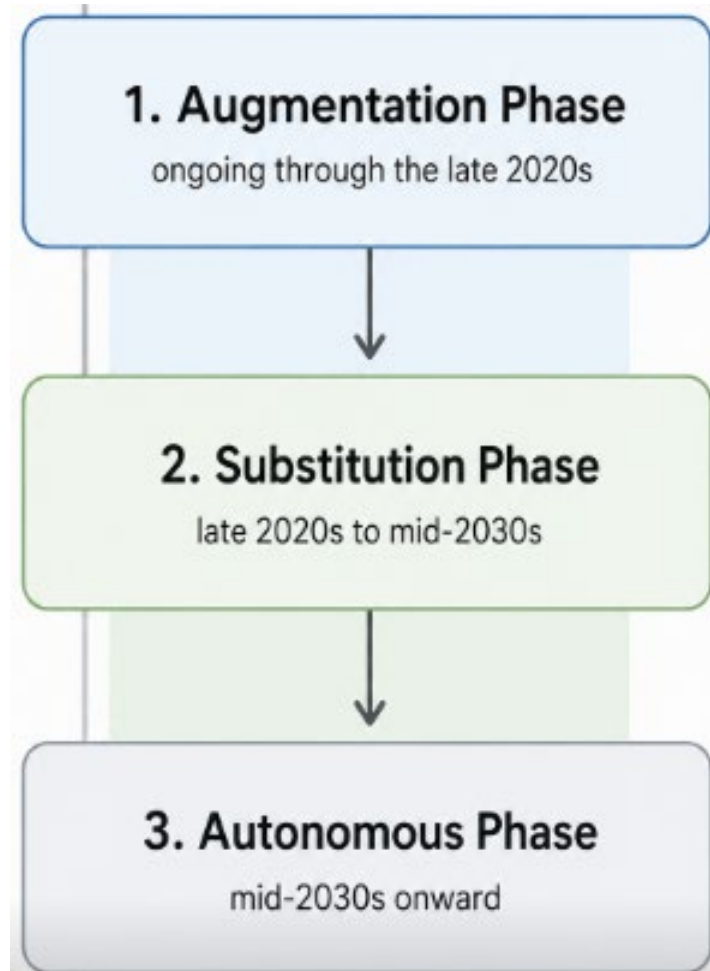
GPT-5.5, released in April 2026, sets a high-water mark for AI, achieving expert-level reasoning that outperforms previous models and **often rivals senior-level human performance in coding and technical tasks**. It excels in speed,, reducing 12 hours of human expert work to just 10 minutes, changing the cost dynamics of AI usage. [MindStudio +4](#)



And finally, full autonomy



And finally, full autonomy

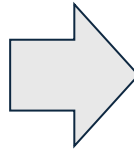
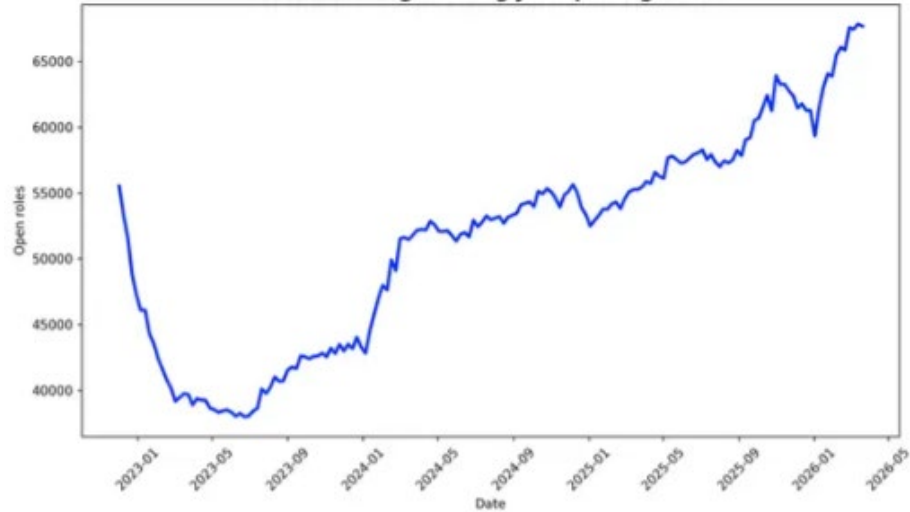


Huge uncertainty for such predictions.

- 10 years in AI years is almost a human century
 - New best model every 2 months
- Current models work best on verifiable tasks
 - Eg best models are coding models
- Architectural breakthroughs are needed
 - Solve continuous learning
 - Solve long term planning
 - Solve memory
- What fuels these autonomous entities?

A more naïve way of looking at this.

Software engineering job openings soar



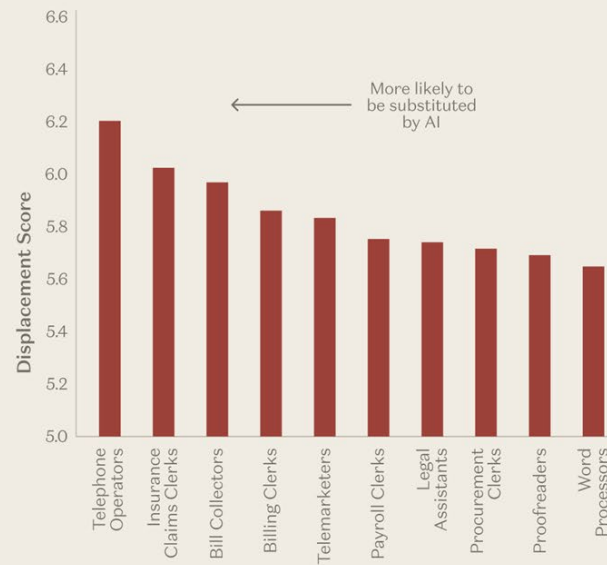
Software gets ~x% cheaper

- Requirements gathering
- Prototyping
- More productive engineers

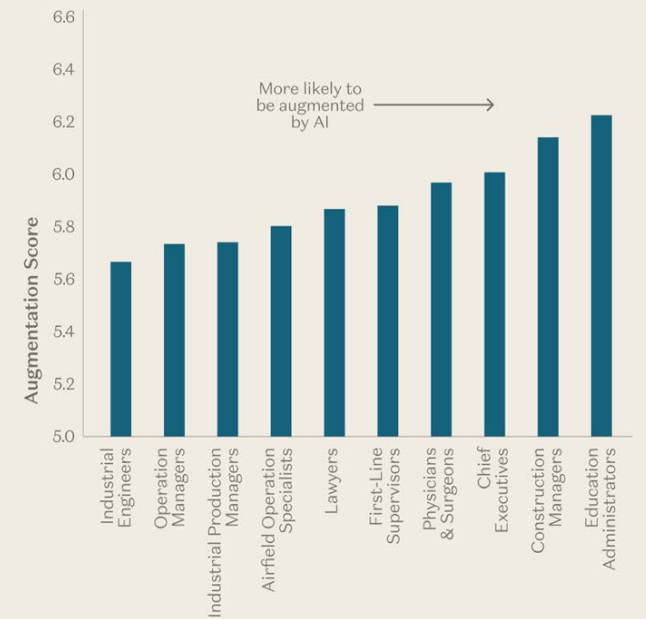
With Great Substitution, Comes Great Augmentation

For every job at-risk of substitution, there are other jobs that stand to benefit

High Exposure to AI Substitution



High Exposure to AI Augmentation



Source: Goldman Sachs Global Investment Research (April 2026)

<https://a16z.com/the-ai-job-apocalypse-is-a-complete-fantasy/>

Lets look at the data:

The actual problem with the labour market.

New job starts among workers age 22-25 in occupations with high and no AI exposure

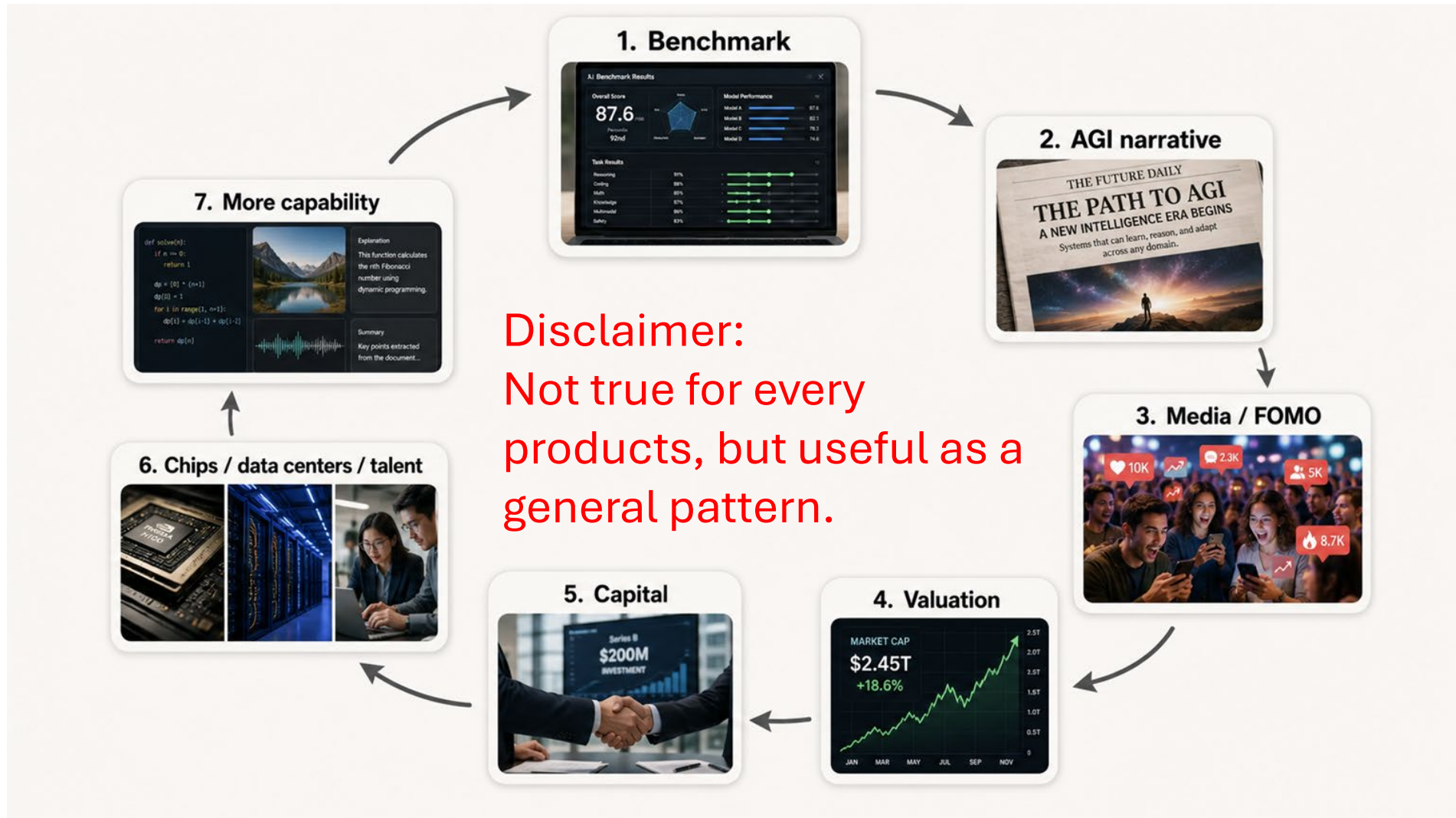


Figure 7: New job starts among workers age 22-25 in occupations with high observed exposure and no AI exposure, Current Population Survey

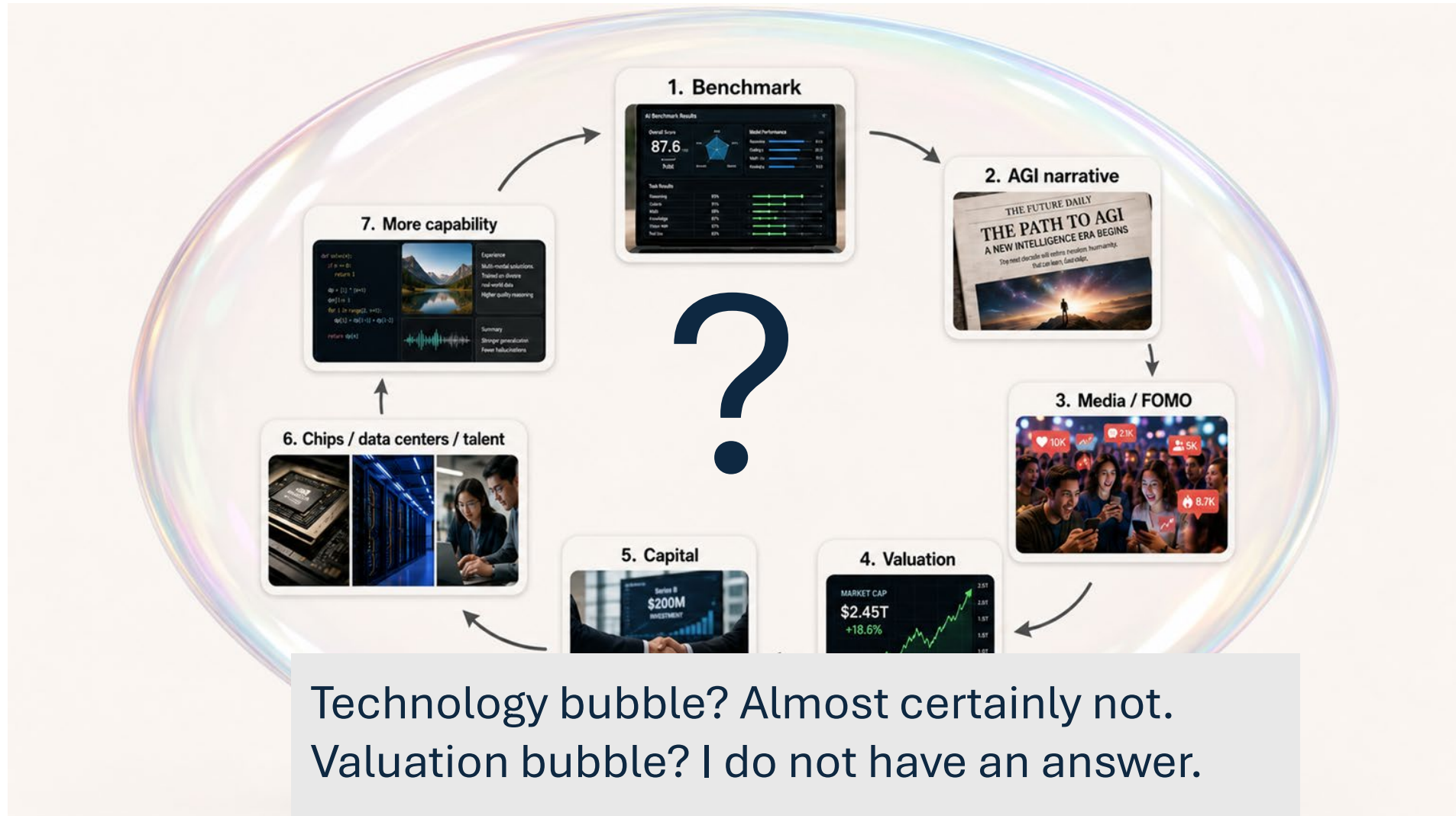
The top panel shows the percent of young workers starting new jobs in high vs. no exposure occupations. The bottom panel measures the gap between these two series in a difference-in-differences framework.

The real question:
What does this
mean for
education?

The expectation loop & the economics of AI



The expectation loop & the economics of AI



Some problems with the expectation loop

Raises at 50B\$



Bloomberg.com

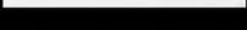
<https://www.bloomberg.com/news/articles/ai-codin...>

AI Coding Startup Cursor Plans New Model to Rival ...

Mar 19, 2026 — **Cursor** has also been in talks to **raise** a new round of financing at a roughly **\$50 billion** valuation, **Bloomberg** News reported this month. [Read more](#)



Composer 2 includes our first continued pretraining run and scaled reinforcement learning

Model	Terminal-Bench 2.0 score
GPT-5.4	75.1% 
Composer 2	61.7 
Opus 4.6	58.0 
Opus 4.5	52.1 
Composer 1.5	47.9 

Trains their in-house model to rival big labs



Kimi.ai  @Kimi_Moonshot · 14 ü



Congrats to the [@cursor_ai](#) team on the launch of Composer 2!

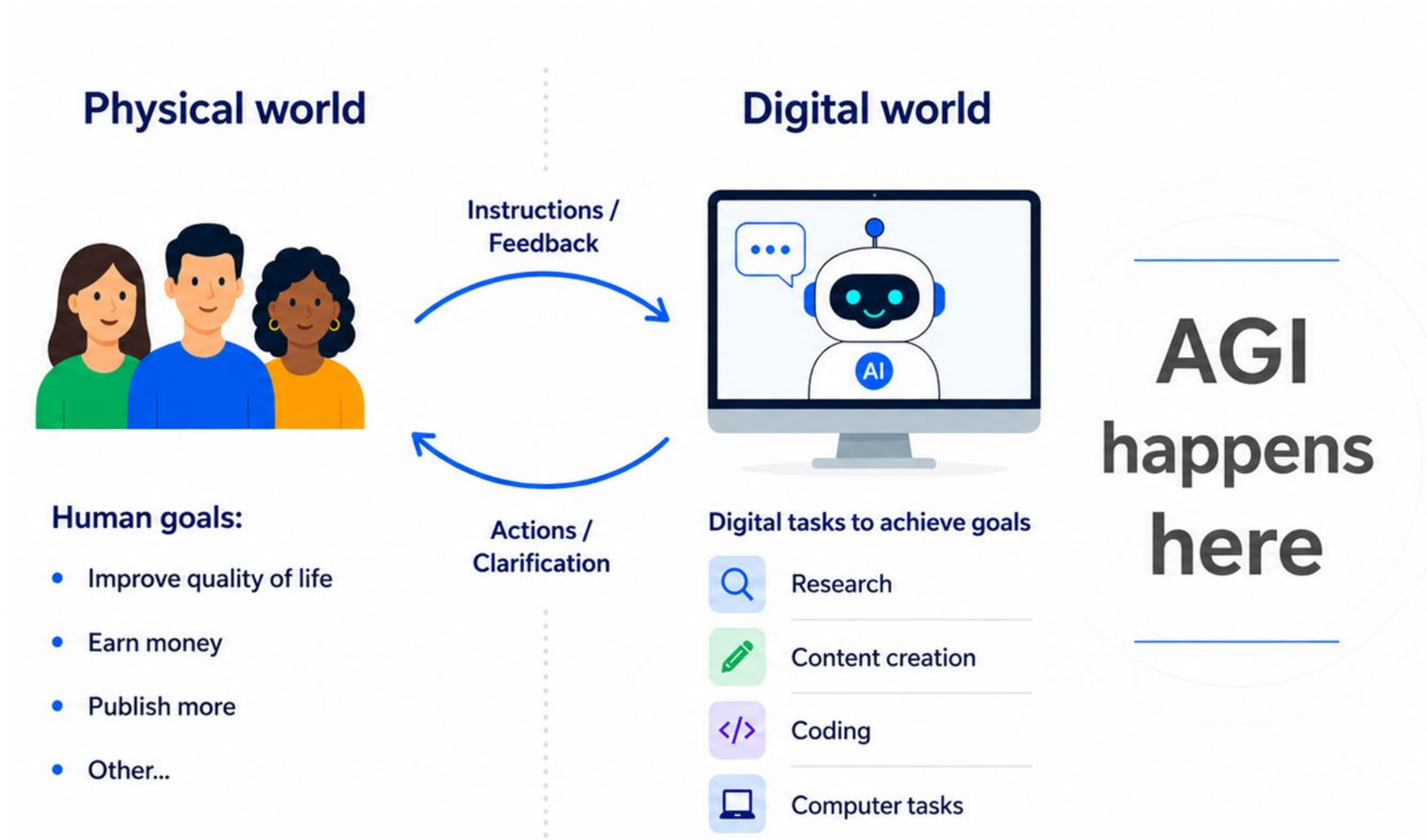
We are proud to see Kimi-k2.5 provide the foundation. Seeing our model integrated effectively through Cursor's continued pretraining & high-compute RL training is the open model ecosystem we love to support.

Note: Cursor accesses Kimi-k2.5 via [@FireworksAI_HQ](#) ' hosted RL and inference platform as part of an authorized commercial partnership.

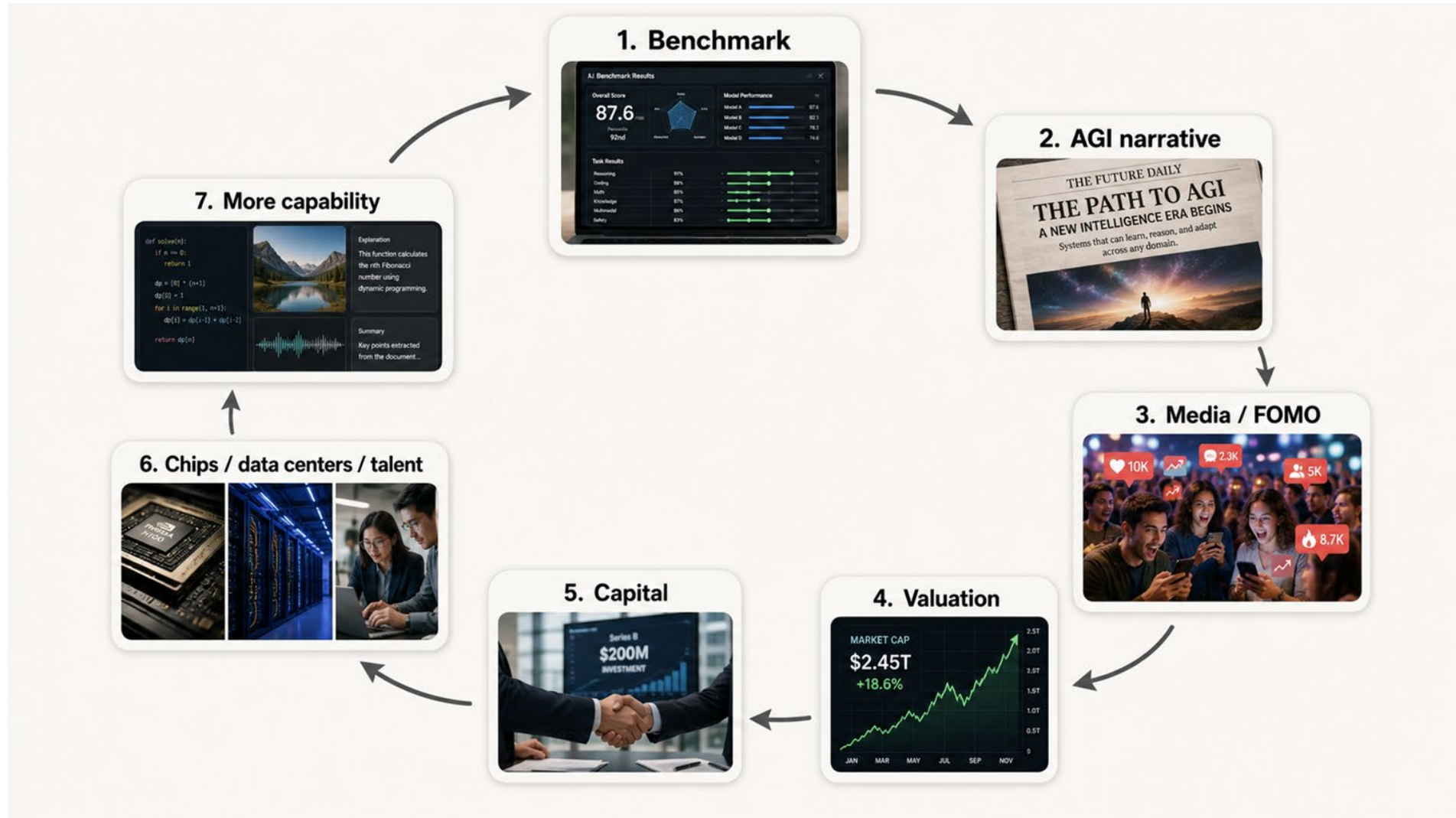
Its fine-tuned Kimi k2.5
Moonshot is valued at ~4.3B

Where the problems begin.

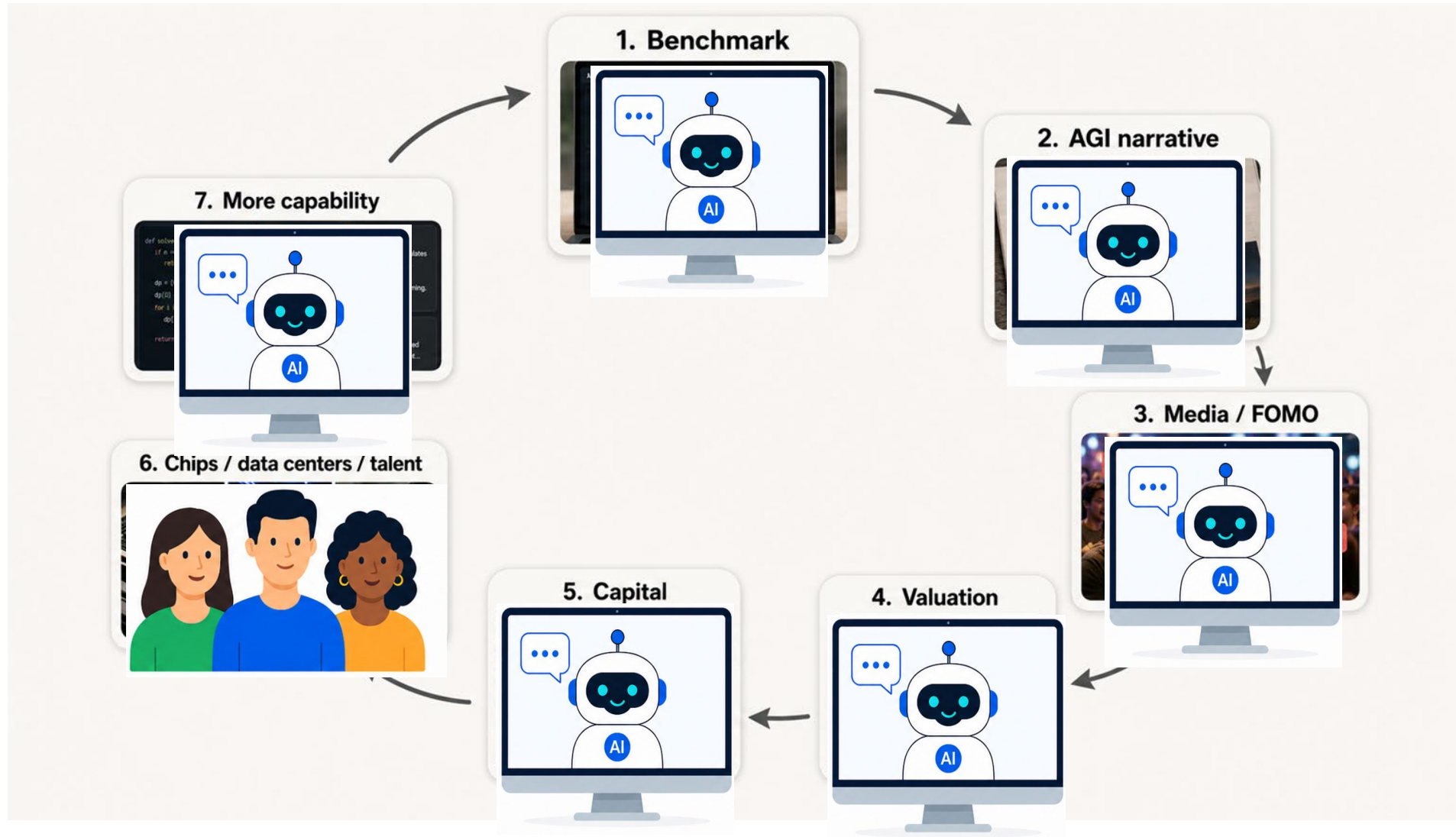
What about the problems here



Where the problems begin.



Where the problems begin.



Where the problems begin.

Physical world



6. Chips / data centers / talent



Reuters

<https://www.reuters.com> › markets › commodities › us-...

[US AI boom faces electric shock](#)

Feb 25, 2026 — Big Tech's race to dominate artificial intelligence **may soon hit a nasty road bump** as U.S. electricity grids struggle to keep pace with the ...



Nature

<https://www.nature.com> › news

[Boycott of major AI conference exposes a growing US–](#)

Apr 14, 2026 — Boycott of major AI conference exposes a growing US–China divide. **China-based researchers make up more than half of all lead authors at NeurIPS.** [Read more](#)



WSJ

<https://www.wsj.com> › tech › spacex-google-in-talks-to...

[SpaceX and Google Are in Talks to Launch Data Centers](#)

18 hours ago — A launch deal would put the two companies in partnership as they gear up to compete on orbital **data centers**, an unproven technology that **SpaceX** ... [Read more](#)



Financial Times

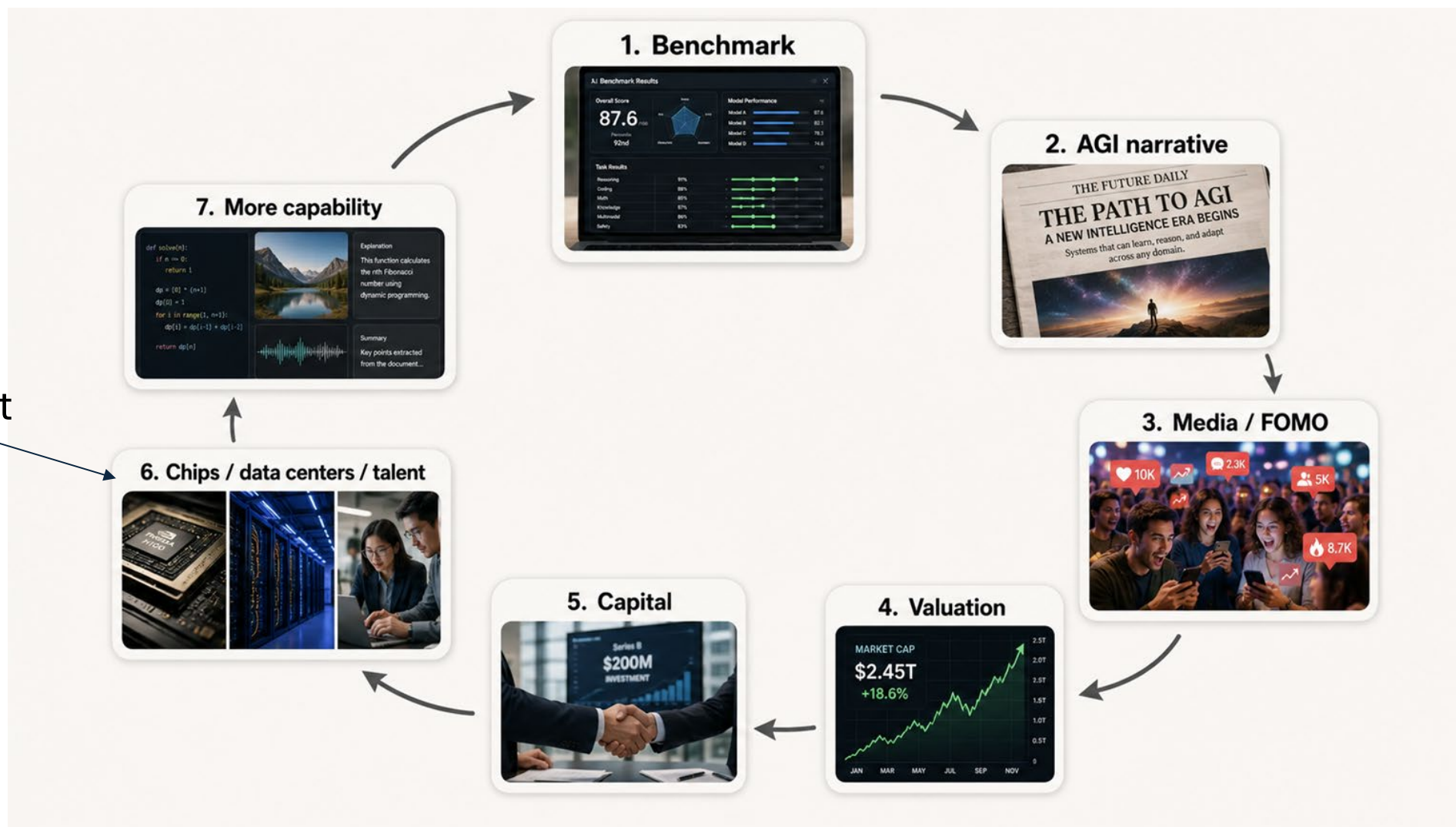
<https://www.ft.com> › ... › Asia-Pacific economy › China

[Nvidia AI chip sales to China stalled by US security review](#)

Feb 3, 2026 — **Nvidia's sales of H200 AI chips to China are still awaiting final approval** from Washington nearly two months after Donald Trump greenlit ... [Read more](#)

Where the problems begin.

Question:
What if this part
slows down?



Where the problems begin.

Question:
What if this part
slows down?



Ownership and inequality

Argument: NVIDIA makes tone of money, **thus** impact is already here

Nvidia's 2025 revenues of \$130 billion are not far off from the \$126 billion GDP of Ethiopia, the second most populous African nation. What makes this comparison intriguing is that Nvidia's revenues were generated by 36,000 employees, resulting in a contribution of \$3.6 million per employee; Ethiopia's 2025 GDP was produced by a labor force of 55 million, resulting in \$2,300 per worker. This mind-

Is this due to Technology?

Or ownership (and influence)?

Ownership and inequality

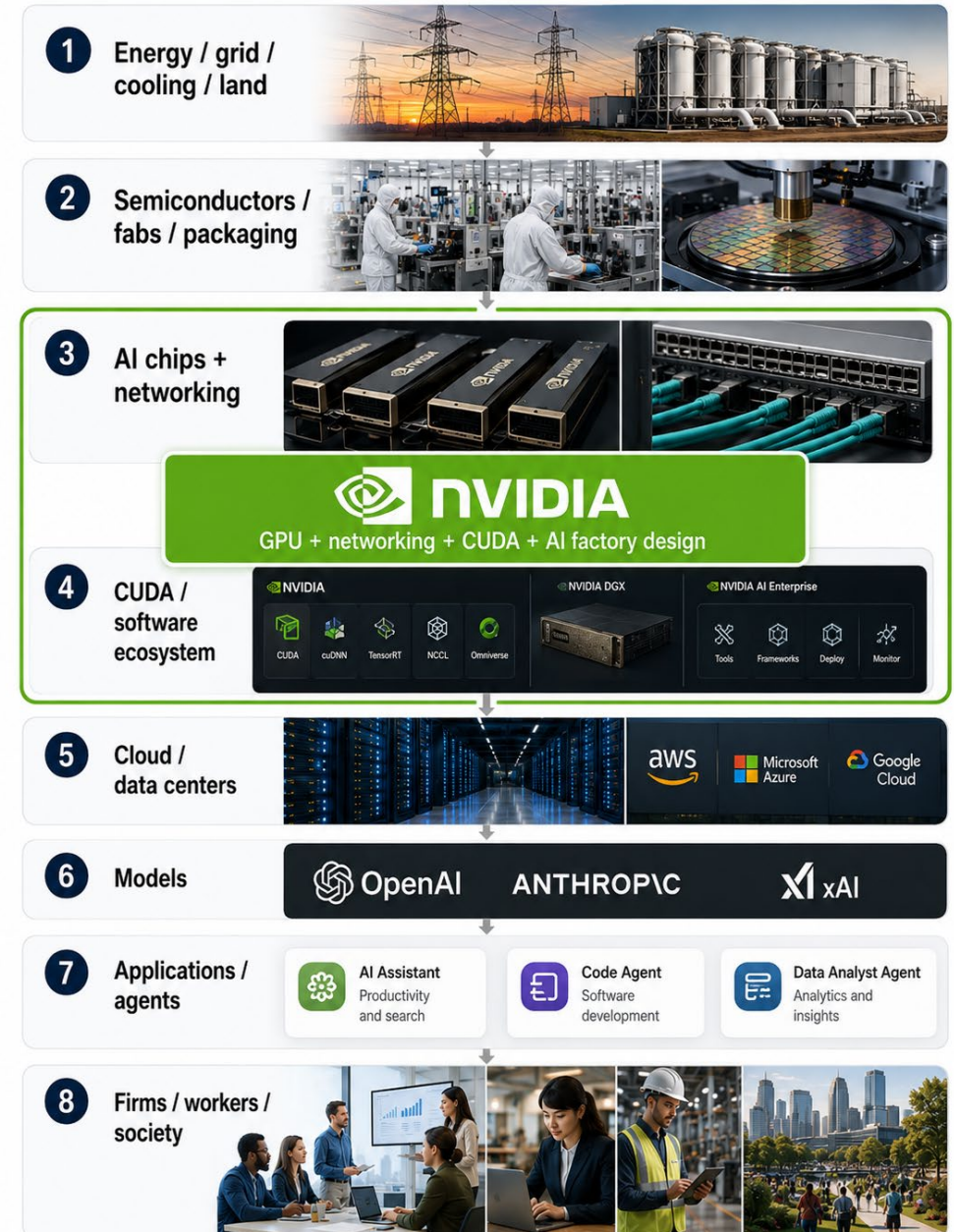
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The AI supply chain



Ownership and inequality

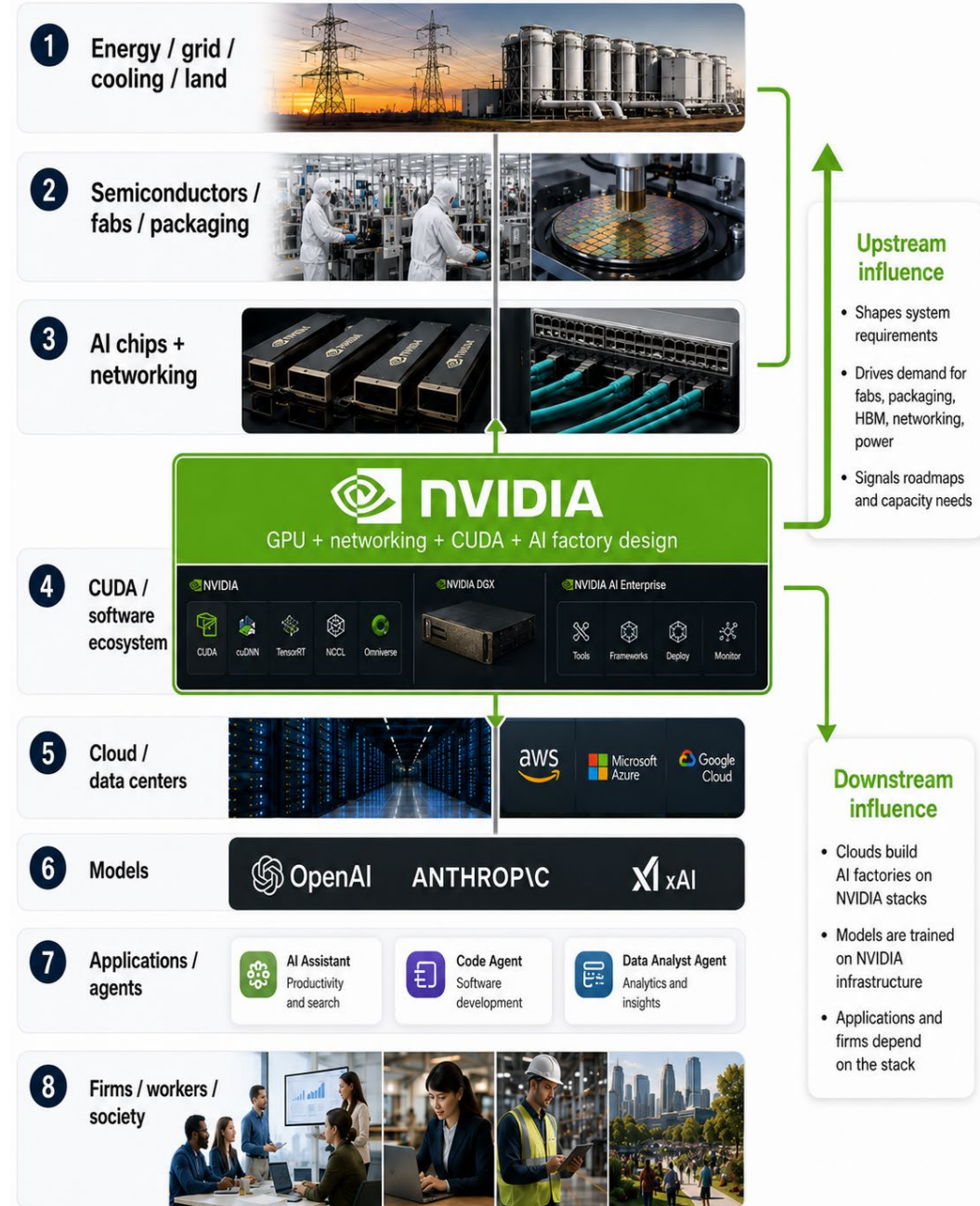
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The AI supply chain

Chips



AMD



ASML

Cloud



Models



ANTHROPIC



Meta



7

Applications /
agents



ANTHROPIC



Meta



8

Firms / workers /
society



Ownership and inequality

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The AI supply chain

Chips



AMD



ASML

Cloud

Hypothetical questions:

Q1: What is the % of these companies in the stock market?

Model

Q2: What happens to the stock market without these companies?

7

Application agents

Q3: What about potential disruptions in the supply chain?

Q4: Who benefits with the actual progress/adoption of AI?

8

Firms / workers / society



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Chips



AMD



ASML

Cloud



Models



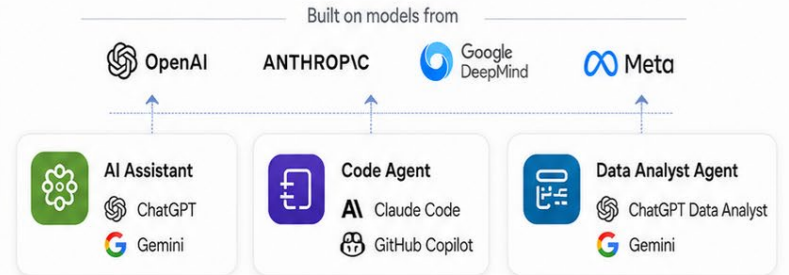
ANTHROPIC



Meta



7
Applications / agents



8
Firms / workers / society



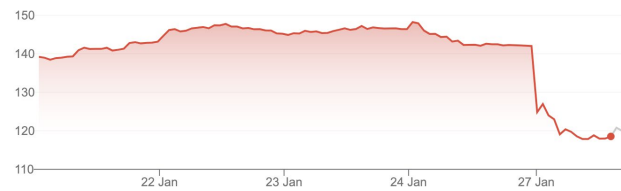
Market Summary > NVIDIA Corp

118.58 USD

-20.64 (-14.83%) **↓** past 5 days

Closed: 27 Jan, 5:07 PM GMT-5 • Disclaimer
After hours 120.66 +2.09 (1.76%)

1D 5D 1M 6M YTD 1Y 5Y Max



DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

DeepSeek-AI

research@deepseek.com

Abstract

Ow



Reuters

<https://www.reuters.com> › business › energy › white-ho...

White House scrambles for gas-price relief as Iran war ...

3 hours ago — ... **prices** even higher this summer. **Energy prices** have spiked since **Iran** cut off access to the Strait of Hormuz, a waterway that normally ... [Read more](#)



Politico

<https://www.politico.com> › news › 2026/05/14 › taiwan...

Taiwan emerges as flash point in Trump-Xi talks

23 hours ago — BEIJING — The sensitive issue of **Taiwan** intruded Thursday on a high-stakes meeting between U.S. President Donald Trump and Chinese leader Xi ... [Read more](#)



Reuters

<https://www.reuters.com> › business › media-telecom › c...

China's Huawei hypes up chip and computing power plans ...

Sep 18, 2025 — The announcement **broke years of secrecy about its chips business** and could increase the stakes in the U.S.-Sino war for tech supremacy.

The AI supply chain

- 1 Energy / grid / cooling / land



- 2 Semiconductors / fabs / packaging



- 3 AI chips + networking



- 4 CUDA / software ecosystem



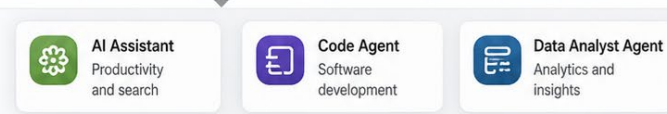
- 5 Cloud / data centers



- 6 Models



- 7 Applications / agents



- 8 Firms / workers / society



Ownership (and influence)?

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Is this due to Technology?

Or ownership (and influence)?

Honestly at this point, the bottleneck does not feel to be something technical.

I am the bottleneck. Me ,my cognitive limitations, and my salary that I can't afford 5 Claude max subscriptions.

I can empty my to-do lists in a night, I can build anything I want.

I am only limited by the quality of the high level abstract ideas I generate.

7:52 μ.μ. ✓

Question 1:

Who owns the technology and who gets access?

... BBC

Anthropic's Claude Fable 5 and Mythos 5 AI suspended over security fears

Anthropic's Claude Fable 5, released publicly this week, sparked concerns about cybersecurity and hacking.



AI Jazeera

US lifts restrictions on Anthropic's powerful AI models Fable and Mythos

AI firm says it will begin restoring access to Claude Fable 5 and Mythos 5 after the removal of export controls.



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Or ownership (and influence)?

Question 1:

Who owns the technology and who gets access?

Question 2:

Who aligns the aligners & who regulates the regulators?

■ Once an AI system exceeds human-level intelligence, if granted the capability of self-improvement it could set its own goals that may conflict with those of humans. This creates an alignment problem: the challenge of designing a system with objectives that remain stable and human-compatible even as the system undergoes radical self-modification and acquires power far beyond that of human capabilities.



BBC

<https://www.bbc.com/news/articles>

Judge rejects Pentagon's attempt to 'cripple' Anthropic

Mar 26, 2026 — A federal judge told the government it could not immediately enforce a ban on Anthropic's tools.



Anthropic

<https://www.anthropic.com/news/where-stand-depar...>

Where things stand with the Department of War

Mar 5, 2026 — Yesterday (March 4) Anthropic received a letter from the Department of War confirming that we have been designated as a supply chain risk to ... [Read more](#)



OpenAI

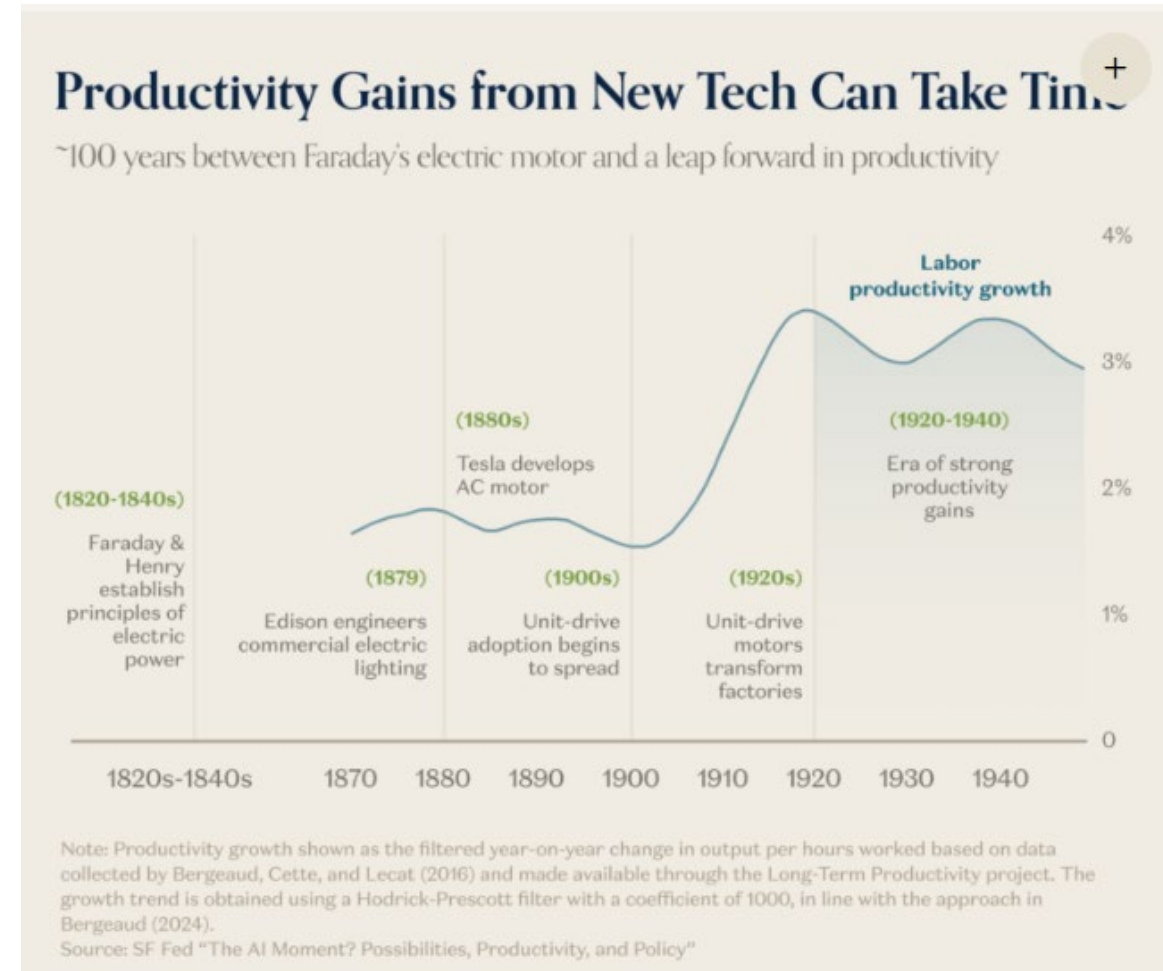
<https://openai.com/index/our-agreement-with-the-de...>

Our agreement with the Department of War

Mar 2, 2026 — Yesterday we reached an agreement with the Pentagon for deploying advanced AI systems in classified environments, which we requested they also ... [Read more](#)

Wrapping up. Why forecasting the impact of AI is hard

1. Trends lean towards augmentation -> Cheap software + more digitalization
2. No striking productivity gains (firm level, not worker level). These will show during consolidation and mass augmentation.
3. Physical boundaries should be considered.
4. No evidence of a significant job market shift (yet)
5. Revenue and valuation might be weak predictors.



And at the end, it might be a choice and not a prediction.

“The question is not what AI will do to us, but what we are going to do with AI”,

Erik Brynjolfsson, Stanford economist.



From student anxieties to vibe-coding, Brynjolfsson frames A.I. as a tool that amplifies intention rather than replaces workers. Photo by Roy Rochlin/Getty Images for Unfinished Live

A quick summary

- AI is not magic. It is a combination of data, compute and smart algorithms
- The core technology is somewhat established. We now build upon it.
- Do not mess capability with impact. AI progress is real. Impact is uneven.
- Productivity gain need systems. Not models.
- Full automation is not yet predictable. We are at early augmentation.
- Cheap software might create more digital job, not less
- Ownership may matter as much as capability
- Forecasting the impact of AI is very hard. It takes time and society adopts.
- The future is not only predicted. It is also designed.

Thank you for the attention 😊



filotas.theodosiou@vives.be



<https://github.com/FilTheo>



<https://www.linkedin.com/in/filotas-theodosiou/>



<https://filtheo.github.io/>